

**P802.1DD/Draft 1.3
September 17, 2025**

Draft Standard for Local and metropolitan area networks— Resource Allocation Protocol

Unapproved draft, prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1
Sponsored by the
LAN/MAN Standards Committee
of the
IEEE Computer Society

This and the following cover pages are not part of the draft. They provide revision and other information for IEEE 802.1 Working Group members and participants in the IEEE Standards Association ballot process, and will be updated as convenient. New participants: Please read these cover pages, they contain information that should help you contribute effectively to this standards development project. Blank pages allow for the addition of cross-references to changed text without forcing renumbering of all pages in the draft. Pages are numbered from 1 (including cover pages) for the convenience of reviewers whose PDF viewers do not easily accommodate different numbering sequences. Pages will of course be renumbered prior to publication.

The text proper of this draft begins with the [Title page](#).

1 Editors' Foreword

2 Throughout this document any notes with a Cyan background (as in this paragraph) are temporary, inserted
3 by the Editors for a variety of purposes. They will be removed prior to SA Ballot and are not part of the
4 normative text. These cover pages will be edited, but will be retained so that information for the stages of SA
5 ballot can be included (including, if appropriate, cross-references to text changed in the course of SA
6 balloting). To avoid changes to page numbering between final WG ballot and initial SA Ballot, the number of
7 pages in this set of cover pages should remain unchanged, with pages intentionally left blank marked as
8 such. The records of participants in the development of the standard will be added at an appropriate time.

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10 All participants in the standardization activities of IEEE 802.1 should be aware of the Working Group Policies
11 and Procedures, and the fact that they have obligations under the IEEE Patent Policy, the IEEE Standards
12 Association (SA) Copyright Policy, and the IEEE SA Participation Policy. For information on these policies see
13 1.ieee802.org/rules/ and the slides presented at the beginning of each of our Working Group and Task Group
14 meeting.

15 As part of our IEEE 802® process, the text of the PAR (Project Authorization Request) and CSD (Criteria for
16 Standards Development) of each project is reviewed regularly to ensure their continued validity. The PAR is
17 summarized in these cover pages and links are provided to the full text of both PAR and CSD. A vote of
18 "Approve" on this draft is also an affirmation that the PAR and CSD for this project are still valid.

19 Comments on this draft are encouraged. NOTE: All issues related to IEEE standards presentation style,
20 formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to
21 publication, after balloting and the process of achieving agreement on the technical content of the standard is
22 complete. Readers are urged to devote their valuable time and energy only to comments that materially affect
23 either the technical content of the document or the clarity of that technical content. Comments should not
24 simply state what is wrong, but also what might be done to fix the problem.

25 Full participation in the work of IEEE 802.1 requires attendance at IEEE 802 meetings. Information on 802.1
26 activities, working papers, and email distribution lists etc. can be found on the 802.1 Website:

27 <http://ieee802.org/1/>

28 Use of the email distribution list is not presently restricted to 802.1 members, and the working group has a
29 policy of considering comments from all who are interested and willing to contribute to the development of the
30 draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity
31 in past projects. The email lists exist primarily to allow the members of the working group to develop
32 standards, and are not a general forum. All contributors to the work of 802.1 should familiarize themselves
33 with the IEEE patent policy and anyone using the email distribution list will be assumed to have done so.
34 Information can be found at <http://standards.ieee.org/db/patents/>

35 Comments on this draft may be sent to the 802.1 email exploder, to the Editor, or to the Chairs of the 802.1
36 Working Group and Time-Sensitive Networking (TSN) Task Group.

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45 acknowledged.

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2 **and should familiarize themselves with that policy, see**
3 <http://standards.ieee.org/about/sasb/patcom/materials.html>

4 As part of our IEEE 802 process, the text of the PAR and CSD (Criteria for Standards Development, formerly
5 referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity.
6 A vote of "Approve" on this draft is also an affirmation by the balloter that the PAR is still valid.

7 **Draft development**

8 During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically
9 coherent drafts from the resolutions of ballot comments and from the other discussions that take place in the
10 working group meetings. Preparation of drafts often exposes inconsistencies in editor's instructions or
11 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
12 and requests by the editors' for contributions on specific issues will be found in the editors' [Introduction to the](#)
13 [current draft](#) and at appropriate points in the draft.

14 The ballot comments received on each draft, and the editors' proposed and final disposition of comments on
15 working group drafts, are part of the audit trail of the development of the standard and are available, along
16 with all the revisions of the draft on the 802.1 website (for address see above).

17 During the early stages of draft development the proposed text can be moved around a great deal, and even
18 minor rearrangement can lead to a lot of 'change', not all of which is noteworthy from the point of the reviewer,
19 so the use of automatic change bars is not very effective. In early drafts change bars may be omitted or
20 applied manually, with a view to drawing the readers attention to the most significant areas of change.
21 Readers interested in viewing every change are encouraged to use Adobe Acrobat to compare the document
22 with their selected prior draft. Note that the FrameMaker change bar feature is useless when it comes to
23 indicating changes to Figures.

1 **iPAR (Project Authorization Request) and CSD**

2 Extracts from the PAR, as approved by IEEE NesCom 26th September 2024:

3 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/11709>

4 and the CSD (Criteria for Standards Development):

5 <https://www.ieee802.org/1/files/public/docs2024/dd-CSD-0724-v01.pdf>

6 **PAR Scope:**

7 This standard specifies protocols, procedures, and managed objects for resource allocation in bridged local
8 area networks for dynamic creation and maintenance of data streams. This standard supports control
9 signaling through data paths and through separate control paths in support of centralized control. This
10 standard makes provisions for backward compatibility with the Stream Reservation Protocol specified in
11 IEEE Std 802.1Q.

12 **PAR Need for the Project:**

13 Successful deployment of current protocol has revealed the need to accommodate a larger number of
14 streams and resource allocation on simultaneously available redundant paths than can be accommodated by
15 the current in-band signaling mechanisms.

16 **CSD Broad market potential:**

17 The original version of the Stream Reservation Protocol (SRP) specified in IEEE Std 802.1Q has been
18 successfully and widely accepted by the professional audio/video, consumer, and automotive markets as an
19 essential tool to realize automatic stream setup with dynamic resource allocation. The success of SRP has
20 expanded the requirements on that protocol beyond that capability.

21 Individuals affiliated with multiple vendors and users for industrial automation, professional audio/video,
22 automotive and other systems requiring a protocol to signal the resource reservation along the end-to-end
23 paths of streams for time-sensitive applications will participate in the development of the project.

24 **CSD Distinct Identity:**

25 No existing IEEE 802 standard provides dynamic resource allocation that supports the required performance
26 and capabilities.

27 This standard expands bridged network capabilities in the area of resource allocation. The rate of addition of
28 other new capabilities to existing IEEE 802.1 standards means that it is more practical to specify expanded
29 resource allocation capabilities in a separate standard focused on that topic. The proposed project replaces
30 the previous IEEE P802.1Qdd Draft Standard for Local and Metropolitan Area Networks — Bridges and
31 Bridged Networks — Amendment: Resource Allocation Protocol.

32 **CSD Technical Feasibility:**

33 The proposed resource allocation methods are similar in principle to those of the Stream Reservation
34 Protocol (SRP) specified in IEEE Std 802.1Q.

35 There is a considerable body of experience in supplying data streams with guarantees for quality of service
36 parameters such as latency, latency variation, or bandwidth. Mechanisms needed for this project are widely
37 used by other protocols already, e.g., IEEE 802.1Q SRP for use in bridged networks and the Resource
38 Reservation Protocol (RSVP, IETF RFC 2205, and IETF RFC 2750) for routers and hosts that use the
39 Internet Protocol.

40 **CSD Economic Feasibility:**

41 The well-established balance between infrastructure and attached stations will not be changed by the
42 proposed standard.

43 All the above cost factors are well-known from the existing resource allocation protocols and registration
44 database distribution methods.

1 Introduction to the current draft

2 This introduction is not part of the draft, and will be revised for SA ballot. A set of cover pages will be
3 retained for use during SA ballot.

4 D1.3

5 In D1.3 the new architecture was updated throughout clause 6. All other parts remained mainly unchanged
6 with regard to the technical changes in architecture and will be adapted after there is consensus about clause
7 6. This draft is intended for further discussions before starting TG ballot.

8 D1.2

9 In D1.2 the architecture was changed to a MRP based architecture as discussed at the May 2025 Interim
10 Session in Rennes. It is intended to discuss the new structure of the document.

11 D1.1

12 D1.0 was not used for TG ballot but as editor's draft.- D1.1 is an editor's draft as well with following changes:

13 Fixes some inconsistencies of D1.0

14 Adds backward compatibility (RAP/MSRP Gateway) skeleton structure.

15 Restructures Token Bucket TSpec clause.

16 D1.0

17 D1.0 is the first draft of P802.1DD and used for the first Task Group ballot.

18 Due to project change (P802.1DD replacing P802.1Qdd), this draft has been created based on the contents
19 as well as the [comment resolution results](#) of [P802.1Qdd/D0.9](#) but using a template for stand-alone IEEE
20 802.1 standards. In addition to the changes required to fit the new format, such as rearrangement of clauses
21 and reworked references to other standards, the major changes from P802.1Qdd D0.9 include the following:

- 22 — [Clause 5.Conformance](#) reworked to define requirements for three types of RAP instance and ten
23 types of RAP systems.
- 24 — [Clause 6.1RAP overview](#) added to give a brief overview of RAP.
- 25 — In [Clause 6.3.1RAP architecture](#), added text and figures for describing the architecture of RAP/MSRP
26 Gateway, an example of RAP Proxy systems, and RAP control of Bridging and FRER functions.
- 27 — [Clause 6.3.6Traffic specification usage rules and types](#) added to define TSpec types and usage rules.
- 28 — In all the state machines, replaced the use of primitives in conditions of state machine transitions with
29 the use of event pointer variables as described in [Clause 6.2.3Event pointer variables](#).
- 30 — [Clause 8.YANG data model for RAP](#) updated to reflect the changes made in [Clause 7.Resource](#)
31 [Allocation Protocol \(RAP\) management](#).

32

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²

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Unapproved draft, prepared by the
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Sponsored by the
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1 **Abstract:** This standard specifies protocols, procedures, and managed objects for resource
2 allocation in bridged local area networks for dynamic creation and maintenance of data streams.
3 This standard supports control signaling through data paths and/or through separate control paths
4 in support of centralized control. This standard makes provisions for backward compatibility with the
5 Stream Reservation Protocol specified in IEEE Std 802.1Q.

6 **Keywords:** Bridged Network, Frame Replication and Elimination for Reliability, FRER, IEEE
7 802.1CB™, IEEE 802.1CS™, IEEE 802.1DD™, IEEE 802.1Q™, Link-local Registration Protocol,
8 LRP, resource allocation, Resource Allocation Protocol, RAP, Time-Sensitive Networking, TSN,
9 Virtual Bridged Local Area Network, VLAN Bridge

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1 Participants

2 <<The following lists will be updated in the usual way prior to publication>>

3 At the time this standard was completed, the IEEE 802.1 working group had the following membership:

4

Glenn Parsons, *Chair*

5

Jessy Rouyer, *Vice Chair*

6

Janos Farkas, *Time-Sensitive Networking Task Group Chair*

7

Martin Mittelberger, *Editor*

8

9 The following members of the individual balloting committee voted on this standard. Balloters may have
10 voted for approval, disapproval, or abstention.

A.N. Other

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12

1

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3 membership:

4

Chair

5

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Past Chair

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8 *Member Emeritus

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10

1 Introduction

2

This introduction is not part of IEEE Std 802.1DD-20XX, IEEE Standard for Local and metropolitan area networks—Resource Allocation Protocol

3 This standard specifies profiles of IEEE 802.1 Time-Sensitive Networking (TSN) and IEEE 802.1 Security
4 standards for aerospace onboard bridged IEEE 802.3 Ethernet networks.

5 This standard was first published as IEEE Std 802.1DD-20XX.

6 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
7 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and
8 to incorporate new related material. Information on the current revision state of this and other IEEE 802
9 standards may be obtained from

10 Secretary, IEEE-SA Standards Board
11 445 Hoes Lane
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14

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Draft Standard for Local and Metropolitan Networks — Resource Allocation Protocol

1. Overview

The standardization of Ethernet communication technology in IEEE Std 802.3TM, specifying transmission over the physical media of individual Local Area Networks (LANs), and in IEEE Std 802.1QTM, specifying Bridges that interconnect IEEE 802[®] LANs,¹ has facilitated widespread deployment of networks that connect significantly more end stations, with significantly greater bandwidth, and at significantly reduced cost compared to prior technology. All these metrics have been improved by several orders of magnitude—reducing costs through the multi-vendor provision of common components (bridges, end station interfaces, integrated circuit and circuit designs, connectors, and software) for a wide range of network applications.

The use of Ethernet communication technology in networks with high-reliability and deterministic latency requirements is further supported by Time-Sensitive Networking (TSN) provisions in IEEE Std 802.1Q, IEEE Std 802.1AS, IEEE Std 802.1CB, IEEE Std 802.1CS, and the security provisions in IEEE Std 802.1AE and IEEE Std 802.1X. The provisions in these standards can be used in various ways, and include options that address different network requirements and parameters that vary by network and application scale. Network design, time to deploy, and component development, selection, validation, and configuration for a particular network can all benefit from consistent choices, across similar networks and network applications, of the provisions, parameters, and options specified in the relevant standards.

This standard specifies protocols, procedures, and managed objects to support the Resource Allocation Protocol (RAP). RAP is designed as an application of the Link-local Registration Protocol (LRP) specified by IEEE Std 802.1CS to provide automatic and dynamic stream setup with resource allocation for time-sensitive applications in need of bounded latency, zero-congestion loss and high availability and reliability.

1.1 Scope

This standard specifies protocols, procedures, and managed objects for resource allocation in bridged local area networks for dynamic creation and maintenance of data streams. This standard supports control signaling through data paths and through separate control paths in support of centralized control. This standard makes provisions for backward compatibility with the Stream Reservation Protocol specified in IEEE Std 802.1Q.

1.2 Purpose

<No purpose is provided in PAR>

¹ IEEE and IEEE 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

1.3 Introduction

To enable decentralized resource allocation for the streams that require using the IEEE 802.1 Time-Sensitive Networking (TSN) features, such as the QoS functions defined by IEEE Std 802.1Q and the seamless redundancy techniques defined by IEEE Std 802.1CB, and to leverage the performance and scalability of the LRP defined by IEEE Std 802.1CS, this standard specifies protocols, procedures, and managed objects for a Resource Allocation Protocol (RAP). RAP provides interoperability with the Multiple Stream Registration Protocol (MSRP) specified in IEEE Std 802.1Q and can also operate in proxy systems to support centralized control. To this end, it

- a) Specifies the requirements for RAP implementations declaring conformance to this standard.
- b) Defines the principles of RAP operation, including the following.
 - 1) RAP architecture
 - 2) RA class and RA advertisement
 - 3) Talker Announcement and Listener Announcement
 - 4) Resource allocation constraints
 - 5) Traffic specification usage rules and types
 - 6) Stream Rank and reservation importance
 - 7) Resource allocation for seamless redundancy
 - 8) Relationship of RAP to MSRP
- c) Defines LRP protocol elements for RAP.
- d) Defines RAP attributes and TLV encoding
- e) Specifies the RAP Endpoint Service Interface (RESI) and the operation of the RAP Endpoint.
- f) Specifies the RAP MAD Internal Service Interface (RISI) and the operation of the RAP MAD.
- g) Defines managed objects for management of RAP.
- h) Defines YANG data models for RAP.
- i) Provides examples of resource allocation using RAP.

1.4 Reference conventions

This standard makes frequent references to specific subclauses in several other IEEE 802.1 standards. To avoid repeating the full names of these referenced standards, the following conventions are used:

- A reference to “subclause x.y in IEEE Std 802.1Q-2022” is of the form: “[Q]:x.y”.
- A reference to “subclause x.y in IEEE Std 802.1CS-2020” is of the form: “[CS]:x,y”.
- A reference to “subclause x.y in IEEE Std 802.1CB-2020” is of the form: “[CB]:x,y”.
- A reference to “subclause x.y in IEEE Std 802.1AB-2016” is of the form: “[AB]:x,y”.
- A reference to “subclause x.y in IEEE Std 802.1AC-2016” is of the form: “[AC]:x,y”.
- A reference to “subclause x.y in this standard” is of the form: “x,y”.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in the text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802[®], IEEE Standard for Local and metropolitan area networks: Overview and Architecture.^{2,3}

IEEE Std 802.1AB[™], IEEE Standard for Local and metropolitan area networks—Station and Media Access Control Connectivity Discovery.

IEEE Std 802.1AC[™], IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1CB[™], IEEE Standard for Local and metropolitan area networks—Frame Replication and Elimination for Reliability.

IEEE Std 802.1CS[™], IEEE Standard for Local and metropolitan area networks—Link-local Registration Protocol.

IEEE Std 802.1Q[™], IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.

IEEE Std 802.3[™], IEEE Standard for Ethernet.

IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

IETF RFC 8343, A YANG Data Model for Interface Management, March 2018.

² IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

³ The IEEE standards or products referred to in this clause are trademarks of The Institute of Electrical and Electronics Engineers, Inc.

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.⁴

This standard makes use of the following term defined in IEEE Std 802:⁵

- end station
- frame
- station

Check if all terms are still used

This standard makes use of the terms defined in IEEE Std 802.1CS:

- Controlled system
- Native system
- Portal
- Proxy system
- record
- target port
- local target port
- LRP application
- neighbor target port

This standard makes use of the terms defined in IEEE Std 802.1CB:

- Compound Stream
- Member Stream

This standard makes use of the following terms defined in IEEE Std 802.1Q:⁶

- Bridge
- Bridge Port
- Bridged Network
- Edge Control Protocol (ECP)
- latency
- Listener
- Multicast
- Port
- Port State
- redundant trees
- Stream
- StreamID
- stream reservation (SR) class
- Talker
- time-sensitive stream
- traffic specification (TSpec)
- traffic class
- Virtual Local Area Network (VLAN)

⁴ *IEEE Standards Dictionary Online* is available at <https://dictionary.ieee.org>.

⁵ This definition is from IEEE Std 802-2014. Notes to definitions are particular to this standard.

⁶ These definitions are from IEEE Std 802.1Q-2022. Notes to definitions are particular to this standard.

- 1 — Virtual Local Area Network (VLAN) Bridge component
- 2 — Virtual Local Area Network (VLAN) Bridged Network

3 The following terms are specific to this standard:

4 **FRER-capable RAP Bridge:** A RAP Native Bridge or a RAP Controlled Bridge that is FRER-capable.

5 **FRER-capable RAP Listener:** A RAP Native end station or a RAP Controlled end station acting as a
6 Listener that is FRER-capable.

7 **FRER-capable RAP Talker:** A RAP Native end station or a RAP Controlled end station acting as a Talker
8 that is FRER-capable.

9 **resource allocation (RA) class:** A priority of a traffic class or a set of traffic classes whose bandwidth and
10 queue resources can be reserved for streams using the Resource Allocation Protocol (RAP).

11 **Resource Allocation Protocol (RAP):** A protocol providing resource reservation for streams.

12 **RAP Bridge Proxy:** A RAP Proxy for Bridges.

13 **RAP Controlled Bridge:** A Virtual Local Area Network (VLAN) Bridge operating as a Controlled system
14 from a RAP Proxy.

15 **RAP Controlled end station:** An end station operating as a Controlled system from a RAP Proxy.

16 **RAP Controlled system:** A RAP Controlled end station or a RAP Controlled Bridge.

17 **RAP End instance:** A RAP instance residing in a RAP Native end station or offloaded from a RAP
18 Controlled end station to a RAP End Station Proxy.

19 **RAP End Station Proxy:** A RAP Proxy for end stations.

20 **RAP instance:** An application instance of the Link-local Registration Protocol (LRP) that provides RAP
21 functionality for an end station or Bridge.

22 **RAP Native Bridge:** A Virtual Local Area Network (VLAN) Bridge operating as a Native system using the
23 Resource Allocation Protocol (RAP).

24 **RAP Native end station:** An end station operating as a Native system using the Resource Allocation
25 Protocol (RAP).

26 **RAP Native system:** A RAP Native Bridge or a RAP Native end station.

27 **RAP Proxy:** A Proxy system using the Resource Allocation Protocol (RAP).

28 **RAP Relay instance:** A RAP instance residing in a RAP Native Bridge or offloaded from a RAP Controlled
29 Bridge to a RAP Bridge Proxy.

30 **RAP/MSRP Gateway:** A Virtual Local Area Network (VLAN) Bridge that supports Multiple Stream
31 Registration Protocol (MSRP) on one or more Bridge Ports and supports Resource Allocation Protocol
32 (RAP) on other Ports, and provides functionality for conversion between these two protocols.

33 **RAP/MSRP Gateway instance:** A RAP instance residing in a RAP Gateway. for a RAP Gateway.
34 providing translation functionality for conversion between the Multiple Stream Registration Protocol
35 (MSRP) and the Resource Allocation protocol (RAP).

36 **stream:** A unidirectional flow of data from one Talker to one or more Listeners that require transmission
37 with a bounded latency.

¹ NOTE—The term “stream”, as used in this standard, is in line with the term “time-sensitive stream” defined in [Q].

4. Abbreviations and acronyms

The following abbreviations and acronyms are used in this standard:⁷

ATS	Asynchronous Traffic Shaping
AV	audio/video
CBS	committed burst size
CFM	Connectivity Fault Management
CID	Company ID ⁸
CQF	cyclic queuing and forwarding
CRC	Cyclic Redundancy Check
EISS	Enhanced Internal Sublayer Service
FDB	Filtering Database
FRER	Frame Replication and Elimination for Reliability (IEEE Std 802.1CB)
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ISS	Internal Sublayer Service
LAN	Local Area Network
LLC	Logical Link Control
LLDP	Link Layer Discovery Protocol (IEEE Std 802.1AB)
LLDPDU	Link Layer Discovery Protocol Data Unit
LRP	Link-local Registration Protocol (IEEE Std 802.1CS)
LRP-DS	Link-local Registration Protocol–Database Synchronization
LRP-DT	Link-local Registration Protocol–Data Transport
LRPDU	Link-local Registration Protocol Data Units
MAC	Medium Access Control
MAD	MRP Attribute Declaration
MMRP	Multiple MAC Registration Protocol
MRP	Multiple Registration Protocol
MRTs	Maximally Redundant Trees
MSRP	Multiple Stream Registration Protocol
MVRP	Multiple VLAN Registration Protocol
OUI	organizationally unique Identifier
PCP	Priority Code Point
ISIS-PCR	Intermediate System to Intermediate System with Path Control and Reservation extensions
PICS	Protocol Implementation Conformance Statement
PSFP	Per-Stream Filtering and Policing
QoS	Quality of Service
R-TAG	Redundancy tag
RA	resource allocation
RAP	Resource Allocation Protocol
RESI	RAP Endpoint Service Interface
RISI	RAP MAD Internal Service Interface
RTID	Resource Allocation Class Template Identifier
SRP	Stream Reservation Protocol
TCP	Transmission Control Protocol

⁷ The abbreviations listed include those defined in standards referenced by this standard and used in this standard.

⁸ See <https://standards.ieee.org/develop/regauth/tut/eui.pdf>.

1	TLV	Type, Length, Value
3	TSpec	traffic specification
5	TSN	Time-Sensitive Networking
7	UNI	User Network Interface
9	VID	VLAN Identifier
11	VLAN	Virtual Local Area Network
13	YANG	Yet Another Next Generation ⁹

⁹ YANG is best viewed as a name, not an acronym.

5. Conformance

This clause specifies mandatory and optional capabilities provided by conformant implementations of this standard.

5.1 Requirements terminology

For consistency with existing IEEE and IEEE 802.1™ standards, requirements are expressed using the following terminology:¹⁰

- a) **Shall** is used for mandatory requirements.
- b) **May** is used to describe implementation or administrative choices (“may” means “is permitted to,” and hence, “may” and “may not” mean precisely the same thing).
- c) **Should** is used for recommended choices (the behaviors described by “should” and “should not” are both permissible but not equally desirable choices).

Protocol Implementation Conformance Statements (PICS) reflect the occurrences of the words “shall,” “may,” and “should” within the standard.

The standard avoids needless repetition and apparent duplication of its formal requirements by using **is**, **is not**, **are**, and **are not** for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementer or administrator, or whose conformance requirement is detailed elsewhere, is described by **can**. Behavior that never occurs in a conformant implementation or system of conformant implementations is described by **cannot**. The word **allow** is used as a replacement for the phrase “Support the ability for,” and the word **capability** means “can be configured to.”

5.2 Protocol Conformance Statement (PCS)

A claim of conformance to this standard attests to the implementation of a system or system component specified in this and referenced standards.

The supplier of an implementation that is claimed to conform to this standard shall provide the information necessary to identify both the supplier and the implementation, and shall complete a copy of the relevant PCS proforma provided in this standard together with the Protocol Implementation Conformance Statements (PICS) for the referenced standards, as identified in the PCS.

5.3 Conformant systems, components, and functionality

This standard specifies conformance requirements for the following components:

- a) RAP Relay instance (5.4)
- b) RAP End instance (5.5)
- c) RAP/MSRP Gateway instance (5.6)

and for the following systems that include instances of the above components:

- d) RAP Bridge (5.7)
- e) FRER-capable RAP Bridge (5.8)
- f) RAP end station (5.9)

¹⁰ Originally derived from ISO/IEC style requirements, and consistent with the terminology specified in the ISO/IEC Directives Part 2:2021, Clause 7 (http://www.iec.ch/members_experts/refdocs).

- 1 g) FRER-capable RAP Talker (5.10)
- 2 h) FRER-capable RAP Listener (5.11)
- 3 i) RAP/MSRP Gateway (5.12)

4 **5.4 RAP Relay instance requirements**

5 **Update this clause according to architectural changes.**

6 A RAP Relay instance shall comprise one RAP Participant (6.7) for each Port and a single RAP Propagator
7 (6.7).

8 A RAP Relay instance implementation conformant to this standard shall:

- 9 a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
- 10 b) Support the RAP Propagator state machine as specified in 6.7.
- 11 c) Support the Application Information TLV as specified in 6.6.6.4.
- 12 d) Support the RAP attributes and their TLV encodings as specified in 6.4.
- 13 e) Support the RAP management as specified in Clause 7.

14 **5.5 RAP End instance requirements**

15 **Update this clause according to architectural changes.**

16 A RAP End instance shall comprise a single RAP Endpoint (6.5) and a single RAP Participant (6.7).

17 A RAP End instance implementation conformant to this standard shall:

- 18 a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
- 19 b) Support the RESI and the RAP Endpoint procedures as specified in 6.5.
- 20 c) Support the Application Information TLV as specified in 6.6.6.4.
- 21 d) Support the RAP attributes and TLV encodings as specified in 6.4.
- 22 e) Support the RAP management as specified in Clause 7.

23 **5.6 RAP/MSRP Gateway instance requirements**

24 **Update this clause according to architectural changes.**

25 A RAP/MSRP Gateway instance shall comprise one RAP Participant (6.7) for each Port that supports the
26 operation of RAP, one RAP/MSRP Converter (6.7) for each Port that supports the operation of MSRP, and a
27 single RAP Propagator (6.7).

28 A RAP /MSRP Gateway instance implementation conformant to this standard shall:

- 29 a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
- 30 b) Support the RAP Propagator state machine as specified in 6.7.
- 31 c) Support the RAP/MSRP Converter as specified in 6.7.
- 32 d) Support the Application Information TLV as specified in 6.6.6.4.
- 33 e) Support the RAP attributes and TLV encodings as specified in 6.4.
- 34 f) Support the RAP management as specified in Clause 7.

1 5.7 RAP Bridge requirements

2 A RAP Bridge implementation conformant to this standard shall:

- 3 a) Include a single VLAN Bridge component conformant to [Q]:5.4.
- 4 b) Conform to the required and optional behaviors for a relay system ([CS]:5).
- 5 c) Include a single RAP Relay instance conformant to the requirements of 5.4.

6 5.8 FRER-capable RAP Bridge requirements

7 A FRER-capable RAP Bridge implementation conformant to this standard shall:

- 8 a) Conform to the required and optional behaviors for a FRER C-component ([CB]:5.15).
- 9 b) Support Active Destination MAC and VLAN Stream identification function ([CB]:6.6).

10 5.9 RAP end station requirements

11 A RAP end station conformant to this standard shall:

- 12 a) Conform to the required behaviors for an end system ([CS]:5).
- 13 b) Support the operation of the Applicant state machine ([CS]:8.3) and the Registrar state machine ([CS]:8.4).
- 14 c) Include a single RAP End instance conformant to the requirements of 5.5.

16 5.10 FRER-capable RAP Talker requirements

17 A FRER-capable RAP Talker implementation conformant to this standard, shall:

- 18 a) Conform to the required, recommended and optional behaviors for a Talker end system ([CB]:5.6, [CB]:5.7, [CB]:5.8).

20 5.11 FRER-capable RAP Listener requirements

21 A FRER-capable RAP Listener implementation conformant to this standard shall:

- 22 a) Conform to the required, recommended and optional behaviors for a Listener end system ([CB]:5.9, [CB]:5.10, [CB]:5.11).

24

1 6. Resource Allocation Protocol (RAP)

2 6.1 RAP overview

3 RAP defines an application of the IEEE 802.1CS Link-local Registration Protocol (LRP) that provides
4 resource allocation in bridged networks for dynamic creation and maintenance of data streams in needs of
5 deterministic QoS (i.e., bounded latency and zero congestion loss) and high availability and reliability.

6 As an LRP application, RAP makes use of the service interface provided by LRP to manage connections and
7 handle attribute declarations and registrations between neighboring devices, while RAP is responsible for
8 processing and propagation of attributes across the network. The definitions of LRP protocol elements
9 required for the operation of RAP are presented in 6.6.6.

10 Resource allocation relies on the capability of each Bridge to support one or more RA classes, each
11 associated with a different priority used by the streams as the PCP value ([Q]:6.9.3) and providing a certain
12 level of guaranteed QoS to each reserved stream. The definitions of RA class parameters and the concept of
13 RA class domain are presented in 6.3.2.

14 As a signaling protocol, RAP performs three types of signaling process, each using a dedicated RAP
15 attribute and serving a different purpose, as described in 6.3.3 for RA Advertisement, in 6.3.4 for Talker
16 Announcement and in 6.3.6 for Listener Attachment. The definitions of these three RAP attributes and their
17 encoding in the TLV format are provided in 6.4.

18 To facilitate decentralized latency and resource computation, RAP adopts the concept of “per-hop per-class
19 guarantees” as the basic principle of resource allocation. This concept allows an examination of
20 deterministic end-to-end QoS to break down into a hop-by-hop check for a set of Bridge-local constraints, as
21 defined in 6.3.5. Used as an essential input for computation of latency bounds and buffer requirements at
22 each hop, a per-hop adjustable Traffic Specification (TSpec), together with the original one generated by the
23 Talker, is carried in each Talker Announcement, as described in 6.3.6.

24 RAP supports the concept of Stream Rank and reservation importance, as described in 6.3.7, in a manner
25 similar to the MSRP in support of stream importance ([Q]:35.2.4.1). This allows the network, in case of
26 lacking resources required for reserving a stream of higher importance, to remove some existing reservations
27 associated with the streams of lower importance.

28 RAP also provides dynamic resource allocation to the streams in need of high availability by use of the
29 Frame Replication and Elimination for Reliability (FRER) specified by IEEE Std 802.1CB, as described in
30 6.3.8. To achieve this, it uses a specific signaling mechanism to propagate the attributes in a set of redundant
31 trees, e.g., Maximally Redundant Trees (MRTs) established using ISIS-PCR ([Q]:45). Upon successful
32 reservation, in addition to the required resources allocated in each participating station, the required FRER
33 functions are configured automatically at the intended locations.

34 RAP enhances MSRP as specified in IEEE Std 802.1Q and thus is compatible to MSRP by design.

1 **6.2 Conventions**

2 This subclause defines the conventions used in the specification of RAP.

3 **6.2.1 State machine diagrams**

4 The state machine diagrams defined in 6.7.7.5 for RAP Participant and in 6.6.2 for RAP Propagator use the
5 conventions defined in [Q]:E with the following extensions:

- 6 a) All transitions and operations are atomic and finish without any progress of time, as soon as the
7 associated conditions of transitions are met. The sole reason for steady states is that none of the
8 conditions of any outgoing transition of a particular state is satisfied.
- 9 b) Service primitives that cannot be processed immediately are queued in their order of invocation for
10 subsequent processing (i.e., no service primitive is lost). In case the conditions of more than one
11 transition are met simultaneously, the taken transition is arbitrarily chosen.

6.3 Principles of RAP operation

6.3.1 RAP architecture

RAP is a signaling protocol that provides end stations with the ability to reserve network resources that will guarantee the transmission and reception of data streams across a network with the requested QoS. These end stations are referred to as Talkers (devices that produce data streams) and Listeners (devices that consume data streams).

A RAP Port instance is an LRP application that implements the RAP functionality. It contains the RAP MAD component (6.6), which provides a RAP Endpoint Service interface (6.5) for the RAP application and communicates with the underlying LRP via the LRP-DS Service Interface (LSI, [CS]:10).

The required extensions to the MRP Attribute Propagation component (MAP, [Q]:10.3) are specified in clause 6.7.

12

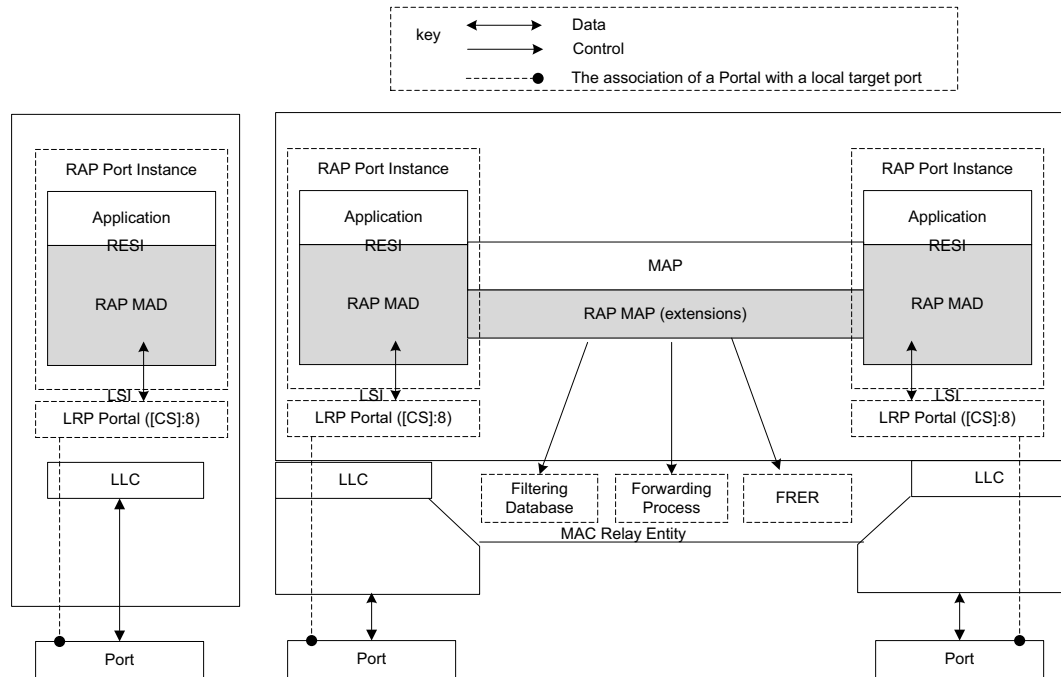


Figure 6-1—Operation of RAP

13

6.3.2 RA class

An RA class represents a traffic class or a set of traffic classes in an end station or Bridge that uses a given transmission selection algorithm, in conjunction with other mechanisms (e.g., PSFP specified in [Q]:8.6.5.1,

1 the traffic scheduling mechanisms specified in [Q]:8.6.8.4) if needed, and a resource reservation method, to
2 provide a bounded latency and zero congestion loss for streams.

3 NOTE—An example of using more than one traffic class for a single RA class is an implementation of the cyclic
4 queuing and forwarding shaper (CQF) using the approach as described in [Q]:T.2. In that case, two transmission queues
5 operate in an coordinated manner and offer the same bounded latency to frames transmitted from either of the two
6 queues.

7 6.3.2.1 RA class ID

8 The RA class ID is an unsigned integer in the range of 0 through 255 that identifies an RA class supported
9 by an end station or Bridge.

10 NOTE—RA class ID is a numeric representation of the RA classes supported by a station. Among a set of RA classes on
11 a given Bridge Port, one RA class that is assigned a numerically higher RA class ID than another RA class, is not
12 necessarily mapped to a numerically higher traffic class than that RA class.

13 6.3.2.2 RA class priority

14 Each RA class supported by an end station or Bridge is associated with a unique priority value in the range 0
15 through 7, termed RA class priority. The RA class priority indicates the received priority of the frames
16 which are to be mapped to the traffic class(es) with which that RA class is associated.

17 NOTE—Mapping of RA class priorities to traffic classes is a matter of port local configuration and can be managed
18 using the Traffic Class Table ([Q]:12.6.3). This standard does not specify default priority to traffic class mapping for a
19 system that supports RA classes.

20 Since there are eight values available for the RA class priority, a station can support up to seven RA classes,
21 in order to ensure that there is at least one traffic class that can support non-stream traffic that is not subject
22 to resource allocation, such as “best effort” traffic.

23 6.3.2.3 RA class template and RTID

24 An RA class template describes a specific method for making up an RA class, including the following:

- 25 — The shaping or scheduling mechanisms employed.
- 26 — The algorithms for computing latency bounds and resources.
- 27 — The encoding of the data in the RaClassTemplateDefinedData field (6.4.2.2.6).

28 An RA class template is identified by an RA Class Template Identifier (RTID), which encodes a 3-octet OUI
29 or CID value identifying the organization that defines that template, followed by a 1-octet index allocated by
30 that organization. The RA class templates currently defined by IEEE 802.1 are listed in Table 6-1.

Table 6-1—IEEE 802.1 RA class templates

RTID	Transmission Selection Algorithm
00-80-C2-00	Strict Priority ([Q]:8.6.8.1)
00-80-C2-01	ATS Transmission Selection ([Q]:8.6.8.5)
99-99-99-99	Credit-Based Shaper ([Q]:8.6.8.5)

31 Editor’s note: Use a proper RTID for CBS in Table 6-3

6.3.2.4 RA class domains and priority regeneration

An RA class domain is a connected subset of end stations and Bridges that support a common RA class (i.e. the same RA class ID, 6.3.2.1) and associate the same priority value with that RA class (i.e. the same RA class priority, 6.3.2.2). Within an RA class domain, any traffic associated with the RA class priority of that domain is treated as stream traffic and subject to resource allocation by RAP. The concept of RA class domains also limits the scope of resource allocation in the sense that a stream can only be established for transmission within the domain of a given RA class.

An RA class domain boundary port is a Port of a Bridge that is part of a given RA class domain and is connected via a LAN to a neighboring station that is not part of that RA class domain. On such a Port, any traffic received with a priority identical to the RA class priority of that RA class domain is subject to priority regeneration ([Q]:6.9.4) according to the administrative values in RA Class Domain Boundary Priority Regeneration Table (7.7) such that the traffic is forwarded by the Bridge using another priority not associated with any RA class.

The detection of RA class domains as well as determining the location of RA class domain boundary ports is controlled by RA Advertisement (6.3.3). Figure 6-2 gives an example of multiple RA Class domains established in a single network, where three domains for RA class ID 1 and three for RA class ID 2 are separately depicted on the left-hand and the right-hand side of the figure, respectively.

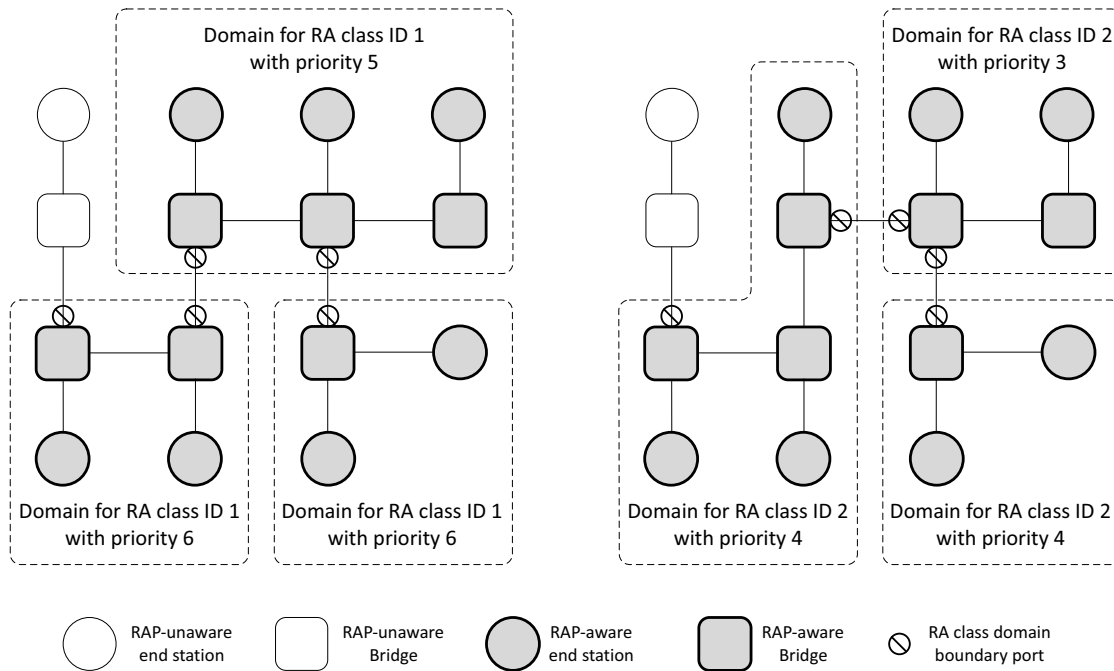


Figure 6-2—Examples of RA class domains and domain boundary ports

Is it still required to depict a “Gateway” System?

6.3.3 RA Advertisement

RA Advertisement refers to declarations of the RA attribute (6.4.2) by a station to each neighboring station. The purposes of RA Advertisement are as follows:

- Advertisement of RA class domain configuration for detection of RA class domains and determination of the location of RA class domain boundary ports, as described in 6.3.2.4.

- 1 — End station learning of the RA class (6.3.2) and Redundancy Context (6.3.8.2) configurations in the
2 network. To allow adaptive participation in resource allocation, an end station can support the ability
3 to adjust its local parameters according to the information received from its neighboring Bridge and
4 then declare its own RA attribute with the adjusted parameters.
- 5 — Advertisement of RA class template configuration. In the process of reserving a stream in a given
6 RA class, the RA class template used by that RA class and corresponding parameter configuration in
7 an upstream station is necessary information used by each downstream station in performing actions
8 such as calculation of the worst-case latency of streams transmitted in that RA class from that
9 upstream station.

10 The following terminology relating to RA Advertisement is used:

- 11 a) **RA declaration:** A declaration of the RA attribute on an end-station or Bridge Port.
- 12 b) **RA registration:** A registration of the RA attribute on an end-station or Bridge Port.
- 13 c) **RA deregistration:** The removal of an RA registration on an end-station or Bridge Port.

14 An RA registration received from a neighboring station on a Bridge Port is not propagated by the Bridge to
15 declare that registered RA attribute to another neighboring on any of other Ports.

16 6.3.4 Talker Announcement

17 The Talker Announcement refers to the process of signaling the Talker Announce attribute (6.4.3) initially
18 declared by the Talker of a stream across a Bridged network to potential Listeners. Each Talker
19 Announcement is identified by a StreamId (6.4.3.1) and VID (6.4.3.5.3) as contained in the Talker
20 Announce attribute.

21 The purposes of the Talker Announcement are as follows:

- 22 — Indication of a Talker's intention of supplying a stream.
- 23 — Advertisement of the stream's characteristics. The Talker Announce attribute contains the
24 information needed by the network to identify the stream and determine the required resources on
25 each Bridge along the stream path(s).
- 26 — Gathering of QoS information from the Bridges along each path such as the accumulated latency
27 and reporting of gathered information to Listeners.
- 28 — Signaling of status information along each potential path about whether a reservation can be made,
29 and if not the information about the cause and the location of the failure.

30 The following terminology relating to the Talker Announcement is used:

- 31 a) **Talker Announce declaration:** A declaration of the Talker Announce attribute on a Port.
- 32 b) **Talker Announce registration:** A registration of the Talker Announce attribute on a Port.
- 33 c) **Talker Announce deregistration:** The removal of a Talker Announce registration on a Port.
- 34 d) **Talker Announce propagation:** Propagation of a Talker Announce registration on an ingress Port
35 of a Bridge to one or more other egress Ports of the Bridge, resulting in a Talker Announce
36 declaration on each such egress Port.
- 37 e) **Talker Announce merge:** Merge of multiple Talker Announce registrations being propagated to a
38 Bridge Port to result in a joint Talker Announce declaration on that Port.
- 39 f) **Originating Talker Announce registration:** A Talker Announce registration from which a given
40 Talker Announce declaration results from is termed an originating Talker Announce registration of
41 that Talker Announce declaration. A Talker Announce declaration can have more than one
42 originating Talker Announce registration in the case of Talker Announce merge.

1 6.3.4.1 Association between Talker Announcement and Listener Attachment

2 A Listener Attachment is associated with a Talker Announcement that carries the same StreamId as that of
3 the Listener Attachment. Depending on the type of the associated Taker Announcement, there are two types
4 of the Listener Attachment, as follows:

- 5 a) **Single-Context Listener Attachment:** A Listener Attachment that is associated with a Single-
6 Context Talker Announcement [item a) in 6.7.1.1].
- 7 b) **Multi-Context Listener Attachment:** A Listener Attachment that is associated with a Multi-
8 Context Talker Announcement [item b) in 6.7.1.1].

9 A Single-Context Listener Attach registration is said to be associated with a Single-Context Talker
10 Announce declaration, if all of the following conditions are met:

- 11 1) The Listener Attach registration is on the same Port as the Talker Announce declaration.
- 12 2) The StreamId (6.4.4.1) of the Listener Attach registration is identical to the StreamId (6.4.3.1)
13 of the Talker Announce declaration.
- 14 3) The VID (6.4.4.2) of the Listener Attach registration is identical to the VID (6.4.3.5.3) of the
15 Talker Announce declaration.

16 A Multi-Context Listener Attach registration is said to be associated with a Multi-Context Talker Announce
17 declaration, if all of the following conditions are met:

- 18 4) The Listener Attach registration is on the same Port as the Talker Announce declaration.
- 19 5) The StreamId (6.4.4.1) of the Listener Attach registration is identical to the StreamId (6.4.3.1)
20 of the Talker Announce declaration.
- 21 6) The set of VLAN Context(s) (6.4.4.4) indicated by the Listener Attach registration is identical
22 to the set of VLAN Context(s) indicated by the Talker Announce declaration (6.4.3.9.2).

23 A Listener Attach declaration is said to be associated with a Talker Announce registration, if all of the
24 following conditions are met:

- 25 7) The Listener Attach declaration is on the same Port as the Talker Announce registration.
- 26 8) The StreamId (6.4.4.1) of the Listener Attach declaration is identical to the StreamId (6.4.3.1)
27 of the Talker Announce registration.
- 28 9) The Listener Attach declaration wholly or partially results from a Listener Attach registration
29 whose associated Talker Announce declaration originates wholly or partially from the Taker
30 Announce registration.

31 A Listener Attach registration with an associated Talker Announce declaration on a Bridge Port indicates a
32 reservation request for transmission of a stream or a Compound or Member Stream through that Port and,
33 upon successful resource allocation, is associated with a dedicated reservation in the Bridge Port. A
34 reservation maintained in a Bridge Port is identified by <StreamId, VID> with the corresponding values
35 contained in the associated Listener Attach registration.

36 6.3.4.2 Listener Attach merge

37 Listener Attach merge is applied, when more than one Listener Attach registration whose associated Talker
38 Announce declaration has a common originating Talker Announce registration is propagated to the same
39 Port, to result in a joint Listener Attach declaration on that Port.

40 In a Single-Context or Multi-Context Listener Attachment for a stream, Listener Attach merge can occur
41 when multiple Listener Attachment processes originating from different Listeners of that stream converge
42 on a Bridge Port, indicating a potential place for multicast forwarding the stream. Additionally, in a Multi-
43 Context Listener Attachment for a Compound Stream, Listener Attach merge can also occur when multiple

1 Listener Attachment processes running in different VLAN Contexts converge on a Bridge or Talker end
2 station Port, indicating a potential place for applying stream splitting (6.3.8.3) to the stream.

3 **6.3.4.3 Listener Attach status**

4 The Listener Attach status of a Listener Attach declaration on a Port is associated with one or more paths,
5 depending on the VLAN Context(s) in which the Listener Attach declaration is made, from the Port to each
6 Listener whose participation in the Listener Attachment has been perceived on that Port, and indicates one
7 of the following:

- 8 a) **Attach Ready:** In the case of the Single-Context Listener Attachment, indicating a reservation has
9 been successfully made on each path to each Listener. In the case of the Multi-Context Listener
10 Attachment, indicating a reservation has been successfully made on at least one path to one Listener,
11 while the reservation status of each path is indicated by the Path status (6.3.5).
- 12 b) **Attach Fail:** A reservation has failed on each path to each Listener;
- 13 c) **Attach Partial Fail:** A reservation has been successfully made on the path to at least one of these
14 Listeners, but has failed on the path to other Listeners. This status is only applicable to the Single-
15 Context Listener Attachment.

16 NOTE—The Attach Partial Fail status is used only in the Single Context Listener Attachment for the same purpose for
17 which the Listener Ready Failed status is used in MSRP as defined in [Q]:35.1.2.2. In the case of the Multi-Context
18 Listener Attachment, there is no distinction made in the Listener Attach status between a full success and a partial
19 success, while the latter case could arise, for example, when reservation has failed on one among all paths to a single
20 Listener, or when reservation on all the paths has succeeded for some Listeners but failed for the others.

21 **6.3.5 Resource allocation constraints**

22 Resource allocation is constrained by limited capacity of end stations and Bridges to ensure bounded latency
23 and zero congestion loss for reserved streams. The subsequent subclauses (6.3.5.1 through 6.3.5.2) describe
24 three constraints taken into account in resource allocation.

25 **6.3.5.1 Bandwidth constraint**

26 The bandwidth constraint of a Bridge is specified in the per-RA class, per-transmission Port maxBandwidth
27 parameter, as defined in item d) of 6.6.4.9. One such parameter configured in a Bridge places an upper
28 bound on the amount of bandwidth that can be reserved for use by an RA class on a transmission Port.

29 The bandwidth constraint also applies to end stations. As in many circumstances the bandwidth availability
30 for streams in an end station is controlled by the applications, no explicit bandwidth constraint parameters
31 are specified by this standard for end stations, and it is a matter of each end station's local decision.

32 A stream reservation request is said to meet the bandwidth constraint of a Bridge or end station, if the total
33 bandwidth needed to transmit that stream and all reserved streams in the same RA class and on the same Port
34 would not exceed the associated upper bound. A Talker Announcement failed due to not meeting the
35 bandwidth constraint is reported with the RAP Failure Code *BandwidthExceeded* defined in Table 6-6.

36 **6.3.5.2 Resource constraint**

37 The resource constraint is associated with the availability of configurable resources for streams in a Bridge
38 or end station's local functions, such as filtering, buffering, shaping and FRER. A stream reservation request
39 is said to meet the resource constraint of a station, if the station has sufficient resources available for in all
40 the functions required for handling that stream. A Talker Announcement failed due to not meeting the
41 resource constraint is reported with the RAP Failure Code *ResourceExceeded* defined in Table 6-6.

1 A measure of resource constraint and its determination method for streams are usually dependent on a
2 particular implementation. No explicit parameters are specified by this standard for resource constraint.

3 6.3.5.3 Latency constraint

4 The latency constraint for a Bridge is specified in the per-RA class, per reception/transmission Port pair
5 maxHopLatency parameter, as defined in item d) of 6.6.4.10. Such a parameter configured in a Bridge is
6 used by the Bridge in resource allocation as a guaranteed latency upper bound provided to each stream
7 reserved in a given RA class when transmitted through the hop from a neighboring station on a given
8 reception Port to a given transmission Port.

9 The latency constraint for an end station is specified in the per-RA class parameter MaxLastHopLatency
10 (6.4.2.2.5) received in an RA registration from a neighboring station. Such a parameter provided to an end
11 station is used by the end station, when acting as a Listener, to impose a latency upper bound to streams
12 when transmitted through the last-hop from that neighboring station to the end station.

13 A stream reservation request is said to meet the latency constraint of a Bridge or an end station (as a
14 Listener), if allowing transmission of this stream would not cause any reserved stream to suffer from a
15 latency exceeding the associated upper bound. A Talker Announcement failed due to not meeting the latency
16 constraint is reported with the RAP Failure Code *LatencyExceeded* defined in Table 6-6.

17 The latency constraint configured on each hop along each path in the network, in conjunction with an
18 accumulation of per-hop latency bounds in AccuMaxLatency (6.4.3.3), builds up the basis for bounded end-
19 to-end latencies in resource allocation.

20 NOTE—The concept of per-hop latency bound requires the ability of each station participating in resource allocation to
21 determine the worst-case maximum latency while checking the latency constraint. The method used for this purpose is
22 specific to each RA class template.

23

24 6.3.6 Traffic specification usage rules and types

25 In resource allocation, the traffic specification (TSpec) carried in the Talker Announcement contains the
26 information about the characteristics of the stream's data traffic.

27 6.3.6.1 Talker TSpec

28 A TSpec received via RESI (6.5.2) from higher layer entities in Talkers and specifying the amount of the
29 stream data transmitted by the Talker into the network. The TSpec contained in this field remains unchanged
30 during the propagation of the Talker Announcement across the network.

31 NOTE 1—Keeping a Talker TSpec unchanged during its propagation is to provide Bridges and Listener end
32 stations with the original information about the stream traffic generated by Talkers. This information can be
33 used, for example, by a specific intermediate Bridge in traffic shaping the stream to restore its original traffic
34 characteristics.

35 There are also two types of TSpec supported by resource allocation, each using a different parameter set to
36 describe the stream traffic characteristics:

- 37 a) **Talker Token Bucket TSpec**, a TSpec comprising the following set of parameters excluding any
38 MAC framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN
39 tag, R-TAG, CRC, interframe gap):
 - 40 1) **MaxPayloadLength**: The maximum frame length of the traffic, in octets, associated with on
41 any transmission Port.
 - 42 2) **MinPayloadLength**: The minimum frame length of the traffic, in octets, associated with on
43 any transmission Port.

- 1 3) **CommittedInformationRate**: the committed information rate, in bits per second.
- 2 4) **CommittedBurstSize**: The committed burst size, in bits sent from the application.
- 3 b) **Interval-based TSpec**, a TSpec comprising the following set of parameters:
 - 4 1) **Interval**: The interval of time, in nanoseconds, taken as a fixed-length observation window to
 - 5 measure the maximum amount of the traffic.
 - 6 2) **MaximumFramePerInterval**: The maximum number of maximum sized frames can be
 - 7 observed from the traffic within one Interval [item 1) above].
 - 8 3) **MaximumFrameSize**: The maximum frame size of the traffic, in octets, excluding any MAC
 - 9 framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN
 - 10 tag, R-TAG, CRC, interframe gap) associated with on any transmission Port.
 - 11 NOTE 3—The Interval-based TSpec type is compatible with the TrafficSpecification defined in [Q]:46.2.3.5,
 - 12 except the TransmissionSelection parameter ([Q]:46.2.3.5.4).

13 6.3.6.2 Network TSpec

14 A TSpec received via RESI (6.6.2) from higher layer entities in Talkers and specifying the amount of the
15 stream data transmitted by the Talker into the network. The TSpec contained in this field remains unchanged
16 during the propagation of the Talker Announcement across the network.

17 NOTE 1—Keeping a Talker TSpec unchanged during its propagation is to provide Bridges and Listener end
18 stations with the original information about the stream traffic generated by Talkers. This information can be
19 used, for example, by a specific intermediate Bridge in traffic shaping the stream to restore its original traffic
20 characteristics.

21 **Token Bucket TSpec**, a TSpec comprising the following set of parameters including the media-dependent
22 overhead ([Q]:12.4.2.2) :

- 23 1) **MaxTransmittedFrameLength**: The maximum frame length of the traffic, in
- 24 octets, associated with a transmission Port for the traffic.
- 25 2) **MinTransmittedFrameLength**: The minimum frame length of the traffic, in octets, associated
- 26 with a transmission Port for the traffic.
- 27 3) **CommittedInformationRate**: the committed information rate, in bits per second.
- 28 4) **CommittedBurstSize**: The committed burst size, in bits.

29 6.3.6.3 Translation of Talker TSpec to Network TSpec

30 When initiating a Talker Announcement at a Talker, higher layer entities can supply either a Talker Token
31 Bucket TSpec or an Interval-based TSpec as the Talker TSpec. The Network TSpec field in the Talker
32 Announce attribute always contains a Token Bucket TSpec derived from the supplied Talker TSpec. When
33 an Interval-based TSpec is supplied as the Talker TSpec, it is translated by the Taker to a Token Bucket
34 TSpec as the Network TSpec as specified in Table 6-2. As the Talker Announcement propagates in the
35 network, each Bridge on the path can adjust the Token Bucket TSpec values contained in the Network TSpec
36 as necessary.

37 6.3.7 Stream Rank and reservation importance

38 RAP supports the concept of the Rank as specified in [Q]:46.2.3.2.1 and gives streams of Rank zero higher
39 precedence than streams of Rank one in a manner that allows Bridges to make a reservation for a stream of
40 Rank zero in case of lacking resources by removing existing reservations made for one or more streams of
41 Rank one. Each reservation removed under such circumstances is termed “the reservation preempted by
42 Rank zero” and uses the RAP Failure Code *ReservationPreempted* as defined in Table 6-6.

43 NOTE—RAP allows reservations of Rank one streams to be preempted only by reservations of Rank zero streams.
44 Reservations of streams with the same Rank, regardless Rank zero or one, are handled on a first-come first-served basis.

Table 6-2—Translation of an Interval-based TSpec to a Token Bucket TSpec

Interval-based TSpec	Token Bucket TSpec
MaximumFrameSize + ALL_OVERHEAD ^a	MaxTransmittedFrameLength
64 + MD_OVERHEAD ^b	MinTransmittedFrameLength
$(\text{MaximumFrameSize} + \text{ALL_OVERHEAD}^a) * \text{MaximumFramePerInterval} * 8 * 1e9 / \text{Interval}$	CommittedInformationRate
$(\text{MaximumFrameSize} + \text{ALL_OVERHEAD}^a) * \text{MaximumFramePerInterval} * 8$	CommittedBurstSize

^a Including all MAC framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN tag, R-TAG, CRC, interframe gap) associated with a transmission Port for the traffic.

^b The media-dependent overhead [Q]:12.4.2.2 associated with a transmission Port for the traffic.

1 The order used by a Bridge in the selection of a reservation, in preference to others, to be preempted by Rank
2 zero is determined based on the reservation importance, i.e., a least important reservation is to be removed
3 first. Among a set of reservations on a Bridge Port, each identified by <StreamId, VID> and with a
4 reservationAge value [item e) in 6.6.4.3], one reservation is considered more important than another,
5 according to the following rules in the shown order:

- 6 a) A Reservation with a numerically larger reservationAge value is more important.
- 7 b) If reservationAge is the same, the one with a numerically smaller StreamId is more important.
- 8 c) If StreamId is the same, the one with a numerically smaller VID is more important.

9 6.3.8 Resource allocation for seamless redundancy

10 RAP supports resource allocation for transmitting streams via redundant paths using the seamless
11 redundancy techniques specified by IEEE Std 802.1CB. The subsequent subclauses describe additional
12 capabilities and operations required by resource allocation for seamless redundancy, while the definitive
13 specification is contained in 6.4, 6.5 and 6.7. For illustrative purposes, some resource allocation examples
14 for seamless redundancy are provided in Annex B.

15 6.3.8.1 FRER-capable stations

16 Resource allocation for seamless redundancy operates in a network where a set of FRER-capable stations,
17 either or both FRER-capable end stations and FRER-capable Bridges, are placed at specific locations to
18 perform FRER operations on Compound and Member Streams with corresponding FRER functions, as
19 shown in Table 6-3.

20 RAP makes use of Multi-Context Talker Announcement and Multi-Context Listener Attachment in resource
21 allocation for seamless redundancy. The potential locations in the network (including end stations) where
22 each stream requesting seamless redundancy is subject to FRER operations along its paths are dependent on
23 the placement of FRER-capable stations and the VLAN topologies configured for use with seamless
24 redundancy, and are automatically determined on a per-stream basis during the stream's Multi-Context
25 Talker Announcement. Resource allocation, including configuration of the FRER functions needed for each
26 stream, is executed during the stream's Multi-Context Listener Attachment.

Table 6-3—FRER functions and operations in FRER-capable stations

		FRER-capable stations		
		FRER-capable RAP Talker	FRER-capable RAP Bridge	FRER-capable RAP Listener
FRER operations	redundancy tagging (6.3.8.2)	sequence generation function ([CB]:7.4.1), Stream encode/decode function ([CB]:7.6), Redundancy tag ([CB]:7.8)		(none)
	stream splitting (6.3.8.3)	Stream splitting function ([CB]:7.7)	(Multicast) ^a	
		Active Destination MAC and VLAN Stream identification function ([CB]:6.6) ^b		
	stream merging (6.3.8.4)	(none)	Active Destination MAC and VLAN Stream identification function ([CB]:6.6) ^b , sequence recovery function ([CB]:7.5),	
	redundancy untagging (6.3.8.2)		Stream encode/decode function ([CB]:7.6), Redundancy tag ([CB]:7.8)	

^a The stream splitting operation in a Bridge relies on the multicast ability of the Forwarding Process to replicate streams, without the need to use Stream splitting function.

^b An end station can, but is not required to explicitly use this function; it is an implementation choice.

6.3.8.2 Redundancy tagging and redundancy untagging

Redundancy tagging refers to a FRER operation in which a FRER-capable RAP Talker or a FRER-capable RAP Bridge uses the particular FRER functions, as shown in Table 6-3, to transform an ordinary stream into a Compound Stream by inserting a Redundancy tag (R-TAG) into the data frames of the stream. Redundancy untagging refers to a FRER operation in which a FRER-capable Bridge or a FRER-capable Listener uses the particular FRER function, as shown in Table 6-3, to transform a Compound Stream back into an ordinary stream by removing the R-TAG carried in the data frames of the Compound Stream.

The potential locations in the network (including end stations) where a stream requesting seamless redundancy is subject to redundancy tagging and redundancy untagging are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

- a) Redundancy tagging is to be performed by the stream's Talker, if that Talker is FRER-capable, and by a FRER-capable Bridge where the stream enters the network otherwise. The Boolean flag RTagStatus (6.4.3.9.1) contained in the Talker Announcement is then set by the station responsible for redundancy tagging. If neither of these conditions for redundancy tagging is fulfilled, the Talker Announcement is failed by the Bridge where the stream enters the network.
- b) Redundancy untagging is to be performed by a FRER-capable Bridge on a transmission Port where all of the Member Streams previously split from the original stream are merged back into a single Compound Stream. In case that no such Bridge Port for stream merging (6.3.8.4) exists on the path towards a given Listener, that Listener is responsible for redundancy untagging, if it is FRER-capable, and fails the Talker Announcement otherwise. The Boolean flag RTagStatus (6.4.3.9.1) contained in the Talker Announcement is cleared by the station responsible for redundancy untagging.

The RAP Failure Code *RTagFailed* defined in Table 6-6 is reported in case of failure to meet the conditions of redundancy tagging or untagging as described above.

6.3.8.3 Stream splitting

Stream splitting refers to a FRER operation in which a FRER-capable Talker or a FRER-capable Bridge uses the particular FRER functions, as shown in Table 6-3, to split a Compound Stream or a Member Stream into multiple Member Streams.

The potential locations in a network (including end stations) where a stream requesting seamless redundancy is subject to stream splitting are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

- a) If the stream's Talker is FRER-capable, it is responsible for stream splitting by making several separate Talker Announce declarations, each with a different VLAN Context from the Redundancy Context chosen by the stream. Otherwise, it makes a Talker Announce declaration that include all the VLAN Contexts in the Redundancy Context, which indicates that the stream is expected to be transformed into a Compound Stream (i.e., redundancy tagging) and then split into Member Streams by a FRER-capable Bridge proxying for this Talker.
- b) Each FRER-capable Bridge participated in the stream's Talker Announcement is responsible for applying stream splitting to a stream indicated by a Talker Announce registration in the Bridge, if following condition is met:
The Talker Announce registration contains more than one VLAN Context and is propagated by the Bridge to at least two other Ports. And at least one of these Ports is not in all of the VLAN Contexts as contained in the Talker Announce registration. This also indicates the fact that the VLAN topologies in which the Talker Announce registration is propagated are disjoint somehow on these Ports.

NOTE—Stream splitting is distinguished from the normal multicast, as the former transforms replicated streams into separate Member Streams with different VIDs. If a FRER-capable Bridge propagates a Multi-Context Talker Announce registration to several Ports each with the same set of VLAN Contexts, it will apply just multicast instead of stream splitting to that Compound Stream.

Meeting the conditions described in item b) above requires a proper placement of FRER-capable stations and VLAN configuration in the network. To prevent incorrect stream splitting operation, a Multi-Context Talker Announcement is failed by a Bridge with the RAP Failure Code *StreamSplitFailed* defined in Table 6-6, in either of the following three circumstances:

- c) A Talker Announce registration in a Bridge that is not FRER-capable meets b).
- d) If the frames cannot be forwarded to the port of the given Talker Announce declaration element.
- e) If the frames cannot be sent with the given VLAN ID configuration at the port of the given Talker Announce declaration element.

6.3.8.4 Stream merging

Stream merging refers to a FRER operation in which a FRER-capable RAP Bridge or a FRER-capable RAP Listener performs the particular FRER function, as shown in Table 6-3, to combine more than one Member Stream into a Compound Stream.

The potential locations in a network (including end stations) where a stream requesting seamless redundancy is subject to stream merging are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

- a) Each FRER-capable RAP Bridge participated in propagation of the stream's Talker Announcement is responsible for stream merging, if the Bridge propagates more than one Talker Announce registration of the stream on different Ports to one or more common destination Ports. Such a

- 1 common destination Port, also termed stream merging Port, is in fact located at a converging point
2 of all or some of the VLAN topologies in the Redundancy Context used by the stream.
- 3 b) A FRER-capable RAP Listener of the stream is responsible for stream merging, if it receives more
4 than one Talker Announce registration of the stream, each containing only a single VLAN Context
5 or partial ones of the Redundancy Context used by the stream.

6 Note that a Bridge Port or a Listener where all of the Member Streams split from the original stream are
7 subject to stream merging into a Compound Stream is also responsible for applying redundancy untagging
8 (6.3.8.2) to the Compound Stream following the stream merging operation.

9 6.3.9 Relationship of RAP to MSRP

10 RAP has conceptual and functional similarities to MSRP. Table 6-4 summarizes the similarities and
11 differences between RAP and MSRP.

Table 6-4—RAP/MSRP differences

	MSRP ([Q]:35)	RAP
Attribute exchange framework	MRP ([Q]:10)	LRP (IEEE Std 802.1CS)
Operational modes	peer-to-peer (MSRPv0) MRP External Control (MSRPv1)	Native, Controlled and Proxy system
Attribute types	[Q]:Table 35-1	Table 6-5
Traffic classes for streams	SR class	RA class
Traffic specification	[Q]:35.2.2.8.4 (MSRPv0) [Q]:35.2.2.10.6 (MSRPv1)	6.4.3.6 6.4.3.7
Queuing and transmission functions for streams	FQTSS in [Q]:34	RA class template based
Support for multiple-path streams	only in MSRPv1 with CNC	6.3.8

6.4 RAP attribute and TLV encoding definitions

This subclause defines the RAP attributes and associated TLV encoding.

6.4.1 General TLV definition

The information of RAP attributes is encoded in a Type-Length-Value (TLV) format, which consists of a 1-octet Type field that indicates the type of that TLV, a 2-octet Length field that indicates the number of octets in the Value field, and then a Value field. The Value field of a TLV encodes zero or more fixed-size parameters, followed by zero or more TLVs of specific types. Figure 6-3 illustrates the general TLV format.

	Octet	Length
Type	1	1
Length	2	2
Value	4	Fixed or variable

Figure 6-3—General TLV format

There are two categories of TLVs defined for RAP, attribute TLVs and sub-TLVs. An attribute TLV encodes a RAP attribute and can include one or more sub-TLVs in the Value field. An sub-TLV can further include one or more sub-TLVs. Attributes TLVs, as top-level TLVs, are never included in a sub-TLV or another attribute TLV. Table 6-5 lists the set of RAP attribute TLVs and sub-TLVs, and the values of the Type and Length field for each of them.

Table 6-5—RAP TLV and sub-TLV Type field and Length field values

TLV name	Type field (1 octet)	Length field (2 octets)	Reference
RA attribute TLV	0x00	variable	6.4.2
Talker Announce attribute TLV	0x01	variable	6.4.3
Listener Attach attribute TLV	0x02	10	6.4.4
RA Class Descriptor sub-TLV	0x20	variable	6.4.2.2
Redundancy Context sub-TLV	0x21	variable	6.4.2.3
Data Frame Parameters sub-TLV	0x22	8	6.4.3.5
Talker Token Bucket TSpec sub-TLV	0x23	16	6.4.3.6
Interval-based TSpec sub-TLV	0x24	8	6.4.3.7
Token Bucket TSpec sub-TLV	0x25	16	6.4.3.8
Redundancy Control sub-TLV	0x26	variable	6.4.3.9
VLAN Context Information sub-TLV	0x27	variable	6.4.3.9.2
Failure Information sub-TLV	0x28	9	6.4.3.10
Organizationally Defined sub-TLV	0x29	variable	6.4.3.11
VLAN Context Status sub-TLV	0x2A	variable	6.4.4.4
Reserved for future standardization	0x03-0x1F, 0x2B-0xFF	-	-

The subsequent subclauses specify the encoding of the Value field for each attribute TLV and sub-TLV, by using the “big-endian” convention, in which higher significance bytes and bits within each byte appear to the left of and above bytes and bits of lower significance. Any fields that are labeled as “Reserved” are transmitted as zero and ignored on receipt. In addition, the term “the stream” used in the specification of Talker Announce attribute TLV and the sub-TLVs thereof refers to a stream for which the Talker Announce attribute is declared.

6.4.2 RA attribute and TLV encoding

The RA attribute is used in RA Advertisements (6.3.3). Figure 6-4 shows the encoding of the Value field of the RA attribute TLV.

	Octet	Length
MaxInterferingFrameSize	1	2
1 or more RA Class Descriptor sub-TLVs	3	variable
zero or more Redundancy Context sub-TLVs	variable	variable

Figure 6-4—Value of RA attribute TLV

6.4.2.1 MaxInterferingFrameSize

A 2-octet unsigned integer, indicating the maximum frame size, in bytes, including media-dependent overhead ([Q]:12.4.2.2), that is allowed to be transmitted through the Port declaring the RA class attribute. The value of this parameter is determined by the operation of the underlying MAC Service on the Port.

6.4.2.2 RA Class Descriptor sub-TLV

One or more RA Class Descriptor sub-TLVs are included in the RA attribute TLV, each encoding in the Value field a set of parameters associated with a distinct RA class (6.3.2) as illustrated in Figure 6-5.

	Octet	Length
RaClassId	1	1
RaClassPriority	2	1
RTID	3	4
TrafficClass	7	1
MaxLastHopLatency	8	4
RaClassTemplateDefinedData	12	variable

Figure 6-5—Value of RA Class Descriptor sub-TLV

6.4.2.2.1 RaClassId

A 1-octet unsigned integer containing the RA class ID (6.3.2.1).

6.4.2.2.2 RaClassPriority

A 1-octet unsigned integer containing the RA class priority (6.3.2.2).

6.4.2.2.3 RTID

A 4-octet RTID (6.3.2.3) identifying the RA class template used by the RA class.

1 6.4.2.2.4 TrafficClass

2 A 1-octet unsigned integer, indicating the traffic class ([Q]:8.6.6) to which the RA class priority is mapped
3 on the Port where the RA attribute is declared.

4 6.4.2.2.5 MaxLastHopLatency

5 A 4-octet unsigned integer, indicating a latency value, in nanoseconds, that specifies a latency constraint
6 (6.3.5.1) for a neighboring end station on the Port where the RA attribute is declared.

7 The latency contained in this field is measured from a point located in this Port, to a point located in the
8 neighboring end station, while the exact measurement points are specific to and defined by the RA class
9 template being used by the RA class.

10 NOTE—This parameter is only intended for use by an end station when acting as a Listener and can be ignored at
11 reception by a Bridge or a Talker-only end station. This also implies that an RA declaration made on a Port to which no
12 Listener end station is expected to be connected can take an arbitrary value in this field.

13 6.4.2.2.6 RaClassTemplateDefinedData

14 The encoding of this field and the semantics associated with its values if any, is specific to the RA class
15 template identified by the value contained in 6.4.2.2.3.

16 6.4.2.3 Redundancy Context sub-TLV

17 One or more Redundancy Context sub-TLVs are included in the RA attribute TLV, each describing a distinct
18 Redundancy Context (6.3.8.2). The Value field contains one or more VLAN Context tuples, each encoding
19 in the VlanContextId field a 12-bit VLAN Context ID, followed by a 4-bit Reserved field, as illustrated in
20 Figure 6-6.

		Octet	Length
VLAN Context tuple 1	VlanContextId	1	12 bits
	Reserved	2	4 bits
...			
VLAN Context tuple n	VlanContextId	2n-1	12 bits
	Reserved	2n	4 bits

Figure 6-6—Value of Redundancy Context sub-TLV

6.4.3 Talker Announce attribute and TLV encoding

The Talker Announce attribute is used in the Talker Announcement (6.3.4). Figure 6-7 shows the encoding of the Value Field of the Talker Announce attribute TLV.

	Octet	Length
StreamId	1	8
StreamRank	9	1
AccuMaxLatency	10	4
AccuMinLatency	14	4
Data Frame Parameters sub-TLV	18	11
Talker Token Bucket TSpec sub-TLV or Interval-based TSpec sub-TLV (Talker TSpec)	29	19 or 11
Token Bucket TSpec sub-TLV (Network TSpec)	variable	19
Redundancy Control sub-TLV (only for Multi-Context Talker Announcement)	variable	variable
0 or 1 Failure Information sub-TLV	variable	variable
0 or more Organizationally Defined sub-TLVs	variable	variable

Figure 6-7—Value of Talker Announce attribute TLV

For support of two types of the Talker Announcement, the encoding of the Talker Announce attribute follows the following rules:

- The Talker Announce attribute used in the Single-Context Talker Announcement [item a) in 6.3.5.1.1] contain no Redundancy Control sub-TLV (6.4.3.9).
- The Talker Announce attribute used in the Multi-Context Talker Announcement [item b) in 6.3.5.1.1] always contains a Redundancy Control sub-TLV (6.4.3.9).

6.4.3.1 StreamId

An 8-octet field encoding the StreamID element as specified in [Q]:46.2.3.1.

6.4.3.2 StreamRank

A 1-octet field encoding a Rank value as specified in [Q]:46.2.3.2.1.

6.4.3.3 AccuMaxLatency

A 4-octet field encoding the AccumulatedLatency element as specified in [Q]:46.2.5.2.

6.4.3.4 AccuMinLatency

A 4-octet unsigned integer, indicating the minimum latency, in nanoseconds, that a single frame of the stream can encounter when transmitted from the Talker along a given path to the Port declaring this Talker Announce attribute.

6.4.3.5 Data Frame Parameters sub-TLV

This sub-TLV contains a set of parameters whose values are carried in the data frames of the stream when transmitted through the declaring Port of this attribute. Figure 6-8 shows the encoding of the Value field of the Data Frame Parameters sub-TLV.

	Octet	Length
DestinationMacAddress	1	6
Priority	7	3 bits
Reserved	7	1 bit
VID	7	12 bits

Figure 6-8—Value of Data Frame Parameters sub-TLV

6.4.3.5.1 DestinationMacAddress

A 6-octet destination MAC address of the data frames of the stream.

6.4.3.5.2 Priority

A 3-bit unsigned integer, indicating the priority to be encoded in the PCP field of the VLAN tag ([Q]:6.9.3), with which the data frames of the stream are tagged. This priority value is used by each receiving Bridge to associate the stream to a local RA class of the same priority.

6.4.3.5.3 VID

A 12-bit VID. The semantics of this field is dependent on the type of the Talker Announcement in which the Talker Announce attribute is used, as follows:

- In the case of the Single-Context Talker Announcement, this field indicates a VID to be encoded in the VLAN tag with which the data frames of the stream are tagged, and also a single VLAN Context used by the Talker Announce attribute.
- In the case of the Multi-Context Talker Announcement, this field is set to the numerically smallest VlanContextId value contained in a VLAN Context Information sub-TLV (6.4.3.9.2).

19

6.4.3.6 Talker Token Bucket TSpec sub-TLV

This sub-TLV contains a Talker Token Bucket TSpec as defined in item a) of 6.3.6. Figure 6-9 shows the encoding of the Value field of the Talker Token Bucket TSpec sub-TLV.

	Octet	Length
MaxPayloadLength	1	2
MinPayloadLength	3	2
CommittedInformationRate	5	8
CommittedBurstSize	13	4

Figure 6-9—Value of Talker Token Bucket TSpec sub-TLV

6.4.3.6.1 MaxPayloadLength

A 2-octet unsigned integer, encoding the MaxPayloadLength parameter [item a).1) of 6.3.6.1].

1 6.4.3.6.2 MinPayloadLength

2 A 2-octet unsigned integer, encoding the MinPayloadLength parameter [item a).2) of 6.3.6.1].

3 6.4.3.6.3 CommittedInformationRate

4 A 8-octet unsigned integer, encoding the CommittedInformationRate parameter [item a).3) of 6.3.6.1].

5 6.4.3.6.4 CommittedBurstSize

6 A 4-octet unsigned integer, encoding the CommittedBurstSize parameter [item a).4) of 6.3.6.1].

7 6.4.3.7 Interval-based TSpec sub-TLV

8 This sub-TLV contains an Interval-based TSpec as defined in item b) of 6.3.6.1. Figure 6-10 shows the
9 encoding of the Value field of the Interval-based TSpec sub-TLV.

	Octet	Length
Interval	1	4
MaximumFramesPerInterval	5	2
MaximumFrameSize	7	2

Figure 6-10—Value of Interval-based TSpec sub-TLV

10 6.4.3.7.1 Interval

11 A 4-octet unsigned integer, encoding the Interval parameter [item b).1) of 6.3.6.1].

12 6.4.3.7.2 MaximumFramesPerInterval

13 A 2-octet unsigned integer, encoding the MaximumFramesPerInterval parameter [item b).2) of 6.3.6.1].

14 6.4.3.7.3 MaximumFrameSize

15 A 2-octet unsigned integer, encoding the MaximumFrameSize parameter [item b).3) of 6.3.6.1].

16 6.4.3.8 Token Bucket TSpec sub-TLV

17 This sub-TLV contains a Token Bucket TSpec as defined in 6.3.6.2. Figure 6-11 shows the encoding of the
18 Value field of the Token Bucket TSpec sub-TLV.

	Octet	Length
MaxTransmittedFrameLength	1	2
MinTransmittedFrameLength	3	2
CommittedInformationRate	5	8
CommittedBurstSize	13	4

Figure 6-11—Value of Token Bucket TSpec sub-TLV

19

20

1

2 **6.4.3.8.1 MaxTransmittedFrameLength**

3 A 2-octet unsigned integer, encoding the MaxTransmittedFramelength parameter [item 1) of 6.3.6.2].

4 **6.4.3.8.2 MinTransmittedFrameLength**

5 A 2-octet unsigned integer, encoding the MinTransmittedFramelength parameter [item 2) of 6.3.6.2].

6 **6.4.3.8.3 CommittedInformationRate**

7 A 8-octet unsigned integer, encoding the CommittedInformationRate parameter [item 3) of 6.3.6.2].

8 **6.4.3.8.4 CommittedBurstSize**

9 A 4-octet unsigned integer, encoding the CommittedBurstSize parameter [item 4) of 6.3.6.2].

10 **6.4.3.9 Redundancy Control sub-TLV**

11 The Redundancy Control sub-TLV contains the information exclusively used by the Multi-Context Talker
12 Announcement. Figure 6-12 shows the encoding of the Value field of the Redundancy Control sub-TLV.

	Octet	Length
RTagStatus	1	1 bit
Reserved	1	7 bits
1 or more VLAN Context Informtion sub-TLVs	2	variable

Figure 6-12—Value of Redundancy Control sub-TLV

13 **6.4.3.9.1 RTagStatus**

14 A Boolean value indicating whether the data frames of the stream are (TRUE) or are not (FALSE) to contain
15 a Redundancy tag (R-TAG, [CB]:7.8) when transmitted through the Port that declares this sub-TLV.

16 **6.4.3.9.2 VLAN Context Information sub-TLV**

17 One or more VLAN Context Information sub-TLVs are included in the Redundancy Control sub-TLV, each
18 describing a distinct VLAN Context (6.3.5.1.1). The Value field encodes in the VlanContextId field a 12-bit
19 VLAN Context ID, followed by a 4-bit Reserved field and then zero or one Failure Information sub-TLV
20 (6.4.3.10), as illustrated in Figure 6-13.

	Octet	Length
VlanContextId	1	12 bits
Reserved	2	4 bits
0 or 1 Failure Information sub-TLV	3	variable

Figure 6-13—Value of VLAN Context Information sub-TLV

6.4.3.10 Failure Information sub-TLV

The Failure Information sub-TLV contains the information about the location and the type of a failure encountered in the Talker Announcement. Figure 6-14 shows the encoding of the Value field of the Failure Information sub-TLV.

	Octet	Length
SystemId	1	8
FailureCode	9	1

Figure 6-14—Value of Failure Information sub-TLV

6.4.3.10.1 SystemId

An 8-octet unsigned integer, indicating the identifier of a Bridge or end station at which the failure occurs. The SystemId value for an end station shall be the 6-octet MAC Address of the end station's port extended to 8 octets by prepending 2 octets of zero. The SystemId value for a Bridge shall be the 6-octet Bridge Identifier ([Q]:13.26.2) of the Bridge extended to 8 octets by prepending 2 octets of zero.

6.4.3.10.2 FailureCode

A 1-octet unsigned integer, containing the value of a RAP Failure Code defined in Table 6-6.

Table 6-6—RAP Failure Codes

Name	Value	Description	Reference
LatencyExceeded	0x01	Latency constraint not met	6.3.5.1
BandwidthExceeded	0x02	Bandwidth constraint not met	6.3.5.1
ResourceExceeded	0x03	Resource constraint not met	6.3.5.2
CrossingDomainBoundary	0x04	Talker Announcement across RA Class domain boundary	6.6.5.1
ResourceAllocationFailed	0x05	Resource allocation failed	6.6.5.17
ReservationPreempted	0x06	Reservation preempted by Rank zero streams	6.6.5.16
RTagFailed	0x07	Requirements for redundancy (un)tagging not fulfilled	6.3.8.2
StreamSplitFailed	0x08	Requirements for Stream splitting not fulfilled	6.3.8.3
MissingTalkerAnnounce	0x09	Missing Talker Announce at a Stream merging point	6.6.4.12
ListenerLackingFrer	0x0A	Listener lacking FRER capability	6.6.8.6
InconsistentRedundancyContext	0x0B	Inconsistent configuration of Redundancy Context	6.3.8.2

6.4.3.11 Organizationally Defined sub-TLV

The Organizationally Defined sub-TLV contains information defined by an organization that owns an OUI or a CID obtained from the IEEE registration authority. Figure 6-15 shows the encoding of the Value field of the Organizationally Defined sub-TLV.

	Octet	Length
OUI/CID	1	3
OrganizationallyDefinedData	4	variable

Figure 6-15—Value of Organizationally Defined sub-TLV

6.4.3.11.1 OUI/CID

A 3-octet OUI or an CID.

6.4.3.11.2 OrganizationallyDefinedData

The encoding of this field and the semantics associated with its values if any, is specific to and defined by the organization that owns the OUI/CID contained in 6.4.3.11.1.

6.4.4 Listener Attach attribute and TLV encoding

The Listener Attach attribute is used in the Listener Attachment (6.3.6). Figure 6-16 shows the encoding of the Value field of the Listener Attach attribute TLV.

	Octet	Length
StreamId	1	8
VID	9	12 bits
ListenerAttachStatus	10	4 bits
VLAN Context Status sub-TLV (only for Multi-Context Listener Attachment)	11	variable

Figure 6-16—Value of Listener Attach attribute TLV

For support of two types of the Listener Attachment, the encoding of the Listener Attach attribute follows the following rules:

- a) The Listener Attach attribute used in the Single-Context Listener Attachment [item a) in 6.3.4.1] contains no VLAN Context Status sub-TLV (6.4.4.4).
- b) The Listener Attach attribute used in a Multi-Context Listener Attachment [item b) in 6.3.4.1] contains one VLAN Context Status sub-TLV (6.4.4.4).

6.4.4.1 StreamId

An 8-octet field encoding the StreamID element as specified in [Q]:46.2.3.1.

6.4.4.2 VID

A 12-bit integer, indicating a VID to be encoded in the VLAN tag with which the data frames of the stream are tagged.

6.4.4.3 ListenerAttachStatus

A 4-bit field, taking one of the following enumerated values to indicate the Listener Attach status (6.3.4.3):

- a) **1: Attach Ready** [item a) in 6.3.4.3];
- b) **2: Attach Fail** [item b) in 6.3.4.3];
- c) **3: Attach Partial Fail** [item c) in 6.3.4.3].

6.4.4.4 VLAN Context Status sub-TLV

The VLAN Context Status sub-TLV contains the information exclusively used by the Multi-Context Listener Attachment. The Value field contains one or more VLAN Context tuples, each encoding in VlanContextId field a 12-bit VLAN Context ID, followed by a 4-bit PathStatus field, as illustrated in Figure 6-17.

		Octet	Length
VLAN Context tuple 1	VlanContextId	1	12 bits
	PathStatus	2	4 bits
...			
VLAN Context tuple n	VlanContextId	2n-1	12 bits
	PathStatus	2n	4 bits

Figure 6-17—Value of VLAN Context Status sub-TLV

6.4.4.4.1 PathStatus

A 4-bit field, taking one of the following enumerated values to indicate the Path status (6.3.5):

- a) **0: Faultless Path** [item a) in 6.3.5];
- b) **1: Faulty Path** [item b) in 6.3.5];
- c) **2: Unresponsive Path** [item c) in 6.3.5].

6.5 RAP Endpoint

6.5.1 RAP Endpoint overview

A RAP Endpoint associated with an end station is responsible for the following:

- Provision of service interface to higher layer entities (termed “RAP Endpoint user”) for controlling of resource allocation by end stations.
- Handling of resource allocation for an end station.
- Generation of attributes for declarations and processing of received attribute registrations.

The operation of a RAP Endpoint is specified by the following:

- RAP Endpoint Service Interface (RESI) in 6.5.2.
- RAP Endpoint procedures in 6.6.8.

6.5.2 RAP Endpoint Service Interface (RESI)

The RESI provided to RAP Endpoint user comprises a set of primitives and associated parameters, which are divided into three groups, as summarized in Table 6-7.

Table 6-7—RAP Endpoint Service Interface primitives

	primitive	reference
RA primitives (6.5.2.1)	DECLARE_RA.request	6.5.2.1.1
	REGISTER_RA.indication	6.5.2.1.2
TA primitives (6.5.2.2)	ANNOUNCE_STREAM.request	6.5.2.2.1
	ANNOUNCE_STREAM.indication	6.5.2.2.2
	TERMINATE_STREAM.request	6.5.2.2.3
	TERMINATE_STREAM.indication	6.5.2.2.4
LA primitives (6.5.2.3)	ATTACH_STREAM.request	6.5.2.3.1
	ATTACH_STREAM.indication	6.5.2.3.2
	DETACH_STREAM.request	6.5.2.3.3
	DETACH_STREAM.indication	6.5.2.3.4

6.5.2.1 RA primitives

The RA primitives specified in 6.5.2.1.1 and 6.5.2.1.2 are used for controlling and monitoring RA Advertisement (6.3.3) in an end station.

The method of using the RA primitives is specific to each end station. For example, if the RAP Endpoint user in an end station is made aware of the RA class and Redundancy Context configurations in the network through a higher layer protocol, it can make an RA declaration without having to wait for an RA registration to be received from a neighboring station. Otherwise, the end station relies on the information received in an RA registration from a neighboring station to make an RA declaration.

1 6.5.2.1.1 DECLARE_RA.request()

2 This primitive is used by the RAP Endpoint user to request the RAP Endpoint to make an RA declaration. It
3 has the following parameters:

- 4 a) **MaxInterferingFrameSize** (6.4.2.1)
- 5 b) One or more sets of the following parameters, each associated with a distinct RA class (6.4.2.2):
 - 6 1) **RAClassId** (6.4.2.2.1)
 - 7 2) **RaClassPriority** (6.4.2.2.2)
 - 8 3) **RTID** (6.4.2.2.3)
 - 9 4) **TrafficClass** (6.4.2.2.4)
 - 10 5) **MaxLastHopLatency** (6.4.2.2.5)
 - 11 6) **RaClassTemplateDefinedData** (6.4.2.2.6)
- 12 c) Zero or more sets of **VLAN Context IDs**, each associated with an Redundancy Context (6.4.2.3).

13 6.5.2.1.2 REGISTER_RA.indication()

14 The REGISTER_RA.indication primitive is used by the RAP Endpoint to inform the RAP Endpoint user
15 about an RA registration. This primitive has the same set of parameters as that defined in 6.5.2.1.1.

16 6.5.2.2 TA primitives

17 The TA primitives specified in the subsequent clauses (6.5.2.2.1 through 6.5.2.3.4) are used for controlling
18 and monitoring Talker Announcement (6.3.4).

19 6.5.2.2.1 ANNOUNCE_STREAM.request()

20 This primitive is used by the RAP Endpoint user in a Talker to request the RAP Endpoint to initiate a Talker
21 Announcement. It has the following parameters:

- 22 a) **StreamId** (6.4.3.1)
- 23 b) **StreamRank** (6.4.3.2)
- 24 c) **AccuMaxLatency** (6.4.3.3)
- 25 d) **AccuMinLatency** (6.4.3.4)
- 26 e) A 3-tuple of {**DestinationMacAddress** (6.4.3.5.1), **Priority** (6.4.3.5.2) and **VID** (6.4.3.5.3)}
- 27 f) Either of the following as TalkerTSpec (6.4.3.6):
 - 28 1) A 4-tuple of {**MaxTransmittedFrameLength** (6.4.3.6.1), **MinTransmittedFrameLength**
29 (**6.4.3.6.2**), **CommittedInformationRate** (6.4.3.6.3) and **CommittedBurstSize** (6.4.3.6.4)},
30 indicating a Token Bucket TSpec (6.4.3.6)
 - 31 2) A 3-tuple of {**Interval** (6.4.3.7.1), **MaximumFramesPerInterval** (6.4.3.7.2) and
32 **MaximumFrameSize** (6.4.3.7.3)}, indicating an Interval-based TSpec (6.4.3.6)
- 33 g) **Redundancy Context ID** (6.3.8.2), if supplied, indicating a Multi-Context Talker Announcement
34 and otherwise, indicating a Single-Context Talker Announcement
- 35 h) A 2-tuple of {**OUI/CID** (6.4.3.11.1) and **OrganizationallyDefinedData** (6.4.3.11.2)}, if supplied,
36 indicating the use of the Organizationally Defined sub-TLV (6.4.3.11)
- 37 i) **Failure Information**, set to either zero or a non-zero RAP Failure Code defined in Table 6-6.

38 The RAP Endpoint shall not accept an ANNOUNCE_STREAM.request with a StreamId value for which it
39 has initiated a Talker Announcement since a previous ANNOUNCE_STREAM.request.

1 **6.5.2.2.2 ANNOUNCE_STREAM.indication()**

2 This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Listener about the result
3 of processing a Talker Announcement received. In addition to all the parameters defined in 6.5.2.2.1, this
4 primitive has the following additional parameter for use only in the case of a Multi-Context Talker
5 Announcement:

- 6 a) A set of 3-tuples, each with {**VlanContextId**, **SystemId**, **FailureCode**} and corresponding to a
7 VLAN Context Information sub-TLV (6.4.3.9.2), where SystemId and FailureCode are provided
8 only when that VlanContextId is reported with a RAP Failure Code in the Talker Announcement.

9 NOTE—As a response to a received ANNOUNCE_STREAM.indication, the RAP Endpoint user of a Listener can, if it
10 is interested in receiving that stream, initiates a Listener Attachment by issuing an ATTACH_STREAM.request
11 (6.5.2.3.1) with the StreamId contained in the received primitive.

12 **6.5.2.2.3 TERMINATE_STREAM.request()**

13 This primitive is used by the RAP Endpoint user in a Talker to request the RAP Endpoint to terminate a
14 Talker Announcement. It has a single parameter StreamId (6.4.3.1).

15 If a Talker Announcement to be terminated is associated with a stream being transmitted by the Talker, it is
16 required that the Talker stops transmitting the stream before issuing a TERMINATE_STREAM.request.

17 NOTE—This requirement can be viewed as part of the fundamental principle of resource allocation, i.e., reservation
18 prior to stream transmission.

19 **6.5.2.2.4 TERMINATE_STREAM.indication()**

20 This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Listener that a Talker
21 Announcement previously received has been removed. It has a single parameter StreamId (6.4.3.1).

22 NOTE—As a response to a received TERMINATE_STREAM.indication, the RAP Endpoint user of a Listener can, if it
23 has initiated a corresponding Listener Attachment, terminate that Listener Attachment by issuing a
24 DETACH_STREAM.request (6.5.2.3.3) with the StreamId contained in the received primitive.

25 **6.5.2.3 LA primitives**

26 The Listener primitives specified in the following subclauses (6.5.2.3.1 through 6.5.2.2.4) are used for
27 controlling and monitoring Listener Attachment (6.3.6).

28 **6.5.2.3.1 ATTACH_STREAM.request()**

29 This primitive is used by the RAP Endpoint user in a Listener to request the RAP Endpoint to initiate a
30 Listener Attachment. It has a single parameter StreamId (6.4.4.1).

31 NOTE—An ATTACH_STREAM.request issued with a given StreamId won't actually cause initiation of a Listener
32 Attachment, if a Talker Announce registration with that StreamId is not present in the RAP Endpoint.

33 **6.5.2.3.2 ATTACH_STREAM.indication()**

34 This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Talker about the result of
35 processing a Listener Attachment received. It has the following parameters:

- 36 a) **StreamId** (6.4.4.1)
37 b) **VID** (6.4.4.2)
38 c) **ListenerAttachStatus** (6.4.4.3)

1 d) A set of 2-tuples, each with {VlanContextId, PathStatus} (6.4.4.4)

2 NOTE—The ListenerAttachStatus value and in the case of a Multi-Context Listener Attachment the set of PathStatus
3 values contained in a received ATTACH_STREAM.indication can be used by the RAP Endpoint user to instruct the
4 Talker whether and when to start transmission of the stream concerned, according to the semantics associated with these
5 status parameters.

6 **6.5.2.3.3 DETACH_STREAM.request()**

7 This primitive is used by the RAP Endpoint user in a Listener to request the RAP Endpoint to terminate a
8 Listener Attachment previously initiated. It has a single parameter StreamId (6.4.4.1).

9 NOTE—The RAP Endpoint user of a Listener can issue a DETACH_STREAM.request to terminate a specific Listener
10 Attachment it has initiated previously at any time as desired, including but not limited to the case when receiving a
11 TERMINATE_STREAM.indication (6.5.2.2.4) that indicates the associated Talker Announcement has been terminated.

12 **6.5.2.3.4 DETACH_STREAM.indication()**

13 This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Talker that a Listener
14 Attachment previously received for a Talker Announcement initiated by the Talker. It has a single parameter
15 StreamId (6.4.3.1).

16 NOTE—A notification of a removed Listener Attachment occurs only when the RAP Endpoint of a Talker has removed
17 any Listener Attach registration previously received as belonging to that Listener Attachment. In this case, it is up to the
18 RAP Endpoint user to determine whether to retain the Talker Announcement concerned.

19 **6.6 RAP MAD**

20 **6.6.1 RAP MAD Overview**

21 RAP MAD is an application specific MAD replacement for RAP that is used instead of the MAD that is
22 described in [Q].

23 The functionality of RAP MAD is different for Endpoints and Bridges.

24 It comprises ingress functions, accesses the MAP and has egress functions.

25 **6.6.2 RAP MAD ingress state machine for Bridges diagrams**

26 The operation of the RAP MAD state machine is specified by the state machine diagram in Figure 6-18 . The
27 state changes are triggered after receiving events from the LSI for the LRP Portal and from the bridge
28 component for the change of the DecOperStatus. The actions executed on entry to each state of the state
29 machine are specified in clause 6.6.4.

30 The primitives received from the LSI are examined and trigger the state transitions in this state machine
31 depending on the received attributes within the data record.

1

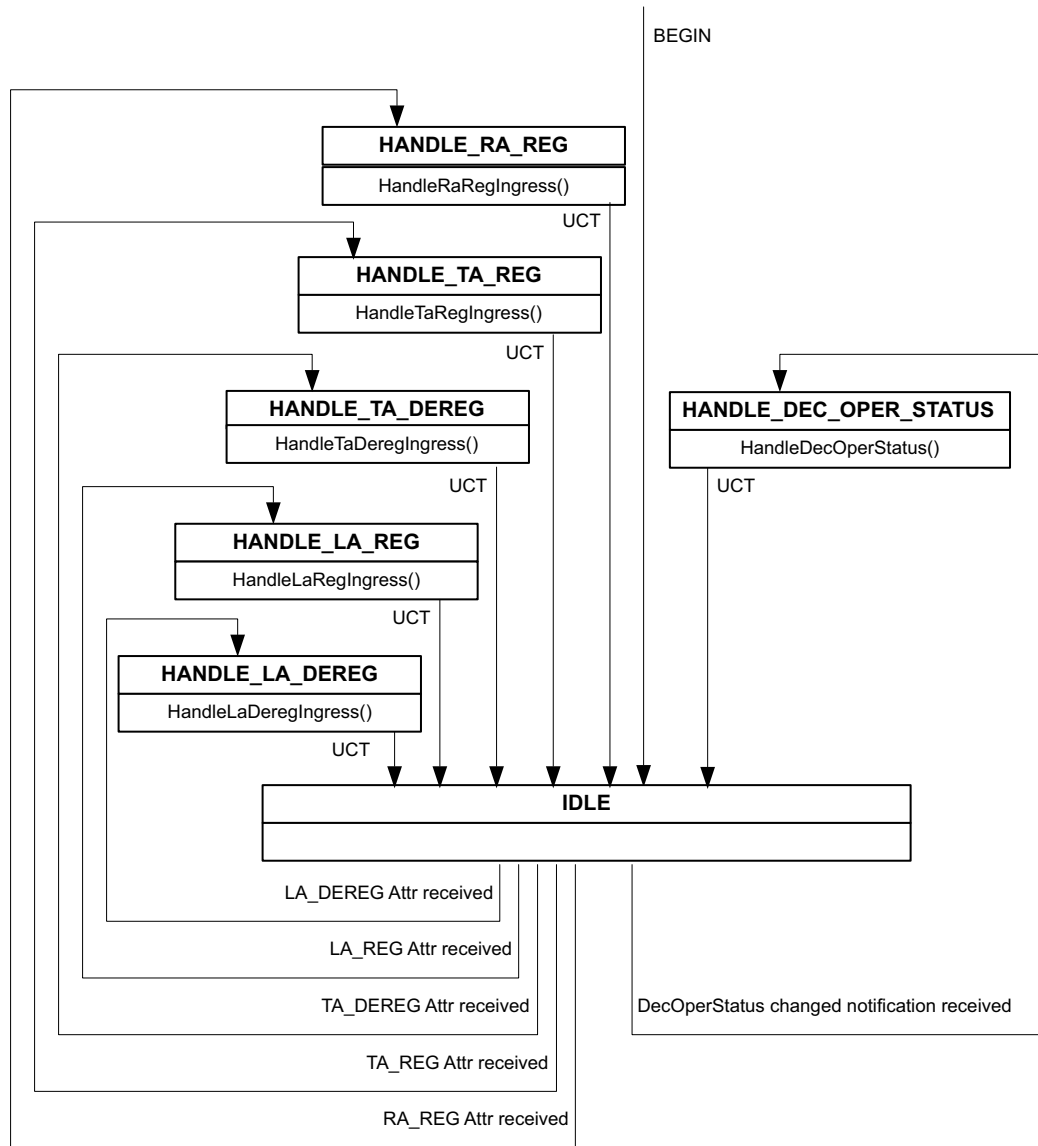


Figure 6-18—RAP MAD Ingress Statemachine

2 6.6.3 RAP MAD Variables

3 6.6.3.1 ingressStatus

4 The ingress status of the Talker Announce registration, taking one of the following enumerated values:

- 5 — TA_RECV_FAIL (0): The Talker Announce registration is failed by an upstream station.
- 6 — TA_INGRESS_SUCCESS (1): The Talker Announce registration is neither failed by an upstream
- 7 station nor failed on ingress of this Bridge.

- 1 — TA_INGRESS_FAIL (2): The Talker Announce registration is failed on ingress of this Bridge with a
2 RAP Failure Code indicated in ingressFailureCode.

3 **6.6.3.2 ingressFailureCode**

- 4 A value of zero, indicating the Talker Announce registration is not failed on ingress of this Bridge, or the
5 value of a RAP Failure Code defined in Table 6-6.

6

7 **6.6.4 RAP MAD ingress procedures**

- 8 RAP MAD ingress functions handle the events, received from a LRP-Portal.

9 **6.6.4.1 HandleRaRegIngress()**

- 10 This procedure handles the event of receiving a registration of a RA Attribute from a LRP-Portal.

- 11 Call the processRaReg() procedure (6.6.7.1).

12 **6.6.4.2 HandleTaRegIngress()**

- 13 This procedure handles the event of receiving a registration of a TA Attribute from a LRP-Portal.

- 14 If TA is valid (isTaValid() (6.6.7.2) == FALSE):

- 15 a) If a TA with the same StreamID and VID was already received at the same port:
16 1) update Stream TA Registration
17 b) otherwise
18 2) store Stream TA Registration.
19 c) Trigger propagateTa() (6.7.7.1).

- 20 otherwise

- 21 d) Stop TA propagation

22 **6.6.4.3 HandleTaDeregIngress()**

- 23 This procedure handles the event of receiving a deregistration of a TA Attribute from a LRP-Portal.

- 24 If the indicated RA was registered at the port trigger HandleTaDeregMAP() (6.7.7.2).

25 **6.6.4.4 HandleLaRegIngress()**

- 26 This procedure handles the event of receiving a registration of a LA Attribute from a LRP-Portal.

- 27 If this LA was already received on this port:

- 28 a) update LA registration

- 29 otherwise

- 30 b) store LA registration

1 if there is a corresponding TA declaration:

2 a) Trigger propagateLa() (6.7.7.3).

3 **6.6.4.5 HandleLaDeregIngress()**

4 This procedure handles the event of receiving a deregistration of a LA Attribute from a LRP-Portal.

5 If this LA registration exists on this port:

6 a) If this LA is part of an active reservation:

7 1) deallocate resources for the ingress port of this event

8 2) updated the reserved bandwidth

9 b) Trigger HandleLaDeregMAP() (6.7.7.4).

10 **6.6.4.6 HandleDecOperStatus()**

11 This procedure handles the event of receiving a change notification about a port's operational state.

12 If the port gets operational:

13 a) Update RA with DECLARE_ATTR.request() (6.6.9.1) for each enabled RA-class on this port

14 b) For each existing TA Registration Declare TA at LRP Portal

15 c) For each existing associated LA Declarations trigger processLaDec() in MAP (6.7.7.11)

16 otherwise:

17 d) For each TA Declaration with an associated LA Registration:

18 1) If there is a reservation for the LA, deallocate the resources and update reserved bandwidth

19 2) For each existing associated LA Declarations trigger processLaDec() in MAP (6.7.7.11)

20 e) Trigger HandleTaDeregMAP() (6.7.7.2)

21 **6.6.5 RAP MAD egress procedures**

22 RAP MAD egress functions handle events after they are processed by MAP.

23 **6.6.5.1 HandleTaDecEgress()**

24 This function handles the egress processing of a TA declaration issued by MAP.

25 If no listener corresponding to the TADec exists, or if no reservation has yet been established:

26 a) the stream reservation feasibility shall be evaluated using the checkReservationConstraints()
27 function (6.6.7.4). The appropriate error code shall be set in the TADec.

28 b) Declare TA at LRP Portal

29 If an associated LA at the portal was already received (according to the criteria defined in 6.3.6.1
30 Association between Talker Announcement and Listener Attachment):

31 c) Trigger processLa() in MAP (6.7.7.13)

32 d) Trigger propagateLaReg() in MAP (6.7.7.10))

33 e) Declare TA in LRP Portal

1 **6.6.5.2 HandleTADeregUpdateTaDecEgress()**

2 This function handles the egress processing of a TA Deregistration issued by MAP to update TA
3 Declarations.

4 Trigger processTaDec() in MAP (6.7.7.14).

5 Update TA with DECLARE_ATTR.request() (6.6.9.1).

6 If there is an associated reservation:

7 a) Trigger processLa() in MAP (6.7.7.13) to update the reservation upstream.

8 **6.6.5.3 HandleTADeregRemoveTaDecEgress()**

9 This function handles the egress processing of a TA Deregistration issued by MAP to memove TA
10 Declarations.

11 Stop declaring this Attribute with WITHDRAW_ATTR.request() (6.6.9.2).

12 Delete the Declaration at the port.

13 If there is an associated reservation:

14 a) Update allocated bandwidth

15 b) Deallocate ressources

16 c) Update LA propagation upstream

17 **6.6.5.4 HandleLaDecEgress()**

18 This function handles the egress processing of a LA declaration issued by MAP.

19 If there is an associated TA Registration:

20 a) Trigger processLaDec() in MAP (6.7.7.11).

21 **6.6.5.5 HandleLaDeregEgress()**

22 This function handles the egress processing of a LA Deregistration issued by MAP.

23 Trigger processLaDec() in MAP (6.7.7.11).

24 **6.6.6 Definition of LRP protocol elements**

25 This subclause specifies the LRP protocol elements that are specific to the operation of RAP.

26 **6.6.6.1 Value of AppId**

27 The general format of AppIds is defined in [CS]:9.2. The AppId that identifies the RAP specified in this
28 standard shall take the values shown in Table 6-7.

29 **6.6.6.2 Use of LLDP**

30 When the IEEE Std 802.1AB Link Layer Discovery Protocol (LLDP) is deployed to advertise and discover
31 target ports, the destination MAC address of the LLDP agent that is selected to carry the AppId (6.6.6.1) in

Table 6-8—The Appld value for RAP

Field	Value	Length (octets)	Offset (octets)
OUI or CID	00-80-C2	3	0
Application Sub-ID	1	1	3

1 an LRP ECP Discovery TLV ([CS]:C.2.1) and/or an LRP TCP Discovery TLV ([CS]:C.2.2), shall be the
2 Nearest Bridge group address (01-80-C2-00-00-0E) as specified in [Q]:Table 8-1, [Q]:Table 8-2, and
3 [Q]:Table 8-3.

4 6.6.6.3 Use of LRP ECP mechanism

5 **Change to more generic text.**

6 When the LRP-DT ECP mechanism ([CS]:6.9.1) is deployed, the destination MAC address of the ECP
7 instance operating on each target port of that system shall be the Nearest Bridge group address (01-80-C2-
8 00-00-0E) as specified in [Q]:Table 8-1, [Q]:Table 8-2, and [Q]:Table 8-3.

9 6.6.6.4 Application Information TLV

10 An Application Information TLV, in the format specified in [CS]:9.4.2.10 and exchanged via LRP Hello
11 LRPDU s ([CS]:9.4.2), is used by each RAP Participant to advertise the information about its capabilities
12 and other operational parameters to its neighbor(s). Figure 6-19 shows the encoding of the Value field of the
13 Application Information TLV:

	Octet	Length
ProtocolVersion	1	1

Figure 6-19—Value of Application Information TLV

- 14 a) **ProtocolVersion:** The ProtocolVersion for the version of RAP defined in this standard takes the
15 hexadecimal value 0x00.

16

1

2 6.6.6.5 Use of LRP-DS service interface

3 Change to more generic text.

4 For the purposes of the operation of a RAP Participant (6.7), a set of primitives and associated parameters
5 chosen from the LRP-DS service interface defined in [CS]:10 are summarized in Table 6-8.

6 The handling of the functions in this table must be described in the next clause

7

Table 6-9—LRP-DS service primitives used by RAP

Primitive	Subclause in IEEE Std 802.1CS-2020	Parameter	Item in IEEE Std 802.1CS-2020
LOCAL_TARGET_PORT.request	10.2.2	(see Table 6-13)	
NEIGHBOR_TARGET_PORT.request	10.2.3		
ASSOCIATE_PORTAL.request	10.2.4	PortalId	10.2.4.1 a)
		IsApproved	10.2.4.1 b)
FIRST_HELLO.indication ^a	10.2.5	PortalId	10.2.5.1 a)
		HelloData	10.2.5.1 b)
PORTAL_STATUS.indication	10.2.6	PortalId	10.2.6.1 a)
		AssociationStatus	10.2.6.1 b)
WRITE_RECORD.request	10.3.1.1	PortalId	10.3.1.1.1 a)
		RecordNumber	10.3.1.1.1 b)
		RecordData	10.3.1.1.1 c)
DELETE_RECORD.request	10.3.2.1	PortalId	10.3.2.1.1 a)
		RecordNumber	10.3.2.1.1 b)
RECORD_WRITTEN.indication	10.3.2.3	PortalId	10.3.2.3.1 a)
		RecordNumber	10.3.2.3.1 b)
		RecordData	10.3.2.3.1 c)

^a It is assumed that a FIRST_HELLO.indication primitive invoked due to receiving a Hello message on a given Port by LRP be passed to a RAP Participant on that Port.

1 6.6.7 RAP MAD internal procedures

2 6.6.7.1 processRaReg()

3 This procedure processes an RA registration on a Port, as follows:

- 4 a) Update neighbor's RAClass information on the associated Port
- 5 b) Check for each local RAClass at receiving Port if this Port is a boundary port and set the
- 6 domainBoundaryPort accordingly

7 Note: a Port is a boundary port if there is no neighbor with the same ClassID and the same priority (PCP)

8 6.6.7.2 isTaValid()

9 This procedure determines whether a Talker Announce registration is valid (TRUE) or not (FALSE). It
10 checks if there is already one or more existing Talker Announce registrations with the same StreamId and
11 compares the new TA registration with the previous TA registrations. The procedure returns FALSE if one or
12 more of the following conditions are met:

- 13 a) the result of the procedure isSingleContext() (6.6.7.3) for the previous and the new TA registration is
14 different
- 15 b) the StreamRank of the previous and the new TA registration is different
- 16 c) the DestinationMacAddress of the previous and the new TA registration is different
- 17 d) the Priority of the previous and the new TA registration is different
- 18 e) the result of the procedure isSingleContext() for the new TA registration is TRUE and the new TA
19 registration has as different VID for the same port
- 20 f) the result of the procedure isSingleContext() for the new TA registration is TRUE and the new TA
21 registration has as different port for the same VID
- 22 g) the result of the procedure isSingleContext() for the new TA registration is FALSE and the set of
23 VLAN Context IDs contained in the Redundancy Control sub-TLV in the new TA registration is
24 neither the same nor a subset of any Redundancy Context configured in redundancyContext
25 (6.7.6.2).

26 Otherwise, the procedure returns TRUE.

27 6.6.7.3 isSingleContext()

28 This procedure returns TRUE if the Talker Announce attribute indicates a Single-Context Talker
29 Announcement according to [item a) in 6.4.3], and returns FALSE if it indicates a Multi-Context Talker
30 Announcement according to the [item b) in 6.4.3]

31 6.6.7.4 checkReservationConstraints()

32 This procedure performs checks to determine whether a stream as indicated by a Talker Announce
33 declaration meets the resource allocation constraints (6.3.5). This subclause specifies only the input and
34 output parameters, along with the rules of handling the stream Rank, but not the detailed algorithms that are
35 specific to the RA class template used by the stream specific or dependent on the particular Bridge
36 implementation.

1 In computing the current available resources, the existing reservations in this Bridge are treated according to
2 the StreamRank value of the TA declaration, as follows:

- 3 a) If StreamRank contains the value zero, only those existing reservations being made for the streams
4 of Rank zero are treated as “reserved”, and all other existing reservations for the streams of Rank
5 one are treated as “not reserved” as they can be preempted as described in 6.3.7.
- 6 b) If StreamRank contains the value one, all of the existing reservations regardless of the stream Rank
7 are treated as “reserved”.

8 The procedure checks each reservation constraint defined in 6.3.5, for which there is no requirement for the
9 order in which one constraint is examined after another. **Note that when the stream is subject to stream
10 merging (6.3.8.4) on a FRER-capable RAP Bridge’s Port, as indicated by pTaDec.origTaReg == NULL, the
11 procedure takes into account only each Member Stream listed in pTaDec.origTaRegList and whose
12 ingressStatus indicates TA_INGRESS_SUCCESS.**

13 In case that one or more of these constraints are not met, the procedure returns the value of the RAP Failure
14 Code associated with one constraint not met, as described in 6.3.5. It returns the value zero, if all of the
15 constraints are met.

16 6.6.8 RAP MAD Endpoint procedures

17 The subsequent subclauses specify a set of procedures for the RAP Endpoint to process request primitives
18 received at RESI (6.5.2) and indication primitives received at RISI (6.6.9).

19 The procedures specified in these subclauses assume that the information about each attribute being declared
20 and each attribute being registered is maintained in a database and accessible for the RAP Endpoint. A
21 description about the data structure and operation methods of such a database is not provided; it is an
22 implementation choice.

23 6.6.8.1 handleDeclareRaRequest()

24 This procedure is invoked when the RAP Endpoint receives at RESI a DECLARE_RA.request (6.5.2.1.1). It
25 constructs the RA attribute (6.4.2) with the parameter values supplied in the primitive and issues a
26 DECLARE_ATTR.request (6.6.9.1) at RISI with the constructed attribute.

27 6.6.8.2 handleRaRegIndication()

28 This procedure is invoked when the RAP Endpoint receives at RISI an ATTR_REG.indication (6.6.9.3) that
29 indicates an RA registration. It issues at RESI a REGISTER_RA.indication (6.5.2.1.2) with the parameter
30 values abstracted from the RA registration supplied in the primitive.

31 6.6.8.3 handleAnnounceStreamRequest()

32 This procedure is invoked when the RAP Endpoint receives at RESI an ANNOUNCE_STREAM.request
33 (6.5.2.2.1). It performs the following actions:

- 34 a) If there exist one or more Talker Announce declarations associated with the same StreamId (6.4.3.1)
35 as indicated in the primitive, terminate the procedure.
- 36 b) Otherwise, construct a Talker Announce attribute TLV (6.4.3) in tTaAttr with the values supplied in
37 items a), b), c), d), e), f) and h) of 6.5.2.2.1.
- 38 c) If a non-zero RAP Failure Code value is supplied in i) of 6.5.2.2.1, construct a Failure Information
39 sub-TLV (6.4.3.10) with that RAP Failure Code and the SystemId value of this end station, and then
40 append it to tTaAttr.

- 1 d) Construct in the TA attribute's NetworkTSpec a Token Bucket TSpec sub-TLV (6.4.3.6) with the
2 values either copied from those supplied in item f).1) of 6.5.2.2.1 for a Token Bucket TSpec, or
3 translated from those supplied in item f).2) of 6.5.2.2.1 for an Interval-based TSpec according to
4 Table 6-2.
- 5 e) If no value is supplied in g) of 6.5.2.2.1, issue a DECLARE_ATTR.request (6.6.9.1) at RISI with TA
6 attribute and then terminate. Otherwise, set RedundancyContextId to the value supplied.
- 7 f) If the end station is not FRER-capable, perform the following:
8 1) Construct a Redundancy Control sub-TLV (6.4.3.9) with RTagStatus (6.4.3.9.1) set FALSE and
9 with a set of VLAN Context Information sub-TLVs (6.4.3.9.2) constructed for each VLAN
10 Context ID belonging to the Redundancy Context identified by RedundancyContextId (see
11 item e), and then append it to TA attribute.
12 2) Set VID in the TA attribute to RedundancyContextId (see item e).
13 3) Issue a DECLARE_ATTR.request (6.6.9.1) at RISI with TA attribute.
- 14 g) Otherwise, i.e., the end station is FRER-capable, for each VLAN Context ID belonging to the
15 Redundancy Context identified by RedundancyContextId (see item e), perform the following:
16 1) Copy the TA attribute to a new TA attribute.
17 2) Construct a Redundancy Control sub-TLV (6.4.3.9) with RTagStatus (6.4.3.9.1) set TRUE and
18 with a single VLAN Context Information sub-TLV (6.4.3.9.2) constructed for the VLAN
19 Context ID, and then append it the new TA attribute.
20 3) Set the VID in the new TA attribute to the VLAN Context ID.
21 4) Issue a DECLARE_ATTR.request (6.6.9.1) at RISI with the new TA attribute.

22 6.6.8.4 handleTerminateStreamRequest()

23 This procedure is invoked when the RAP Endpoint receives a TERMINATE_STREAM.request (6.5.2.2.3)
24 at RESI with a StreamId value. It performs the following actions:

- 25 a) If there is no Talker Announce declaration associated with StreamId, terminate.
- 26 b) Otherwise, for each Talker Announce declaration associated with StreamId, issue at RISI a
27 WITHDRAW_ATTR.request (6.6.9.2) with the corresponding Talker Announce attribute and then
28 remove the Talker Announce declaration.

29 6.6.8.5 handleTaReg()

30 This procedure is invoked when the RAP Endpoint receives at RISI an ATTR_REG.indication (6.6.9.3) that
31 indicates a Talker Announce registration with a TA attribute. It performs the following actions:

- 32 a) If the TA attribute indicates a Multi-Context Talker Announcement and the RedundancyControl
33 value in the TA attribute contains only part of the VLAN Contexts of a Redundancy Context
34 declared by this end station, call **handleTaMerge()** (6.6.8.6) and terminate.
- 35 b) Copy the TA attribute to a new TA attribute.
- 36 c) If the TA attribute indicates **Announce Success**, perform checks of resource allocation constraints
37 (6.3.5) and then the following:
38 1) If all the resource allocation constraints are met, update both AccuMaxLatency and
39 AccuMinLatency in the new TA attribute.
40 2) Otherwise, fail TA attribute with the RAP Failure Code associated with a failure encountered
41 during the constraint checking.
- 42 d) Issue at RESI an ANNOUCE_STREAM.indication (6.5.2.2.2) with its parameters set to the
43 corresponding values contained in the new TA attribute.

1 6.6.8.6 **handleTaMerge()**

2 This procedure is invoked by the **handleTaReg()** procedure (6.6.8.5) to process a Multi-Context Talker
3 Announcement identified by a StreamId and subject to Talker Announce merge (6.3.5.3) in the end station. It
4 performs the following actions:

- 5 a) Search for each existing Talker Announce registration associated with the StreamId. If no such TA
6 registration is found, issue a TERMINATE_STREAM.indication (6.5.2.2.4) with the StreamId and
7 then terminate. Otherwise, proceed.
- 8 b) Construct a new Talker Announce attribute and fill each of the following fields with the common
9 value contained in the corresponding field of each TA registration found in a): StreamId,
10 StreamRank, DestinationMacAddress and Priority.
- 11 c) If one of the following conditions is met, fail the new TA attribute with the indicated RAP Failure
12 Code and then proceed to g) below:
13 1) *ListenerLackingFrer*, if this end station is not FRER-capable.
14 2) *RTagFailed*, if there is at least one TA attribute whose RTagStatus contains FALSE.
- 15 d) Copy each VLAN Context Information sub-TLV (6.4.3.9.2) contained in each TA attribute found in
16 a) to the RedundancyControl value in the new TA attribute, and set the VID in the new TA attribute
17 to the numerically smallest VLAN Context ID copied.
- 18 e) If each TA registration found in a) indicates **Announce Fail**, fail the new TA attribute with the
19 numerically smallest RAP Failure Code contained in RedundancyControl value in the new TA
20 attribute and then proceed to g).
- 21 f) If the new TA attribute indicates **Announce Success**, perform checks of resource allocation
22 constraints (6.3.5) on the new TA attribute and then the following:
23 1) If all the resource allocation constraints are met, update both AccuMaxLatency and
24 AccuMinLatency in the new TA attribute.
25 2) Otherwise, fail the new TA attribute with the RAP Failure Code associated with a failure
26 encountered during the constraint checking.
- 27 g) Issue at RESI an ANNOUCE_STREAM.indication (6.5.2.2.2) with its parameters set to the
28 corresponding values contained in new TA attribute.

29 6.6.8.7 **handleTaDereg()**

30 This procedure is invoked when the RAP Endpoint receives at RISI an ATTR_DEREG.indication (6.6.9.4)
31 that indicates a Talker Announce deregistration. It performs the following actions:

- 32 a) Remove the Talker Announce registration in pTaAttr.
- 33 b) If there is no remaining Talker Announce registration associated with the associated StreamId, issue
34 at RESI a TERMINATE_STREAM.indication (6.5.2.2.4) with the StreamId.
- 35 c) Otherwise, call **handleTaMerge()** for the StreamId.

36 6.6.8.8 **handleAttachStreamRequest()**

37 This procedure is invoked when the RAP Endpoint receives at RESI an ATTACH_STREAM.request
38 (6.5.2.3.1) with a StreamId value. It performs the following actions:

- 39 a) Search for each Talker Announce registration associated with the StreamId. If no such TA
40 registration is found, terminate; otherwise proceed.
- 41 b) If there is no reservation associated with the StreamId in the end station, and the Talker
42 Announcement identified by the StreamId is not failed, make a reservation for the StreamId.
- 43 c) For each TA attribute found in a), issue at RISI a DECLARE_ATTR.request (6.6.9.1) with a
44 Listener Attach attribute constructed as follows:
45 1) Set the StreamId value in the LA attribute to the StreamId.

- 2) Set the VID value in the LA attribute to the VID in the TA attribute.
- 3) Set the ListenerAttachStatus in the LA attribute to **Attach Ready**, if there is a reservation associated with the StreamId, and **Attach Fail**, otherwise.
- 4) If the TA attribute indicates a Multi-Context Talker Announcement, construct a VLAN Context Status sub-TLV (6.4.4.4) with each VLAN Context Id contained in the TA attribute and the associated PathStatus set to **Faultless Path** if the LA attribute indicates **Attach Ready**, and **Faulty Path** if the LA attribute indicates **Attach Fail**, and then append it to the LA attribute.

6.6.8.9 handleDetachStreamRequest()

This procedure is invoked when the RAP Endpoint receives at RESI a DETACH_STREAM.request (6.5.2.3.3) with a StreamId value. It performs the following actions:

- a) If there is no Listener Attach declaration associated with the StreamId, terminate.
- b) If there is a reservation associated with the StreamId, remove the reservation.
- c) For each Listener Attach declaration associated with the StreamId, issue at RISI a WITHDRAW_ATTR.request (6.6.9.2) with the corresponding Listener Attach attribute and then remove the Listener Attach declaration.

6.6.8.10 handleLaReg()

This procedure is invoked when the RAP Endpoint receives at RISI an ATTR_REG.indication (6.6.9.3) that indicates a Listener Attach registration. It performs the following actions:

- a) Search for a Talker Announce declaration with which the LA attribute is associated, in accordance with 6.3.4.1. If no such TA declaration is found, terminate; otherwise, proceed.
- b) If the LA attribute indicates either **Attach Ready** or **Attach Partial Fail**, the TA attribute indicates **Announce Success**, and there is no reservation associated with the LA attribute, then make a reservation for the LA attribute.
- c) Else if the LA registration indicates **Attach Fail** and there is a reservation associated with the LA registration, then remove the reservation.
- d) If there is more than one Talker Announce declaration whose StreamId is the same as the LA registration's StreamId, call **handleLaMerge()** with this StreamId; otherwise, issue at RESI an ATTACH_STREAM.indication (6.5.2.3.2) with its parameters set to the corresponding values contained in the LA registration.

6.6.8.11 handleLaMerge()

This procedure is invoked by the **handleLaReg()** procedure (6.6.8.10) to process a Multi-Context Listener Attachment identified by a StreamId and subject to Listener Attach merge (6.3.4.2) in the end station. It performs the following actions:

- a) Search for each Listener Attach registration associated with the StreamId.
- b) Construct a Listener Attach attribute as follows:
 - 1) Set the StreamId in the LA attribute to the StreamId.
 - 2) Set the VID in the attribute to the Redundancy Context Id used by this Listener Attachment.
 - 3) Set the ListenerAttachStatus in the LA attribute to **Attach Ready**, if at least one La attribute found in a) indicates **Attach Ready**, and **Attach Fail**, otherwise.
 - 4) Construct a VLAN Context Status sub-TLV (6.4.4.4) with each VlanContextId and PathStatus pair contained in each the LA attribute found in a), and append it to new LA attribute.
- c) Issue at RESI an ATTACH_STREAM.indication (6.5.2.3.2) with its parameters set to the corresponding values contained in the new LA attribute.

6.6.8.12 handleLaDeReg()

This procedure is invoked when the RAP Endpoint receives an ATTR_DEREG.indication (6.6.9.4) at RISI that indicates a Listener Attach deregistration. It performs the following actions:

- a) Remove the reservation associated with the LA attribute, if any, and then remove the Listener Attach registration in the LA attribute.
- b) If there is no remaining Listener Attach registration associated with the StreamId, issue at RESI a DETACH_STREAM.indication (6.5.2.3.4) with the StreamId.
- c) Otherwise, call **handleLaMerge()** for the StreamId.

6.6.9 RAP MAD Internal Service Interface (RISI)

Check wording

The RISI interface provides a set of primitives (Table 6-12) for interaction of RAP MAD and the bridge internal propagation functionality..

Table 6-10—RAP MAD Internal Service Interface primitives

primitive name	reference
DECLARE_ATTR.request()	6.6.9.1
WITHDRAW_ATTR.request()	6.6.9.2
ATTR_REG.indication()	6.6.9.3
ATTR_DEREG.indication()	6.6.9.4
DEC_OPER_STATUS.indication()	6.6.9.5

6.6.9.1 DECLARE_ATTR.request()

This primitive is used by RAP MAD to make a declaration for an attribute.

6.6.9.2 WITHDRAW_ATTR.request()

This primitive is used invoked by RAP MAD to withdraw an existing declaration for an attribute.

6.6.9.3 ATTR_REG.indication()

This primitive is used by RISI to inform RAP MAD about registration of an attribute.

6.6.9.4 ATTR_DEREG.indication()

This primitive is used by RISI to inform RAP MAD about deregistration of an attribute.

6.6.9.5 DEC_OPER_STATUS.indication()

This primitive is used by RISI to notify RISI of whether its attribute declaration function is operational or not.

1 6.7 MAP Extensions

2 6.7.1 Talker Announce propagation

3 A set of rules used in Talker Announce propagation are defined in the following subclauses. These rules
4 refer to specific Forwarding Process functions ([Q]:8.6) in VLAN Bridges, which can be configured by use
5 of relevant protocols or management entities specified in [Q].

6 6.7.1.1 VLAN Context

7 Each Talker Announcement indicates one or more VLAN Contexts for use in Talker Announce propagation
8 by Bridges. A VLAN Context in a Bridge is identified by a VID, termed VLAN Context ID, and
9 corresponds to a VLAN topology formed by the set of Ports that includes each Bridge Port for which the
10 following are true:

- 11 1) The Port is in the Forwarding state and part of the active topology that supports that VID.
- 12 2) The Port is a member of the member set ([Q]:8.8.10) for that VID.

13 Depending on the number of VLAN Contexts used, there are two types of the Talker Announcement, as
14 follows:

- 15 a) **Single-Context Talker Announcement:** The Talker Announcement propagated in a single VLAN
16 Context and used in resource allocation for transmission of a stream along a single path from the
17 Talker to each Listener.
- 18 b) **Multi-Context Talker Announcement:** The Talker Announcement propagated in more than one
19 VLAN Context and used in resource allocation for transmission of a stream with seamless
20 redundancy along multiple paths from the Talker to each Listener.

21 One or more VLAN Contexts contained in a Talker Announce registration are used by a Bridge to determine
22 a set of potential egress Ports in propagating a Talker Announce registration on an ingress Port. From a
23 network-wide perspective, a specific VLAN Context used by a Talker Announcement constrains the extent
24 of the Talker Announcement to a corresponding VLAN topology of the network.

25 NOTE—An appropriate VLAN configuration in the network for the set of VLAN Contexts used in resource allocation is
26 required to control the reach of Talker Announcement. As one way to configure VLAN membership in Bridges for
27 VLAN Contexts, each Listener desiring to receive a given Talker Announcement can use MVRP ([Q]:11.2) to issue an
28 MVRP VLAN membership declaration for each VLAN Context ID to be used by that Talker Announcement. In such
29 cases, the knowledge of the VLAN Context(s) concerned needs to be made available in advance to the Listener, e.g., by
30 use of a specific higher layer protocol.

31 6.7.1.2 Stream DA Pruning

32 In addition to VLAN Context, Stream DA Pruning applies an additional constraint to Talker Announce
33 propagation by reference to the status of MAC Address Registrations Entries ([Q]:8.8.4) in the FDBs of the
34 Bridges. Stream DA Pruning can be enabled or disabled on a per-Bridge Port basis.

35 When Stream DA Pruning is enabled on a Port, a Talker Announce registration is not propagated to that
36 Port, if the DestinationMacAddress (6.4.3.5.1) value contained in the Talker Announce attribute is not found
37 in the MAC Address Registration Entries for that Port.

38 NOTE—To propagate a Talker Announcement along a path containing one or more Bridges with Stream DA Pruning
39 enabled on the egress port, a registration of the Stream Destination MAC Address used by the Talker Announcement on
40 each such Port is required. As one way to register MAC addresses, a Listener on that path can use MMRP ([Q]:10) to
41 issue an MMRP declaration for that MAC address. In such cases, the knowledge of the MAC address concerned needs to
42 be made available in advance to the Listener, e.g., by use of a specific higher layer protocol.

6.7.1.3 Ingress Blocking

Ingress Blocking prohibits propagation of a Talker Announce registration on an ingress Port of a Bridge to any other egress Ports, if all of the following conditions are met:

- a) Ingress Filtering ([Q]:8.6.2) is enabled on the Port.
- b) The Port is not in the member set (8.8.10) associated with the VID (6.4.3.5.3) of the Talker Announce registration.

NOTE—In addition to the need for VLAN Context on egress Ports, there is also a need for an appropriate VLAN configuration on each potential Talker Announce registration Port in the network based on the required reach of Talker Announcements. For example, in case that it is not desired to block a Talker Announcement on an ingress Port of any Bridge with Ingress Filtering enabled, the Talker can use MVRP ([Q]:11.2) to issue an MVRP VLAN membership declaration for the VID concerned. B.4 shows another example of VLAN configuration in a case where ingress blocking a Talker Announcement on specific Bridge Ports is intended.

6.7.1.4 Talker Announcement across RA class domains

For the purpose of providing ??? RAP allows Talker Announcement propagation across boundaries of RA class domains (6.3.2.4), but fails the Talker Announcement once it is propagated to a Bridge located outside the RA class domain from which it originates, with the RAP Failure Code *CrossingDomainBoundary* defined in Table 6-6.

6.7.2 Talker Announce status

The Talker Announce status of a Talker Announce declaration made on a given Port is associated with an ordinary stream in the case of the Single-Context Talker Announcement, or a Compound or a Member Stream in the case of the Multi-Context Talker Announcement, to be transmitted on that Port, indicating whether the stream could be reserved from the Talker to that Port along a single path or one or more paths, respectively, in the VLAN Context(s) indicated in the Talker Announce declaration, as follows:

- a) **Announce Success:** The stream can be reserved on the Port, as it has not encounter any problems on a single path from the Talker to that Port, if the stream is an ordinary stream, or on at least one path from the Talker to that Port, if the stream is a Compound Stream or Member Stream. The Announce Success status is represented by the absence of a Failure Information sub-TLV (6.4.3.10) in the Value field of the Talker Announce attribute TLV (6.4.3).
- b) **Announce Fail:** The stream can not be reserved on the Port, as it has encounter problems on each path from the Talker to that Port. The Announce Fail status is represented by the presence of a Failure Information sub-TLV (6.4.3.10) in the Value field Talker Announce attribute TLV (6.4.3).

6.7.3 Listener Attachment

Listener Attachment refers to the process of signaling the Listener Attach attribute (6.4.4) initially declared by one or more Listeners desiring to receive a stream across a Bridge network to the stream's Talker. Each Listener Attachment is identified by a StreamId (6.4.4.1) and VID (6.4.3.5.3) as contained in the Listener Attach attribute.

The purposes of Listener Attachment are as follows:

- Indication of the intention of one or more Listeners of receiving a stream.
- Triggering of resource allocations (or deallocation) on each Bridge along potential stream path(s).
- Signaling of reservation status hop-by-hop in an upstream direction from each participating Listener back to the Talker.

1 The following terminology relating to Listener Attachment is used:

- 2 a) **Listener Attach declaration:** A declaration of the Listener Attach attribute on a Port.
- 3 b) **Listener Attach registration:** A registration of the Listener Attach attribute on a Port.
- 4 c) **Listener Attach deregistration:** The removal of a Listener Attach registration on a Port.
- 5 d) **Listener Attach propagation:** Propagation of a Listener Attach registration from a Bridge Port to
- 6 one or more other Bridge Ports.
- 7 e) **Listener Attach merge:** Merge of more multiple Listener Attach registrations propagated to a
- 8 Bridge Port to result in a joint Listener Attach declaration on that Port.
- 9 f) **Associated Talker Announce declaration:** A Talker Announce declaration with which a given
- 10 Listener Attach registration is associated is termed the associated Talker Announce declaration of
- 11 that Listener Attach registration.
- 12 g) **Originating Listener Attach registration:** A Listener Attach registration from which a given
- 13 Listener Attach declaration results is termed an originating Listener Attach registration of that
- 14 Listener Attach declaration. A Listener Attach declaration can have more than one originating
- 15 Listener Attach registration in the case of Listener Attach merge.
- 16 h) **Associated Talker Announce registration:** A Talker Announce registration with which a given
- 17 Listener Attach declaration is associated is termed the associated Talker Announce registration of
- 18 that Listener Attach declaration.

19 6.7.4 Redundancy

20 6.7.4.1 Redundancy Context

21 Resource allocation for seamless redundancy operates in a network in which one or more instances of
 22 redundant trees are established and maintained by other protocols such as ISIS-PCR ([Q]:45). Each Multi-
 23 Context Talker Announcement for a Compound Stream operates within a particular instance of redundant
 24 trees. The group of VLAN topologies associated with an instance of redundant trees comprising two or more
 25 trees forms the set of correlated VLAN Contexts, termed “the Redundancy Context”. A Redundancy
 26 Context is identified by the numerically smallest VLAN Context ID in the set, termed “the Redundancy
 27 Context ID”. Multiple Redundancy Contexts, each with distinct set of VLAN Context IDs, can exist in a
 28 network due to the existence of multiple instances of redundant trees.

29 NOTE—For example, there can be one instance of redundant trees for each Bridge located at the edge of the network
 30 and rooted at that Bridge, with the intention that each Compound Stream using the redundant trees rooted at a Bridge
 31 where it enters the network be established for transmission along the multiple paths that are disjoint from each other.

32 The per-Bridge parameter redundancyContext (6.7.6.2), configurable as a managed object defined in Table
 33 7-8 and containing the configuration of each existing Redundancy Context in the network, provides the
 34 information needed by operations such as Talker Announce merge as described in 6.3.5.3. Proper operation
 35 of resource allocation for seamless redundancy requires a consistent Redundancy Context configuration
 36 among all the Bridges involved in resource allocation for seamless redundancy. An inconsistent Redundancy
 37 Context configuration, e.g. resulting from misconfiguration, can cause incorrect operation. To facilitate
 38 inconsistency detection, each Bridge advertises its redundancyContext value in the RA Advertisement
 39 (6.3.3) and maintains the per-Port array portRedundancyContext in 6.6.4.8. To prevent incorrect operation, a
 40 Multi-Context Talker Announcement is failed once it is registered on a Bridge Port whose
 41 portRedundancyContext table entry has a consistency value of FALSE, and reports the RAP Failure Code
 42 InconsistentRedundancyContext defined in Table 6-6.

43 6.7.4.2 Talker Announce merge

44 Talker Announce merge can only occur in the Multi-Context Talker Announcement in resource allocation
 45 for seamless redundancy (6.3.8) and is associated with the FRER operation stream merging in FRER-
 46 capable Bridges or Listeners, as described in 6.3.8.4.

Table 6-11—FRER functions and operations in FRER-capable stations

		FRER-capable stations		
		FRER-capable RAP Talker	FRER-capable RAP Bridge	FRER-capable RAP Listener
FRER operations	redundancy tagging (6.3.8.2)	sequence generation function ([CB]:7.4.1), Stream encode/decode function ([CB]:7.6), Redundancy tag ([CB]:7.8)		(none)
	stream splitting (6.3.8.3)	Stream splitting function ([CB]:7.7)	(Multicast) ^a	
		Active Destination MAC and VLAN Stream identification function ([CB]:6.6) ^b		
	stream merging (6.3.8.4)	(none)	Active Destination MAC and VLAN Stream identification function ([CB]:6.6) ^b , sequence recovery function ([CB]:7.5),	
	redundancy untagging (6.3.8.2)		Stream encode/decode function ([CB]:7.6), Redundancy tag ([CB]:7.8)	

^a The stream splitting operation in a Bridge relies on the multicast ability of the Forwarding Process to replicate streams, without the need to use Stream splitting function.

^b An end station can, but is not required to explicitly use this function; it is an implementation choice.

1 Talker Announce merge is performed in a FRER-capable Bridge or Listener among a set of Multi-Context
2 Talker Announce registrations that are propagated to the same Port and contain the same StreamId (6.4.3.1)
3 value, to result in a joint Multi-Context Talker Announce declaration on that Port for that StreamId and
4 containing a merged set of VLAN Contexts comprising each one used in the propagation of those Talker
5 Announce registrations.

6 6.7.4.3 Path status

7 A Multi-Context Listener Attachment also signals the Path status on a per VLAN Context basis. The Path
8 status of a Listener Attach declaration on a Port for a given VLAN Context is associated with a single path
9 within that VLAN Context from the Port to each Listener whose participation in the Listener Attachment has
10 been perceived on that Port, and indicates one of the following:

- 11 a) **Faultless Path:** Resource allocation has succeeded without encountering any problems on the path.
- 12 b) **Faulty Path:** Resource allocation has encountered one or more problems on the path and failed
13 wholly or partially on the path.
- 14 c) **Unresponsive Path:** No Listener Attachment with respect to the path has been received.

15 NOTE—In the Multi-Context Listener Attachment, resource allocation can succeed partially on a path as a result of the
16 fact that the path goes through one or more places of stream merging in the network, as described in 6.3.8.4.

17 6.7.5 RAP MAP state machine

18 The events that control RAP MAP extensions are specified in the state machine diagram in Figure 6-20. The
19 state changes are triggered by the bridge component.

1

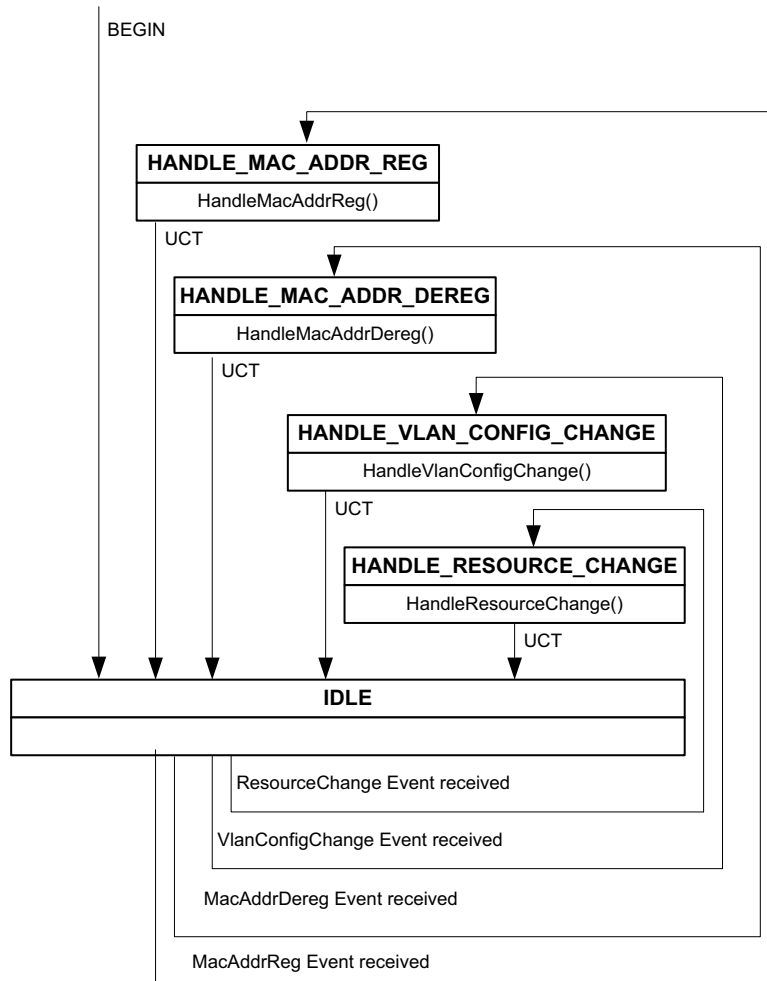


Figure 6-20—RAP MAP state machine

2 6.7.6 RAP MAP Variables

3 6.7.6.1 portPairRaClass

4 The portPairRaClass variable is an array in which each element is associated with an RA class supported by
5 the Bridge and a reception Port transmission Port pair, and consists of the following member variables:

- 6 a) **receptionPortRef**: The portRef (6.7.7.6.1) value of the associated reception Port.
- 7 b) **transmissionPortRef**: The portRef (6.7.7.6.1) value of the associated transmission Port.
- 8 c) **raClassId**: The RA class ID (6.3.2.1) of the associated local RA class.
- 9 d) **maxHopLatency**: An unsigned integer, indicating a latency value, in nanoseconds, that specifies a
10 latency constraint as described in 6.3.5.1. The latency is measured from a point located in an
11 upstream station connected via a LAN to the reception Port, to a point located in the transmission

1 Port, while the exact measurement points are specific to and defined by the RA class template being
2 used by the RA class.

3 The portPairRaClass variable is associated with a key <receptionPortRef, transmissionPortRef, raClassId>.

4 **6.7.6.2 redundancyContext**

5 The redundancyContext variable is an array in which each element represents a Redundancy Context
6 supported by the Bridge and consists of the following member variables:

- 7 a) **id**: The Redundancy Context ID of the Redundancy Context, as defined in 6.3.8.2.
- 8 b) **vlanContextList**: The set of VLAN Context IDs associated with the Redundancy Context.

9 The redundancyContext variable is associated with a key <id>.

10 **6.7.6.3 frerCapable**

11 A Boolean value, indicating whether the Bridge is a FRER-capable RAP Bridge (TRUE) or not (FALSE).

12 **6.7.6.4 maxProcessingDelay**

13 An unsigned integer, indicating the maximum delay, in nanoseconds, that a frame can experience during the
14 forwarding process of the Bridge until it is placed into an outbound queue.

15 **6.7.6.5 minProcessingDelay**

16 An unsigned integer, indicating the minimum delay, in nanoseconds, that a frame can experience during the
17 forwarding process of the Bridge until it is placed into an outbound queue.

18 **6.7.7 RAP MAP Procedures**

19 **6.7.7.1 propagateTa()**

20 This procedures is triggered by RAP MAD if it receives a TA Registration event.

21 Determine the egress ports from the context from the TA.

22 Call processTa() (6.7.7.9) for each egress port and trigger HandleTADecEgress() (6.6.5.1) at the
23 corresponding MAD Instance.

24 **Add some wording about the interface to the MAP functionality**

25 **6.7.7.2 HandleTaDeregMAP()**

26 This procedures is triggered by RAP MAD if it receives a TA Deregistration event.

27 For each TA Declaration that is associated with the corresponding TA Registration: If the TA Declaration is
28 based on a TA merge operation for redundancy (origTaRegList is empty):

- 29 a) Remove the TA Registration from the origTaRegList (a List which contains all TAREgs which are
30 the reasons for declaring this TA)
- 31 b) if origTAList is empty:
 - 32 1) Trigger HandleTADeregRemoveTaDecEgress() (6.6.5.3) at the corresponding MAD Instance
- 33 c) otherwise

1 2) Trigger HandleTADeregUpdateTaDecEgress() (6.6.5.2) at the corresponding MAD Instance

2 Add some wording about the interface to the MAP functionality

3 **6.7.7.3 propagateLa()**

4 This procedures is triggered by RAP MAD if it receives a LA Registration event.

5 If there is a corresponding TA for this LA:

6 a) Call processLa() (6.7.7.13)

7 b) Call propagateLaReg() (6.7.7.10)

8 c) for each egress port:

9 1) Trigger HandleLaDecEgress() (6.6.5.4) at the corresponding MAD Instance

10 d) Call processTaDec() (6.7.7.14)

11 Add some wording about the interface to the MAP functionality

12 **6.7.7.4 HandleLaDeregMAP()**

13 This procedures is triggered by RAP MAD if it receives a LA Deregistration event.

14 If this LA is part of an active reservation:

15 a) set LA ingressStatus to NOT_PROPAGATED

16 For each upstream propagation port of the LA:

17 a) Trigger HandleLaDeregEgress() (6.6.5.5) at the corresponding MAD Instance.

18 Add some wording about the interface to the MAP functionality

19 **6.7.7.5 HandleMacAddrReg()**

20 This procedure is triggered by the RAP MAP state machine if it receives a MacAddrReg event.

21 Add Info what to do

22 **6.7.7.6 HandleMacAddrDereg()**

23 This procedure is triggered by the RAP MAP state machine if it receives a MacAddrDereg event.

24 If DA pruning is enabled then execute for each TA declaration:

25 a) if the MAC address is the Destination MAC Address of the TA declaration and there is an associated
26 LA registration:

27 1) deallocate the resources and update reserved bandwidth

28 2) for each existing associated LA Declaration trigger processLaDec() in MAP (6.7.7.11)

29 3) Trigger HandleTADeregMAP() (6.7.7.2)

30 Add some wording about the interface to the MAP functionality

31 **6.7.7.7 HandleVlanConfigChange()**

32 This procedure is triggered by the RAP MAP state machine if it receives a VlanConfigChange event.

1 Add Info what to do

2 **6.7.7.8 HandleResourceChange()**

3 This procedure is triggered by the RAP MAP state machine if it receives a ResourceChange event.

4 Add Info what to do

5 **6.7.7.9 processTa()**

6 This procedure propagates a Talker Announce registration to a Port, by associating it with a new or existing
7 Talker Announce declaration in taDec, as follows:

8 If a TA Declaration (TADec) is already present on the port to which the TA Registration (TAReg) is to be
9 propagated, return from the procedure without doing anything.

10 If the stream is not to be merged at this bridge, set the VLAN Identifier (VID) to the numerically lowest VID
11 among all known VLAN contexts associated with this stream.

12 If the stream is to be merged, set the VID to the numerically lowest VID among all VLAN contexts within
13 the associated redundancy context.

14 **6.7.7.10 propagateLaReg()**

15 This procedure propagates a given Listener Attach registration, by associating it with one or more new or
16 existing Listener Attach declaration elements in laDec, as follows:

17 The propagateLAReg procedure is responsible for the propagation of a Listener Advertisement (LA) in
18 accordance with If a corresponding Talker Advertisement (TA) exists on any port, the LA shall be
19 propagated to that port.

20 For streams classified as multi-context, propagate the LA to all ports within the associated redundancy
21 context where a TA Registration (TAReg) for the stream is known.

22 Note: Propagation is constrained to occur only within a defined map context or redundancy context. This
23 constraint may be generalized as a prerequisite for MAP-based operations.

24 If no Listener Advertisement Declaration instance exists on the egress port, a new LAdec instance shall be
25 instantiated.

26 If a LA from a different source port already exists on the egress port, and a new LA is being propagated to
27 the same port, the two LAs shall be merged in accordance with the merging policy defined for LA
28 propagation. the defined listener attachment rules.

29 **6.7.7.11 processLaDec()**

30 This procedure processes a Listener Attach declaration, as follows:

31 Complete and validate the following instruction text here

1 For each LA Registration for this LA Declaration, if the ingressStatus is NOT_PROPAGATED:

2 a) delete the LA Registration

3 If there are no failed Listeners and no ready Listeners:

4 a) WITHDRAW_ATTR for this LA Declaration

5 b) delete the LA Declaration

6 otherwise:

7 ...

8 6.7.7.12 cancelTaRegPropagation(pTaReg)

9 This procedure stops propagation of a Talker Announce registration referenced by a taReg element
10 (pTaReg), as follows:

11 add instruction text here

12 6.7.7.13 processLa(pLaReg)

13 This procedure processes a Listener Attach registration referenced by a laReg element (pLaReg), as follows:

14 Call handleLaReg() (6.6.8.10).

15 add instruction text here

16 6.7.7.14 processTaDec()

17 This procedure defines the processing of a Talker Announce Declaration (TADec).

18 If the Talker Announce declaration is not subject to Talker Announce merge (6.3.5.3):

- 19 a) If the ingress status of the TADec is already set to TA_RECV_FAIL, no further action shall be taken.
- 20 b) If the ingress status is TA_INGRESS_FAIL, the corresponding error code shall be set in the TADec.
- 21 c) If the ingress status is TA_INGRESS_SUCCESS, the accumulated latency and the network traffic
22 specification (tspec) shall be set by applying the setAccuLatencies() and setNetworkTSpec()
23 functions to the TADec.
- 24 d) The RedundancyControl.RTagStatus field shall be set according to the rTagging value specified in
25 the associated TA Registration (TAREg).
- 26 e) If the TADec represents a multi-context talker, the VLAN Identifier (VID) shall be set to the
27 numerically lowest VID among all VLAN contexts in which the stream is to be propagated.
- 28 f) For multi-context talkers, the VLANContextInformation field of the TADec shall be restricted to
29 include only those VLAN contexts that are to be propagated at the current bridge.

30 else:

31 The following is the former function processTaDecMerge() and has to be simplified

- 32 a) Set each of the following fields in pTaDec.attr to the value of the corresponding field contained in
33 one of the taReg element listed in pTaDec.origTaRegList: StreamId (6.4.3.1), StreamRank (6.4.3.2),
34 DestinationMacAddress (6.4.3.5.1) and Priority (6.4.3.5.2).
- 35 b) Set pTaDec.attr.VID to the VID value used in the key of pTaDec in taDec, as determined when
36 creating pTaDec by the processTa() procedure (6.6.5.2).

- 1 c) Set the Redundancy Control sub-TLV (6.4.3.9) in pTaDec.attr.RedundancyControl as follows:
 - 2 1) tRedundancyContextId = **getRedundancyContextId**(pTaDec.attr.VID);
 - 3 2) tMergedVlanContextList = **getMergedVlanContextList**(tRedundancyContextId;
4 pTaDec.portRef
 - 5 3) If **isRedundancyUntagging**(pTaDec) == TRUE, set pTaDec.RedundancyControl.RTagStatus
6 to FALSE and set pTaDec.rUntagging to TRUE; otherwise, set pTaDec.rUntagging to FALSE.
 - 7 4) For each element tTaReg in pTaDec.origTaRegList, copy each VLAN Context Information
8 sub-TLV (6.4.3.9.2) in tTaReg.attr whose VlanContextId value is listed in
9 tMergedVlanContextList to pTaDec.RedundancyControl. If a copied VLAN Context
10 Information sub-TLV does not contain a Failure Information sub-TLV (6.4.3.10) and the
11 tTaReg from which the sub-TLV is copied indicates in ingressStatus TA_INGRESS_FAIL,
12 construct a Failure Information sub-TLV using the RAP Failure Code value contained in
13 tTaReg.ingressFailureCode and the SystemId value of this Bridge, and then append it to the
14 copied sub-TLV.
 - 15 5) For each VLAN Context listed in tMergedVlanContextList but not indicated in any VLAN
16 Context Information sub-TLV copied to pTaDec.RedundancyControl in b).4) above, construct
17 a VLAN Context Information sub-TLV for that VLAN Context with a Failure Information sub-
18 TLV that contains the value of the RAP Failure Code *MissingTalkerAnnounce* defined in Table
19 6-6 and the SystemId value of this Bridge, and then append the constructed VLAN Context
20 Information sub-TLV to pTaDec.attr.RedundancyControl.
- 21 d) If pTaDec.origTaRegList contains no element tTaReg whose ingressStatus indicates
22 TA_INGRESS_SUCCESS, perform the following:
 - 23 1) Set tEgressFailureCode to the numerically smallest RAP Failure Code contained in the VLAN
24 Context Information sub-TLV(s) in pTaDec.attr.RedundancyControl.
 - 25 2) Call **failTaDec**(pTaDec, tEgressFailureCode).
- 26 e) Otherwise, i.e., pTaDec.origTaRegList contains at least one element tTaReg whose ingressStatus
27 indicates TA_INGRESS_SUCCESS, perform the following:
 - 28 1) Call **setAccuLatencies**(pTaDec) and then **setNetworkTSpec**(pTaDec).
 - 29 2) tEgressFailureCode = **checkReservationConstraints**(pTaDec). If tEgressFailureCode != 0,
30 call **failTaDec**(pTaDec, tEgressFailureCode).

31

32 6.7.8 Events

33 6.7.8.1 Listener Attach propagation

34 A Listener Attachment follows the path(s) traversed by its associated Talker Announcement in reversed
35 direction.

36 A Listener Attach registration on a Bridge Port with an associated Talker Announce declaration is
37 propagated to each other Port that contains an originating Talker Announce registration of the associated
38 Talker Announce declaration. A Listener Attach registration on a Bridge Port without an associated Talker
39 Announce declaration is not propagated by the Bridge.

40 6.7.8.2 VLAN_CONFIG_CHANGE.indication()

41 This primitive is invoked to inform the RAP Propagator state machine about changes in the VLAN
42 configuration. Such a change can result from a spanning tree ([Q]:7.3) or VLAN topology ([Q]:7.4)
43 reconfiguration.

1 **6.7.8.3 MAC_ADDR_REG.indication(iPortRef, iMacAddr)**

2 This primitive is invoked to inform the RAP Propagator state machine about registration of a MAC address
3 (iMacAddr) on a Port (iPortRef).

4 **6.7.8.4 MAC_ADDR_DEREG.indication(iPortRef, iMacAddr)**

5 This primitive is invoked to inform the RAP Propagator state machine about deregistration of a MAC
6 address (iMacAddr) on a Port (iPortRef).

7 **6.8 RAP Generic Functions**

8 Check which functions are required .

9

7. Resource Allocation Protocol (RAP) management

Check if this matches the new structure.

This subclause defines the following managed objects that allows administrative configuration of RAP:

- a) RAP Participant Table (7.1)
- b) RAP Propagator Bridge Table (7.2)
- c) RAP Propagator Port Table (7.3)
- d) RA Class Bridge Table (7.4)
- e) RA Class Port Table (7.5)
- f) RA Class Port Pair Table (7.6)
- g) Priority Regeneration Override Table (7.7)
- h) Redundancy Context Table (7.8)
- i) Talker Announce Registration Port Table (7.9)
- j) Talker Announce Declaration Port Table (7.10)
- k) Listener Attach Registration Port Table (7.11)
- l) Listener Attach Declaration Port Table (7.12)

7.1 RAP MAD Table

There is one RAP MAD Table per RAP MAD instance (6.7). The table contains a set of parameters that supports the operation of RAP MAD for an end station or Bridge Port, as detailed in Table 7-1.

Table 7-1—RAP MAD Table

Name	Data type	Operations supported ^a	References
participantEnabled	Boolean	RW	6.7.7.6.2

^a R = read only access; RW = Read/Write access.

7.2 RAP MAP Bridge Table

There is one RAP MAP Bridge Table. The table contains a set of parameters that supports the operation of RAP for a Bridge as a whole, as detailed in Table 7-2.

Table 7-2—RAP Propagator Bridge Table

Name	Data type	Operations supported ^a	References
frerCapable	Boolean	R	6.7.6.3
maxProcessingDelay	unsigned integer	R	6.7.6.4
minProcessingDelay	unsigned integer	R	6.7.6.5

^a R = read only access; RW = Read/Write access.

1 7.3 RAP MAP Port Table

2 There is one RAP MAP Port Table per Bridge Port. The table contains a set of parameters that supports the
3 operation of the RAP Propagator for a Bridge Port, as detailed in Table 7-3

4 Validate if parts are in MAD.

Table 7-3—RAP MAP Port Table

Name	Data type	Operations supported ^a	References
streamDaPruningEnabled	Boolean	RW	6.6.4.7 b)
maxInterferingFrameSize	unsigned integer	R	6.6.4.7 e)
maxPropagationDelay	unsigned integer	R	6.6.4.7 g)
minPropagationDelay	unsigned integer	R	6.6.4.7 h)

^a R = read only access; RW = Read/Write access.

5 7.4 RA Class Bridge Table

6 There is one RA Class Bridge Table per RAP Propagator (6.7). Each table row contains a set of parameters
7 that defines an RA class (6.3.2) for a Bridge as a whole, as detailed in Table 7-4.

Table 7-4—RA Class Bridge Table row elements

Name	Data type	Operations supported ^a	References
raClassId	unsigned integer [0..255]	RW	6.3.2.1
raClassPriority	unsigned integer [0..7]	RW	6.3.2.2
rtid	integer	RW	6.3.2.3
templatedDefinedData	octet string	RW	6.4.2.2.6

^a R = read only access; RW = Read/Write access.

1 7.5 RA Class Port Table

2 There is one RA Class Port Table per Bridge Port for a RAP Propagator (6.7). Each table row contains a set
3 of parameters associated with an RA class configured in the RA Class Bridge Table (7.4) and a Bridge Port,
4 as detailed in Table 7-5.

Table 7-5—RA Class Port Table row elements

Name	Data type	Operations supported ^a	References
raClassId	unsigned integer [0..255]	RW	6.6.4.9 b)
domainBoundaryPort	Boolean	R	6.6.4.9 c)
maxBandwidth	unsigned integer	RW	6.6.4.9 d)
allocatedBandwidth	unsigned integer	R	6.6.4.9 e)
maxLastHopLatency	unsigned integer	RW	6.6.4.9 f)

^a R = read only access; RW = Read/Write access.

5 7.6 RA Class Port Pair Table

6 There is one RA Class Port Pair Table per reception transmission Port pair of a Bridge for a RAP Propagator
7 (6.7). Each table row is associated with an RA class present in the RA Class Bridge Table (7.4) and contains
8 a set of parameters for a pair of a reception Port and a transmission Port, as detailed in Table 7-6.

Table 7-6—RA Class Port Pair Table row elements

Name	Data type	Operations supported ^a	References
receptionPort	Port Number	RW	6.6.4.10 a)
transmissionPort	Port Number	RW	6.6.4.10 b)
raClassId	unsigned integer [0..255]	RW	6.6.4.10 c)
maxHopLatency	unsigned integer	RW	6.6.4.10 d)

^a R = read only access; RW = Read/Write access.

1 7.7 RA Class Domain Boundary Priority Regeneration Table

2 There is one RA Class Domain Boundary Priority Regeneration Table per Bridge Port for a RAP Propagator
3 (6.7). The values contained in this table are used in the detection of RA class domains to control the Bridge's
4 priority regeneration function ([Q]:6.9.4) on an RA class domain boundary port, as described in 6.3.2.4.
5 Each table row contains a set of parameters, as detailed in Table 7-7.

Table 7-7—RA Class Domain Boundary Priority Regeneration Table row elements

Name	Data type	Operations supported ^a	References
receivedPriority	unsigned integer [0..7]	RW	7.7.1
regeneratedPriority	unsigned integer [0..7]	RW	7.7.2

^a R = read only access; RW = Read/Write access.

6 7.7.1 receivedPriority

7 This parameter contains a received priority that is associated with an RA class configured in the RA Class
8 Bridge Table (7.4).

9 7.7.2 regeneratedPriority

10 This parameter contains the regenerated priority for the given received priority contained in 7.7.1.

11 7.8 Redundancy Context Table

12 There is one RAP Redundancy Context Table per RAP Propagator (6.7). Each table row contains a set of
13 parameters that defines a Redundancy Context (6.3.8.2) supported by the Bridge, as detailed in Table 7-8.

14 7.9 Talker Announce Registration Port Table

15 There is one Talker Announce Registration Port Table per Bridge Port. Each table row in the table associated
16 with a Port represents a Talker Announce registration [6.3.4 item b)] on that Port, and contains a set of

Table 7-8—Redundancy Context Table row elements

Name	Data type	Operations supported ^a	References
redundancyContextId	unsigned integer [1..4094]	RW	6.3.8.2, 6.7.6.2
vlanContextList	a list of unsigned integer [1..4094]	RW	6.3.8.2, 6.7.6.2

^a R = read only access; RW = Read/Write access.

¹ parameters as detailed in Table 7-9. Rows in the table can be created, updated and deleted dynamically as

² Talker Announce registrations are created, updated and deleted on the Port.

Table 7-9—Talker Announce Registration Port Table row elements

Name	Data type	Operations supported ^a	References
streamId	octet string(size(8))	R	6.4.3.1
streamRank	unsigned integer [0..1]	R	6.4.3.2
accumulatedMaxLatency	unsigned integer	R	6.4.3.3
accumulatedMinLatency	unsigned integer	R	6.4.3.4
destinationMacAddress	MAC address	R	6.4.3.5.1
priority	unsigned integer [0..7]	R	6.4.3.5.2
vid	unsigned integer [1..4094]	R	6.4.3.5.3
talkerTokenBucketTSpec	Boolean	R	7.9.1
talkerTokenBucketTSpecValue	octet string	R	7.9.2
talkerMsrpTSpec	Boolean	R	7.9.3
talkerMsrpTSpecValue	octet string	R	7.9.4
networkTSpec	Boolean	R	7.9.5
networkTSpecValue	octet string	R	7.9.6
redundancyControl	Boolean	R	7.9.7
redundancyControlValue	octet string	R	7.9.8
failureInfo	Boolean	R	7.9.9
failureInfoValue	octet string	R	7.9.10
orgDefinedInfo	Boolean	R	7.9.11
orgDefinedInfoValue	octet string	R	7.9.12
invalid	Boolean	R	6.6.4.1 c)
ingressBlocked	Boolean	R	6.6.4.1 d)

^a R = read only access; RW = Read/Write access.

1 **7.9.1 talkerTokenBucketTSpec**

2 This parameter contains a Boolean value that indicates whether a Token Bucket TSpec sub-TLV (6.4.3.6) is
3 contained in the TalkerTSpec field (6.4.3.6) of the Talker Announce attribute (TRUE) or not (FALSE).

4 **7.9.2 talkerTokenBucketTSpecValue**

5 When talkerTokenBucketTSpec (7.9.1) is TRUE, this parameter contains an octet string copied from the
6 whole of the Value field of a Token Bucket TSpec sub-TLV (6.4.3.6) contained in the TalkerTSpec field
7 (6.4.3.6) of the Talker Announce attribute.

8 When talkerTokenBucketTSpec is FALSE, this parameter contains the empty octet string.

9 **7.9.3 talkerMsrpTSpec**

10 This parameter contains a Boolean value that indicates whether an Interval-based TSpec sub-TLV (6.4.3.7)
11 is contained in the TalkerTSpec field (6.4.3.6) of the Talker Announce attribute.

12 **7.9.4 talkerMsrpTSpecValue**

13 When talkerMsrpTSpec (7.9.3) is TRUE, this parameter contains an octet string copied from the whole of
14 the Value field of an Interval-based TSpec sub-TLV (6.4.3.7) contained in the TalkerTSpec field (6.4.3.6) of
15 the Talker Announce attribute.

16 When talkerMsrpTSpec is FALSE, this parameter contains the empty octet string.

17 **7.9.5 networkTSpec**

18 This parameter contains a Boolean value that indicates whether a Token Bucket TSpec sub-TLV (6.4.3.6) is
19 contained in the NetworkTSpec field (6.4.3.9) of the Talker Announce attribute (TRUE) or not (FALSE).

20 **7.9.6 networkTSpecValue**

21 When networkTSpec (7.9.5) is TRUE, this parameter returns an octet string copied from the whole of the
22 Value field of a Token Bucket TSpec sub-TLV (6.4.3.6) contained in the NetworkTSpec field (6.4.3.9) of the
23 Talker Announce attribute.

24 When networkTSpec is FALSE, this parameter contains the empty octet string.

25 **7.9.7 redundancyControl**

26 This parameter contains a Boolean value that indicates whether a Redundancy Control sub-TLV (6.4.3.9) is
27 contained in the Talker Announce attribute (TRUE) or not (FALSE).

28 **7.9.8 redundancyControlValue**

29 When redundancyControl (7.9.7) is TRUE, this parameter contains an octet string copied from the whole of
30 the Value field of a Redundancy Control sub-TLV (6.4.3.9) contained in the Talker Announce attribute.

31 When redundancyControl is FALSE, this parameter contains the empty octet string.

1 7.9.9 failureInfo

2 This parameter contains a Boolean value that indicates whether a Failure Information sub-TLV (6.4.3.9.2) is
3 contained in the Talker Announce attribute (TRUE) or not (FALSE).

4 7.9.10 failureInfoValue

5 When failureInfo (7.9.9) is TRUE, this parameter contains an octet string copied from the whole of the
6 Value field of a Failure Information sub-TLV (6.4.3.9.2) contained in the Talker Announce attribute.

7 When failureInfo is FALSE, this parameter contains the empty octet string.

8 7.9.11 orgDefinedInfo

9 This parameter contains a Boolean value that indicates whether an Organizationally Defined sub-TLV
10 (6.4.3.10) is contained in the Talker Announce attribute (TRUE) or not (FALSE).

11 7.9.12 orgDefinedInfoValue

12 When orgDefinedInfo (7.9.11) is TRUE, this parameter contains an octet string copied from the whole of the
13 Value field of an Organizationally Defined sub-TLV (6.4.3.10) contained in the Talker Announce attribute.

14 When orgDefinedInfo is FALSE, this parameter contains the empty octet string.

15 7.10 Talker Announce Declaration Port Table

16 There is one Talker Announce Declaration Port Table per Bridge Port for a RAP Propagator (6.7). Each table
17 row in the Table associated with a Port represents a Talker Announce declaration [6.3.4 item a)] on that Port,
18 and contains each of the parameters as specified in Table 7-9 except the invalid and ingressBlocked
19 parameters. Rows in the table can be created, updated and deleted dynamically as Talker Announce
20 declarations are created, updated and deleted on the Port.

21 7.11 Listener Attach Registration Port Table

22 There is one Listener Attach Registration Port Table per Bridge Port for a RAP Propagator (6.7). Each table
23 row in the Table associated with a Port represents a Listener Attach registration [6.3.6 item b)] on that Port
24 and contains a set of parameters as detailed in Table 7-10. Rows in the table can be created, updated and
25 deleted dynamically as the Listener Attach registrations are created, updated and deleted on the Port.

26 7.11.1 vlanContextStatus

27 This parameter contains a Boolean value that indicates whether a VLAN Context Status sub-TLV (6.4.4.4) is
28 contained in the Listener Attach attribute (TRUE) or not (FALSE).

29 7.11.2 vlanContextStatusValue

30 When vlanContextStatus (7.11.1) is TRUE, this parameter contains an octet string copied from the whole of
31 the Value field of a VLAN Context Status sub-TLV (6.4.4.4) contained in the Listener Attach attribute.

Table 7-10—Listener Attach Registration Port Table row elements

Name	Data type	Operations supported ^a	References
streamId	octet string(size(8))	R	6.4.4.1
vid	unsigned integer [1..4094]	R	6.4.4.2
listenerAttachStatus	Enumerated	R	6.4.4.3
vlanContextStatus	Boolean	R	7.11.1
vlanContextStatusValue	octet string	R	7.11.2
reservationAge	unsigned integer	R	6.6.4.3 e)
ingressStatus	unsigned integer	R	6.6.4.3 f)

^a R = read only access; RW = Read/Write access.

1 7.12 Listener Attach Declaration Port Table

2 There is one Listener Attach Declaration Port Table per Bridge Port for a RAP Propagator (6.7). Each table
3 row in the Table associated with a Port represents a Listener Attach declaration [6.3.6 item a)] on that Port
4 and contains each of the parameters as specified in Table 7-10 except the reservationAge and ingressStatus
5 parameters. Rows in the table can be created, updated and deleted dynamically as Listener Attach
6 declarations are created, updated and deleted on the Port.

1 8. YANG data model for RAP

2 Check if this matches the new structure.

3 YANG (IETF RFC 7950) is a data modeling language used to model configuration data and state data for
4 remote network management protocols. YANG-based remote network management protocols include the
5 Network Configuration Protocol (NETCONF) (IETF RFC 6242) and RESTCONF (IETF RFC 8040). Each
6 remote network management protocol uses a specific encoding on the wire, such as XML or JSON. A
7 YANG module specifies the organization and rules for the management data, and a mapping from YANG to
8 the specific encoding enables the data to be understood correctly by both client and server.

9 This clause specifies the YANG data model for RAP.

10 This clause:

- 11 a) Introduces the organization of the data models (8.1)
- 12 b) Shows the relationship with other standards (8.2)
- 13 c) Provides an overview of the hierarchy of the data models using a representation similar to the
14 Unified Modeling Language (UML) (8.3)
- 15 d) Summarizes the structure of the YANG data model (8.4)
- 16 e) Reviews security considerations (8.5)
- 17 f) Provides a schema tree as an overview of the YANG module (8.6)
- 18 g) Specifies the YANG module (8.7)

19 8.1 The YANG framework

20 The YANG framework applies hierarchy in the following areas:

- 21 a) The Uniform Resource Name (URN), as specified in IEEE Std 802d-2017.
- 22 b) The YANG objects form a hierarchy of configuration and operational data structures that define the
23 YANG model.

24 Clause 7 specifies the information model for management of this standard. The data model for a specific
25 management mechanism is derived from the information model. Since YANG-based protocols are an
26 example of a management mechanism, the YANG data model of this clause is derived from Clause 7.

27 8.2 Relationship to other YANG modules

28 The RAP YANG model augments the VLAN Bridge component model as defined in [Q].

8.3 RAP YANG model

This clause uses a UML-like representation to provide an overview of the hierarchy of the RAP YANG data model.

For all figures, data that is imported from outside this standard is shown in white, and data in augments of those YANG modules is shown in gray.

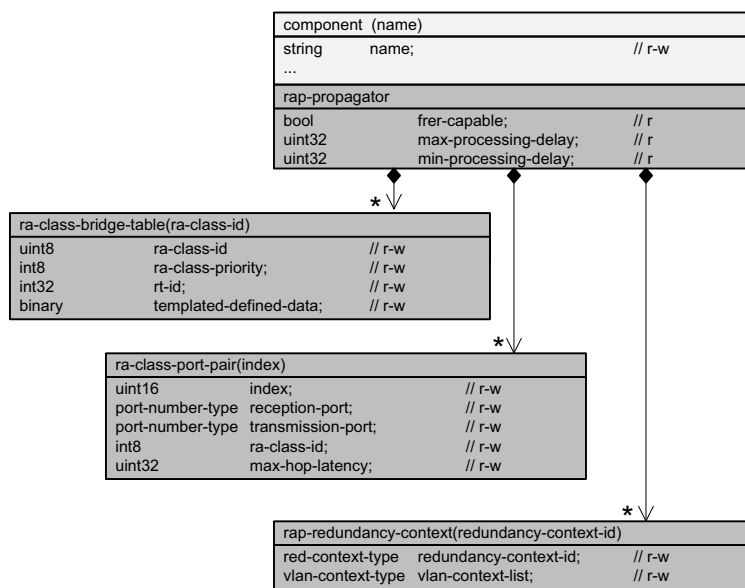


Figure 8-1—Bridge component augmentation

1

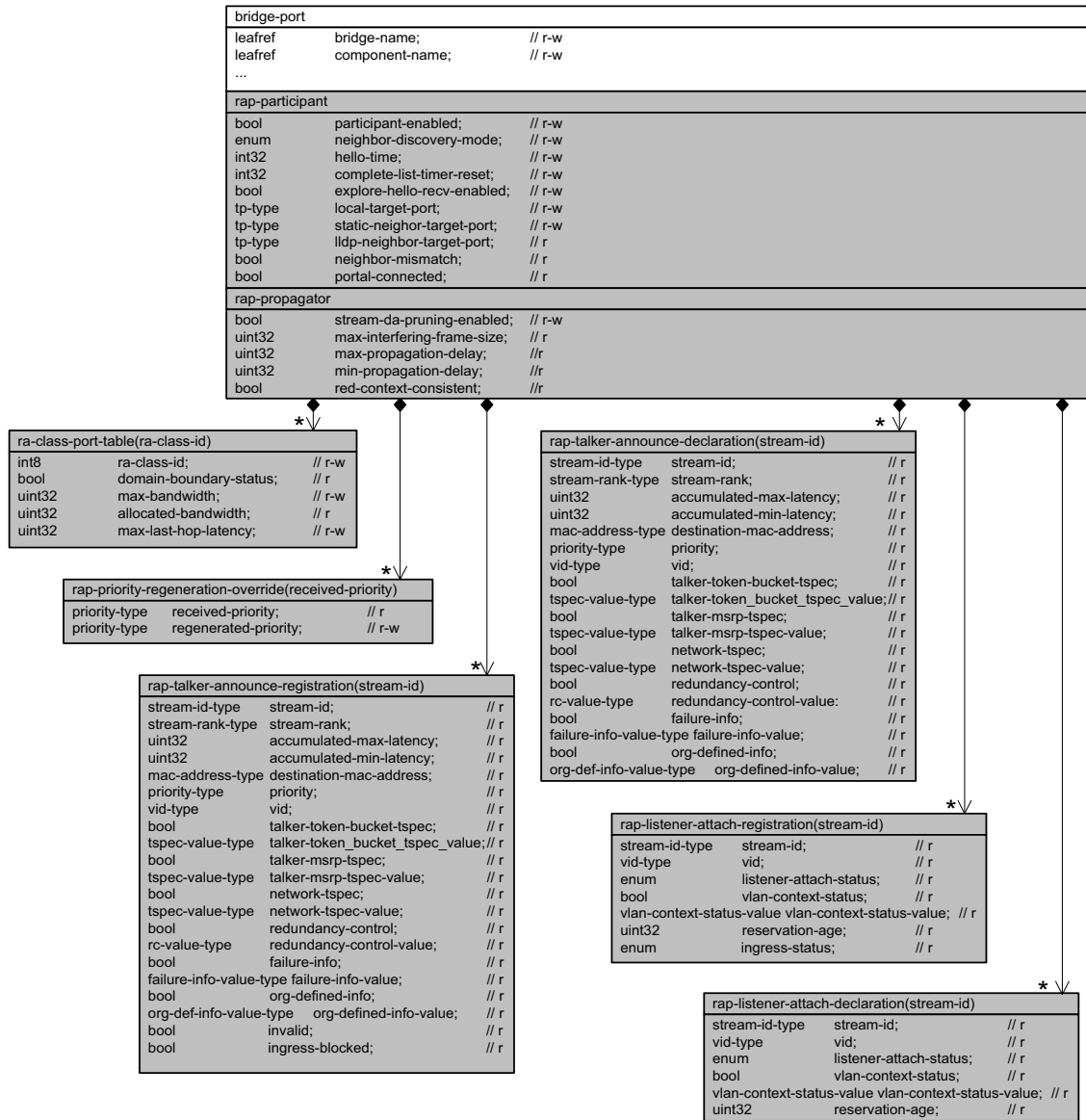


Figure 8-2—Bridge port augmentation

2 8.4 Structure of the YANG model

3 The RAP YANG model is provided as a single YANG module.

4 In the YANG module definitions, if any discrepancy between the “description” text and the corresponding
5 definition in any other part of this standard occurs, the definitions outside this clause take precedence.

6

1 8.5 Security Considerations

2 The YANG module specified in this clause is designed to be accessed via a network configuration protocol, such as NETCONF (IETF RFC 6242 [B13a]) and RESTCONF (IETF RFC 8040). NETCONF and RESTCONF protocols provide the means to secure communication between client and server, using secure transport layers such as Secure Shell (SSH) (IETF RFC 6242) and Transport Layer Security (TLS) (IETF RFC 7589).

7 It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those principals (users) that have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

12 8.6 YANG data scheme tree definitions

13 The schema tree in this clause is provided as an overview of the RAP YANG module. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

15 8.6.1 Tree diagram for ieee802-dot1dd-rap.yang

```

16 module: ieee802-dot1dd-rap
17
18   augment /dot1q:bridges/dot1q:bridge/dot1q:component:
19     +--rw rap-propagator
20       | +--ro frer-capable?          boolean
21       | +--ro max-processing-delay?  uint32
22       | +--ro min-processing-delay?  uint32
23     +--rw ra-class-bridge-table* [ra-class-id]
24       | +--rw ra-class-id            uint8
25       | +--rw ra-class-priority?     int8
26       | +--rw rt-id?                int32
27       | +--rw templated-defined-data? binary
28     +--rw ra-class-port-pair* [index]
29       | +--rw index                  uint16
30       | +--rw reception-port?        dot1qtypes:port-number-type
31       | +--rw transmission-port?     dot1qtypes:port-number-type
32       | +--rw ra-class-id?           int8
33       | +--rw max-hop-latency?       uint32
34     +--rw rap-redundancy-context* [redundancy-context-id]
35       +--rw redundancy-context-id    uint16
36       +--rw vlan-context-list* [vlan-context]
37         +--rw vlan-context           uint16
38   augment /if:interfaces/if:interface/dot1q:bridge-port:
39     +--rw rap-participant
40       | +--rw participant-enabled?    boolean
41       | +--rw neighbor-discovery-mode? enumeration
42       | +--rw hello-time?            int32
43       | +--rw complete-list-timer-reset? int32
44       | +--rw explore-hello-recv-enabled? boolean
45       | +--rw local-target-port
46       | | +--rw chassis-id?          ieeetypes:chassis-id-type
47       | | +--rw port-id?             ieeetypes:port-id-type
48       | | +--rw ecp-capable?         boolean
49       | | +--rw tcp-capable?         boolean
50       | | +--rw tcp-port?            uint16
51       | | +--rw addr-ip-v4?          inet:ipv4-address
52       | | +--rw addr-ip-v6?          inet:ipv6-address
53       | +--rw static-neighbor-target-port
54       | | +--rw chassis-id?          ieeetypes:chassis-id-type
55       | | +--rw port-id?             ieeetypes:port-id-type
56       | | +--rw ecp-capable?         boolean
57       | | +--rw tcp-capable?         boolean
58       | | +--rw tcp-port?            uint16
59       | | +--rw addr-ip-v4?          inet:ipv4-address
60       | | +--rw addr-ip-v6?          inet:ipv6-address
61       | +--ro lldp-neighbor-target-port

```



```

1 | | +--ro chassis-id?      ieeetypes:chassis-id-type
2 | | +--ro port-id?       ieeetypes:port-id-type
3 | | +--ro ecp-capable?   boolean
4 | | +--ro tcp-capable?   boolean
5 | | +--ro tcp-port?      uint16
6 | | +--ro addr-ip-v4?    inet:ipv4-address
7 | | +--ro addr-ip-v6?    inet:ipv6-address
8 | | +--ro neighbor-mismatch? boolean
9 | | +--ro portal-connected? boolean
10 | +--rw rap-propagator
11 | | +--rw stream-da-pruning-enabled? boolean
12 | | +--ro max-interfering-frame-size? uint32
13 | | +--ro max-propagation-delay?    uint32
14 | | +--ro min-propagation-delay?    uint32
15 | | +--ro red-context-consistent?   boolean
16 | +--rw ra-class-port-table* [ra-class-id]
17 | | +--rw ra-class-id          uint8
18 | | +--ro domain-boundary-status? boolean
19 | | +--rw max-bandwidth?       uint32
20 | | +--ro allocated-bandwidth?  uint32
21 | | +--rw max-last-hop-latency? uint32
22 | +--rw rap-priority-regeneration-override* [received-priority]
23 | | +--rw received-priority    dot1qtypes:priority-type
24 | | +--rw regenerated-priority? dot1qtypes:priority-type
25 | +--ro rap-talker-announce-registration* [stream-id vid]
26 | | +--ro stream-id            tsn:stream-id-type
27 | | +--ro stream-rank?         uint8
28 | | +--ro accumulated-max-latency? uint32
29 | | +--ro accumulated-min-latency? uint32
30 | | +--ro destination-mac-address? ieeetypes:mac-address
31 | | +--ro priority?            dot1qtypes:priority-type
32 | | +--ro vid                  uint16
33 | | +--ro talker-token-bucket-tspec? boolean
34 | | +--ro talker-token_bucket_tspec_value? binary
35 | | +--ro talker-msrp-tspec?    boolean
36 | | +--ro talker-msrp-tspec-value? binary
37 | | +--ro network-tspec?        boolean
38 | | +--ro network-tspec-value?  binary
39 | | +--ro redundancy-control?   boolean
40 | | +--ro redundancy-control-value? binary
41 | | +--ro failure-info?         boolean
42 | | +--ro failure-info-value?   binary
43 | | +--ro org-defined-info?     boolean
44 | | +--ro org-defined-info-value? binary
45 | | +--ro invalid?              boolean
46 | | +--ro ingress-blocked?      boolean
47 | +--ro rap-talker-announce-declaration* [stream-id vid]
48 | | +--ro stream-id            tsn:stream-id-type
49 | | +--ro stream-rank?         uint8
50 | | +--ro accumulated-max-latency? uint32
51 | | +--ro accumulated-min-latency? uint32
52 | | +--ro destination-mac-address? ieeetypes:mac-address
53 | | +--ro priority?            dot1qtypes:priority-type
54 | | +--ro vid                  uint16
55 | | +--ro talker-token-bucket-tspec? boolean
56 | | +--ro talker-token_bucket_tspec_value? binary
57 | | +--ro talker-msrp-tspec?    boolean
58 | | +--ro talker-msrp-tspec-value? binary
59 | | +--ro network-tspec?        boolean
60 | | +--ro network-tspec-value?  binary
61 | | +--ro redundancy-control?   boolean
62 | | +--ro redundancy-control-value? binary
63 | | +--ro failure-info?         boolean
64 | | +--ro failure-info-value?   binary
65 | | +--ro org-defined-info?     boolean
66 | | +--ro org-defined-info-value? binary
67 | | +--ro invalid?              boolean
68 | | +--ro ingress-blocked?      boolean
69 | +--ro rap-listener-attach-registration* [stream-id vid]
70 | | +--ro stream-id            tsn:stream-id-type
71 | | +--ro vid                  uint16
72 | | +--ro listener-attach-status? enumeration

```

```

1 | +--ro vlan-context-status?    boolean
2 | +--ro vlan-context-status-value?  binary
3 | +--ro reservation-age?      uint32
4 | +--ro ingress-status?      enumeration
5 +--ro rap-listener-attach-declaration* [stream-id vid]
6   +--ro stream-id            tsn:stream-id-type
7   +--ro vid                  uint16
8   +--ro listener-attach-status?  enumeration
9   +--ro vlan-context-status?    boolean
10  +--ro vlan-context-status-value?  binary
11

```

12

13 8.7 YANG module ieee802-dot1dd-rap

```

14 module ieee802-dot1dd-rap {
15   yang-version 1.1;
16   namespace "urn:ieee:std:802.1DD:yang:ieee802-dot1dd-rap";
17   prefix rap;
18
19   import ietf-interfaces {
20     prefix if;
21   }
22   import ieee802-dot1q-types {
23     prefix dot1qtypes;
24   }
25   import ieee802-dot1q-bridge {
26     prefix dot1q;
27   }
28   import ieee802-dot1q-tsn-types {
29     prefix tsn;
30   }
31   import ieee802-types {
32     prefix ieeeetypes;
33   }
34   import ietf-inet-types {
35     prefix inet;
36   }
37
38   organization
39     "IEEE 802.1 Working Group";
40   contact
41     "WG-URL: http://www.ieee802.org/1/
42      WG-EMail: stds-802-1-l@ieee.org
43
44      Contact: IEEE 802.1 Working Group Chair
45      Postal: C/O IEEE 802.1 Working Group
46              IEEE Standards Association
47              445 Hoes Lane
48              Piscataway, NJ 08854
49              USA
50
51      E-mail: stds-802-1-chairs@ieee.org";
52   description
53     "This module provides management of 802.1Q Bridge components that
54      support the Resource Allocation Protocol (RAP).
55
56      Copyright (C) IEEE (202x).
57
58      This version of this YANG module is part of IEEE Std 802.1DD;
59      see the standard itself for full legal notices.";
60
61   revision 2024-11-13 {
62     description
63       "Published as part of IEEE Std 802.1Qdd-202x.
64
65       The following reference statement identifies each referenced
66       IEEE Standard as updated by applicable amendments.";
67     reference

```

```

1  "IEEE Std 802.1DD Resource Allocation Protocol:
2  IEEE Std 802.1Q Bridges and Bridged Networks:
3  IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4  IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
5  IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
6  IEEE Std 802.1Qdy-2024.";
7  }
8
9  grouping target-port {
10   leaf chassis-id {
11     type ieee-types:chassis-id-type;
12     description
13       "Chassis component associated with the local system.";
14     reference
15       "8.5.2.3 of IEEE Std 802.1AB-2016";
16   }
17   leaf port-id {
18     type ieee-types:port-id-type;
19     description
20       "Port component associated with a given port in the local
21       system.";
22     reference
23       "8.5.3.3 of IEEE Std 802.1AB-2016";
24   }
25   leaf ecg-capable {
26     type boolean;
27     description
28       "A Boolean value indicating whether the target
29       port supports the LRP-DT ECG mechanism (TRUE) or not (FALSE).";
30   }
31   leaf tcp-capable {
32     type boolean;
33     description
34       "A Boolean value indicating whether that the target
35       port supports the LRP-DT TCP mechanism (TRUE) or not (FALSE).";
36   }
37   leaf tcp-port {
38     type uint16;
39     description
40       "A 2-byte TCP port number for the target port.";
41     reference
42       "C.2.2.6.1 of IEEE Std 802.1CS-2020";
43   }
44   leaf addr-ip-v4 {
45     type inet:ipv4-address;
46     description
47       "A 4-byte IPv4 address for the target port or NULL.";
48     reference
49       "Item 1) in C.2.2.6.2 of IEEE Std 802.1CS-2020";
50   }
51   leaf addr-ip-v6 {
52     type inet:ipv6-address;
53     description
54       "A 16-byte IPv6 address for the target port or NULL.";
55     reference
56       "Item 2) in C.2.2.6.2 of IEEE Std 802.1CS-2020";
57   }
58 }
59
60 grouping rap-talker-announce {
61   leaf stream-id {
62     type tsnn:stream-id-type;
63     description
64       "An 8-octet field encoding the StreamID element as specified in
65       46.2.3.1 of IEEE Std 802.1Q.";
66     reference
67       "6.5.3.1 of IEEE Std 802.1DD";
68   }
69   leaf stream-rank {
70     type uint8;
71     description
72       "A 1-octet field encoding a Rank value as specified in

```

```

1      46.2.3.2.1 of IEEE Std 802.1Q.";
2      reference
3      "6.5.3.2 of IEEE Std 802.1DD";
4  }
5  leaf accumulated-max-latency {
6      type uint32;
7      description
8      "A 4-octet field encoding the AccumulatedLatency element as
9      specified in 46.2.5.2 of IEEE Std 802.1Q.";
10     reference
11     "6.5.3.3 of IEEE Std 802.1DD";
12 }
13 leaf accumulated-min-latency {
14     type uint32;
15     description
16     "A 4-octet unsigned integer, indicating the minimum latency, in
17     nanoseconds, that a single frame of the stream can encounter when
18     transmitted from the Talker along a given path to the Port
19     declaring this attribute.";
20     reference
21     "6.5.3.4 of IEEE Std 802.1DD";
22 }
23 leaf destination-mac-address {
24     type ieee-types:mac-address;
25     description
26     "A 6-octet destination MAC address of the data frames of the
27     stream.";
28     reference
29     "6.5.3.5.1 of IEEE Std 802.1DD";
30 }
31 leaf priority {
32     type dot1q-types:priority-type;
33     description
34     "A 3-bit unsigned integer, indicating the priority to be encoded in
35     the PCP field of the VLAN tag, with which the data frames of the
36     stream are tagged. This priority value is used by each receiving
37     Bridge to associate the stream to a local RA class of the same
38     priority.";
39     reference
40     "6.5.3.5.2 of IEEE Std 802.1DD";
41 }
42 leaf vid {
43     type uint16 {
44         range "1..4094";
45     }
46     description
47     "A 12-bit VID. The semantics of this field is dependent on the type
48     of a Talker Announcement in which the Talker Announce attribute is
49     used, as follows:
50     a) In the case of a Single-Context Talker
51     Announcement, this field indicates a VID to be encoded in the VLAN
52     tag with which the data frames of the stream are tagged, and also
53     a single VLAN Context used by the Talker Announce attribute.
54     b) In the case of a Multi-Context Talker Announcement, this field
55     is set to the numerically smallest VlanContextId value contained
56     in a VLAN Context Information sub-TLV.";
57     reference
58     "6.5.3.5.3 of IEEE Std 802.1DD";
59 }
60 leaf talker-token-bucket-tspec {
61     type boolean;
62     description
63     "This parameter returns a Boolean value that indicates whether a
64     Token Bucket TSpec sub-TLV is contained in the TalkerTSpec field
65     of the Talker Announce attribute (TRUE) or not (FALSE).";
66     reference
67     "7.9.1 of IEEE Std 802.1DD";
68 }
69 leaf talker-token_bucket_tspec_value {
70     type binary;
71     description
72     "When talkerTokenBucketTSpec is TRUE, this parameter returns an

```

```

1      octet string that is copied from the whole of the Value field of a
2      Token Bucket TSpec sub-TLV contained in the TalkerTSpec field of
3      the Talker Announce attribute.";
4      reference
5      "7.9.2 of IEEE Std 802.1DD";
6  }
7  leaf talker-msrp-tspec {
8      type boolean;
9      description
10     "This parameter returns a Boolean value that indicates whether a
11     MSRP TSpec sub-TLV is contained in the TalkerTSpec field of the
12     Talker Announce attribute.";
13     reference
14     "7.9.3 of IEEE Std 802.1DD";
15 }
16 leaf talker-msrp-tspec-value {
17     type binary;
18     description
19     "When talkerMsrpTSpec is TRUE, this parameter returns an octet
20     string that is copied from the whole of the Value field of a MSRP
21     TSpec sub-TLV contained in the TalkerTSpec field of the Talker
22     Announce attribute.";
23     reference
24     "7.9.4 of IEEE Std 802.1DD";
25 }
26 leaf network-tspec {
27     type boolean;
28     description
29     "This parameter returns a Boolean value that indicates whether a
30     Token Bucket TSpec sub-TLV is contained in the NetworkTSpec field
31     of the Talker Announce attribute (TRUE) or not (FALSE).";
32     reference
33     "7.9.5 of IEEE Std 802.1DD";
34 }
35 leaf network-tspec-value {
36     type binary;
37     description
38     "When networkTSpec is TRUE, this parameter returns an octet string
39     that is copied from the whole of the Value field of a Token Bucket
40     TSpec sub-TLV contained in the NetworkTSpec field of the Talker
41     Announce attribute.";
42     reference
43     "7.9.6 of IEEE Std 802.1DD";
44 }
45 leaf redundancy-control {
46     type boolean;
47     description
48     "This parameter returns a Boolean value that indicates whether a
49     Redundancy Control sub-TLV is contained in the Talker Announce
50     attribute (TRUE) or not (FALSE).";
51     reference
52     "7.9.7 of IEEE Std 802.1DD";
53 }
54 leaf redundancy-control-value {
55     type binary;
56     description
57     "When redundancyControl is TRUE, this parameter returns an octet
58     string that is copied from the whole of the Value field of a
59     Redundancy Control sub-TLV contained in the Talker Announce
60     attribute.";
61     reference
62     "7.9.8 of IEEE Std 802.1DD";
63 }
64 leaf failure-info {
65     type boolean;
66     description
67     "This parameter returns a Boolean value that indicates whether a
68     Failure Information sub-TLV is contained in the Talker Announce
69     attribute (TRUE) or not (FALSE).";
70     reference
71     "7.9.9 of IEEE Std 802.1DD";
72 }

```

```

1  leaf failure-info-value {
2      type binary;
3      description
4          "When failureInfo is TRUE, this parameter returns an octet string
5              that is copied from the whole of the Value field of a Failure
6              Information sub-TLV contained in the Talker Announce attribute.";
7      reference
8          "7.9.10 of IEEE Std 802.1DD";
9  }
10 leaf org-defined-info {
11     type boolean;
12     description
13         "This parameter returns a Boolean value that indicates whether an
14             Organizationally Defined sub-TLV is contained in the Talker
15             Announce attribute (TRUE) or not (FALSE).";
16     reference
17         "7.9.11 of IEEE Std 802.1DD";
18 }
19 leaf org-defined-info-value {
20     type binary;
21     description
22         "When orgDefinedInfo is TRUE, this parameter returns an octet
23             string that is copied from the whole of the Value field of an
24             Organizationally Defined sub-TLV contained in the Talker Announce
25             attribute.";
26     reference
27         "7.9.12 of IEEE Std 802.1DD";
28 }
29 leaf invalid {
30     type boolean;
31     description
32         "A Boolean indicating whether the Talker Announce registration is
33             invalid (TRUE) or not (FALSE), as determined by the isTaInvalid()
34             procedure (6.8.5.3)";
35     reference
36         "item c) in 6.8.4.1 of IEEE Std 802.1DD";
37 }
38 leaf ingress-blocked {
39     type boolean;
40     description
41         "A Boolean indicating whether the Talker Announce registration is
42             ingress blocked (TRUE) or not (FALSE), as determined by the
43             isTaIngressBlocked() procedure (6.8.5.4)";
44     reference
45         "item d) in 6.8.4.1 of IEEE Std 802.1DD";
46 }
47 }
48
49 grouping rap-listener-attach {
50     leaf stream-id {
51         type tsn:stream-id-type;
52         description
53             "An 8-octet field encoding the StreamID element as specified in
54                 46.2.3.1 of IEEE Std 802.1Q.";
55         reference
56             "6.5.4.1 of IEEE Std 802.1D";
57     }
58     leaf vid {
59         type uint16 {
60             range "1..4094";
61         }
62         description
63             "A 12-bit integer, indicating a VID to be encoded in the VLAN tag
64                 with which the data frames of the stream are tagged.";
65         reference
66             "6.5.4.2 of IEEE Std 802.1DD";
67     }
68     leaf listener-attach-status {
69         type enumeration {
70             enum attach-ready {
71                 value 1;
72             }

```

```

1      enum attach-fail {
2          value 2;
3      }
4      enum attach-partial-fail {
5          value 3;
6      }
7  }
8  description
9      "An enumeration indicating the listener attach status";
10 reference
11     "6.5.4.3 of IEEE Std 802.1DD";
12 }
13 leaf vlan-context-status {
14     type boolean;
15     description
16         "This parameter returns a Boolean value that indicates whether a
17         VLAN Context Status sub-TLV is contained in the Listener Attach
18         attribute (TRUE) or not (FALSE).";
19     reference
20         "7.11.1 of IEEE Std 802.1DD";
21 }
22 leaf vlan-context-status-value {
23     type binary;
24     description
25         "When vlanContextStatus is TRUE, this parameter returns an octet
26         string that is copied from the whole of the Value field of a VLAN
27         Context Status sub-TLV contained in the Listener Attach
28         attribute.";
29     reference
30         "6.5.4.4 of IEEE Std 802.1DD";
31 }
32 }
33
34 augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
35     description
36         "Augment Bridge with RAP configuration.";
37     reference
38         "6 of IEEE Std 802.1DD.";
39     container rap-propagator {
40         leaf frer-capable {
41             type boolean;
42             config false;
43             description
44                 "A Boolean value, indicating whether the Bridge is a
45                 FRER-capable Bridge (TRUE) or not (FALSE).";
46             reference
47                 "6.8.4.11 of IEEE Std 802.1DD.";
48         }
49         leaf max-processing-delay {
50             type uint32;
51             config false;
52             description
53                 "An unsigned integer, indicating the maximum delay, in
54                 nanoseconds, that a frame can experience during the
55                 forwarding process of the Bridge until it is placed into an
56                 outbound queue.";
57             reference
58                 "6.8.4.12 of IEEE Std 802.1DD.";
59         }
60         leaf min-processing-delay {
61             type uint32;
62             config false;
63             description
64                 "An unsigned integer, indicating the minimum delay, in
65                 nanoseconds, that a frame can experience during the
66                 forwarding process of the Bridge until it is placed into an
67                 outbound queue.";
68             reference
69                 "6.8.4.13 of IEEE Std 802.1DD.";
70         }
71     }
72     list ra-class-bridge-table {

```

```

1  key "ra-class-id";
2  description
3    "Each table row contains a set of parameters that defines a
4    single RA class";
5  leaf ra-class-id {
6    type uint8 {
7      range "0 .. 255";
8    }
9    description
10     "The RA class ID is an integer in the range of 0 through 255
11     that identifies an RA class supported by an end station or
12     Bridge.";
13    reference
14     "6.3.2.1 of IEEE Std 802.1DD.";
15  }
16  leaf ra-class-priority {
17    type int8 {
18      range "0 .. 7";
19    }
20    description
21     "Each RA class supported by an end station or Bridge is
22     associated with a unique priority value in the range 0
23     through 7, termed RA class priority. The RA class priority
24     indicates the received priority of the frames which are to
25     be mapped to the traffic class(es) with which that RA class
26     is associated.";
27    reference
28     "6.3.2.2 of IEEE Std 802.1DD.";
29  }
30  leaf rt-id {
31    type int32;
32    description
33     "An RA class template is identified by an RA Class Template
34     Identifier (RTID), which encodes a 3-octet OUI or CID value
35     identifying the organization that defines that template,
36     followed by a 1-octet index allocated by that
37     organization.";
38    reference
39     "6.3.2.3 of IEEE Std 802.1DD.";
40  }
41  leaf templated-defined-data {
42    type binary;
43    description
44     "The encoding of this field and the semantics associated
45     with its values if any, is specific to the RA class
46     template identified by the value contained in rt-id";
47    reference
48     "6.5.2.2.6 of IEEE Std 802.1DD.";
49  }
50 }
51 list ra-class-port-pair {
52   key "index";
53   description
54     "Each table row corresponds to an RA class configured in the
55     RA Class Bridge Table and contains a set of parameters
56     associated with a reception Port and a transmission Port";
57   leaf index {
58     type uint16;
59     description
60       "The index for the list";
61   }
62   leaf reception-port {
63     type dot1qtypes:port-number-type;
64     description
65       "The port number of the associated reception Port.";
66     reference
67       "Item a) in 6.8.4.9 of IEEE Std 802.1DD.";
68   }
69   leaf transmission-port {
70     type dot1qtypes:port-number-type;
71     description
72       "The port number of the associated transmission Port.";

```



```

1      reference
2      "Item b) in 6.8.4.9 of IEEE Std 802.1DD.";
3  }
4  leaf ra-class-id {
5      type int8;
6      description
7      "The RA class ID of the associated local RA class. ";
8      reference
9      "Item c) in 6.8.4.9 of IEEE Std 802.1DD.";
10 }
11 leaf max-hop-latency {
12     type uint32;
13     description
14     "An unsigned integer, containing the administratively
15     configured maximum latency value, in nanoseconds, that is
16     used as an upper bound latency in the latency constraint
17     imposed on each stream reserved in the RA class, received
18     on the reception Port, and transmitted on the transmission
19     Port. The latency is measured from a point located in an
20     upstream station connected via a LAN to the reception Port,
21     to a point located in the transmission Port, where the
22     exact measurement points are specific to and defined by the
23     RA class template being used by the RA class.";
24     reference
25     "Item d) in 6.8.4.9 of IEEE Std 802.1DD.";
26 }
27 }
28 list rap-redundancy-context {
29     key "redundancy-context-id";
30     description
31     "Each table row contains a set of parameters associated with a
32     Redundancy Context supported by the Bridge.";
33     leaf redundancy-context-id {
34         type uint16 {
35             range "1..4096";
36         }
37         description
38         "The Redundancy Context ID of the Redundancy Context";
39         reference
40         "Item a) in 6.8.4.10 of IEEE Std 802.1DD.";
41     }
42     list vlan-context-list {
43         key "vlan-context";
44         description
45         "The list of VLAN Context IDs associated with the Redundancy
46         Context.";
47         reference
48         "Item b) in 6.8.4.10 of IEEE Std 802.1DD.";
49         leaf vlan-context {
50             type uint16 {
51                 range "1..4096";
52             }
53         }
54     }
55 }
56 }
57
58 augment "/if:interfaces/if:interface/dot1q:bridge-port" {
59     description
60     "Augment Bridge Port with RAP configuration";
61     reference
62     "6 of IEEE Std 802.1DD.";
63     container rap-participant {
64         leaf participant-enabled {
65             type boolean;
66             description
67             "A Boolean variable indicating whether the operation of the
68             RAP Participant state machine is administratively enabled
69             (TRUE) or not (FALSE).";
70             reference
71             "6.7.4.2 of IEEE Std 802.1DD.";
72         }

```

```

1 leaf neighbor-discovery-mode {
2   type enumeration {
3     enum lldp-discovery {
4       value 1;
5     }
6     enum static-configuration {
7       value 2;
8     }
9     enum exploratory-hello {
10      value 3;
11    }
12  }
13  description
14    "An administratively assigned value, indicating the operation
15    mode in which neighbor discovery is performed on the local
16    target Port, and taking one of the following enumerated
17    values:
18    1) LLDP_DISCOVERY: The information about a neighbor target
19    port to be passed to the underlying LRP is obtained through
20    the exchange of LRP Discovery TLVs (Annex C of IEEE Std
21    802.1CS-2020) by use of LLDP, and contained in
22    lldpNeighborTargetPort (51.7.4.7.4).
23    2) STATIC_CONFIGURATION: The information about a neighbor
24    target port to be passed to the underlying LRP is statically
25    configured by the management and contained in
26    staticNeighborTargetPort (51.7.4.7.3).
27    3) EXPLORATORY_HELLO: No neighbor target port information
28    needs to be passed to the underlying LRP.";
29  reference
30    "6.7.4.3 of IEEE Std 802.1DD.";
31 }
32 leaf hello-time {
33   type int32;
34   default "30";
35   description
36     "An administratively assigned integer value, in the range 30
37     through 65535, for the Hello Time parameter in a Local Target
38     Port request issued by the RAP Participant state machine to
39     the underlying LRP.";
40   reference
41     "6.7.4.4 of IEEE Std 802.1DD.";
42 }
43 leaf complete-list-timer-reset {
44   type int32;
45   description
46     "An administratively assigned integer value, in the range x
47     through y, for the cplCompleteListTimerReset parameter in a
48     Local Target Port request issued by the RAP Participant state
49     machine to the underlying LRP. The default value is z.";
50   reference
51     "6.7.4.5 of IEEE Std 802.1DD.";
52 }
53 leaf explore-hello-recv-enabled {
54   type boolean;
55   description
56     "An administratively assigned Boolean value for the
57     imPplExploreRecv parameter in a Neighbor Target Port request
58     issued by the RAP Participant state machine to the underlying
59     LRP.";
60   reference
61     "6.7.4.6 of IEEE Std 802.1DD.";
62 }
63 container local-target-port {
64   uses target-port;
65   description
66     "Contains the administratively configured parameters of the
67     local target port.";
68   reference
69     "6.7.4.7.2 of IEEE Std 802.1DD.";
70 }
71 container static-neighbor-target-port {
72   uses target-port;

```

```

1      description
2        "Contains the administratively configured parameters of a
3        neighbor target port to which the local target port is to be
4        connected.";
5      reference
6        "6.7.4.7.3 of IEEE Std 802.1DD.";
7    }
8    container lldp-neighbor-target-port {
9      uses target-port;
10     config false;
11     description
12       "Contains the configuration parameters of a neighbor target
13       port discovered by LLDP.";
14     reference
15       "6.7.4.7.4 of IEEE Std 802.1DD.";
16   }
17   leaf neighbor-mismatch {
18     type boolean;
19     config false;
20     description
21       "A Boolean variable, set TRUE when detecting a mismatch between
22       the local target port and the neighbor target port to be
23       connected.";
24     reference
25       "6.7.4.8 of IEEE Std 802.1DD.";
26   }
27   leaf portal-connected {
28     type boolean;
29     config false;
30     description
31       "A Boolean value indicating whether a Portal association for the
32       Portal as indicated in portal-id (51.7.4.11) has been
33       established by the underlying LRP (TRUE) or not (FALSE).";
34     reference
35       "6.7.4.12 of IEEE Std 802.1DD.";
36   }
37 }
38 container rap-propagator {
39   leaf stream-da-pruning-enabled {
40     type boolean;
41     description
42       "A Boolean indicating whether Stream DA Pruning (51.3.4.1.2)
43       is administratively enabled (TRUE) or disabled (FALSE) on the
44       Port.";
45     reference
46       "Item b) in 6.8.4.7 of IEEE Std 802.1DD.";
47   }
48   leaf max-interfering-frame-size {
49     type uint32;
50     config false;
51     description
52       "An unsigned integer, indicating the maximum frame size, in
53       bytes, including media-dependent overhead (12.4.2.2), that is
54       allowed to be transmitted through the Port. The value of this
55       parameter is determined by the operation of the underlying
56       MAC Service.";
57     reference
58       "Item e) in 6.8.4.7 of IEEE Std 802.1DD.";
59   }
60   leaf max-propagation-delay {
61     type uint32;
62     config false;
63     description
64       "An unsigned integer, indicating the maximum latency, in
65       nanoseconds, a frame can experience when transmitted from the
66       underlying physical medium on the Port to a reception port
67       connected via a LAN to the Port.";
68     reference
69       "Item g) in 6.8.4.7 of IEEE Std 802.1DD.";
70   }
71   leaf min-propagation-delay {
72     type uint32;

```

```

1      config false;
2      description
3          "An unsigned integer, indicating the minimum latency, in
4              nanoseconds, a frame can experience when transmitted from the
5              underlying physical medium on the Port to a reception port
6              connected via a LAN to the Port.";
7      reference
8          "Item h) in 6.8.4.7 of IEEE Std 802.1DD.";
9  }
10 leaf red-context-consistent {
11     type boolean;
12     config false;
13     description
14         "A Boolean indicating whether the Redundancy Context
15         configuration in this Bridge is consistent with that in a
16         neighboring station on the Port (TRUE) or not (FALSE). ";
17     reference
18         "Item i) in 6.8.4.7 of IEEE Std 802.1DD.";
19 }
20 }
21 list ra-class-port-table {
22     key "ra-class-id";
23     description
24         "Each table row contains a set of parameters that defines a
25         single RA class";
26     leaf ra-class-id {
27         type uint8 {
28             range "0 .. 255";
29         }
30         description
31             "The RA class ID of the associated local RA class.";
32         reference
33             "Item b) in 6.8.4.8 of IEEE Std 802.1DD.";
34     }
35     leaf domain-boundary-status {
36         type boolean;
37         config false;
38         description
39             "A Boolean indicating whether the Port is a domain boundary
40             port for the RA class (TRUE) or not (FALSE).";
41         reference
42             "Item c) in 6.8.4.8 of IEEE Std 802.1DD.";
43     }
44     leaf max-bandwidth {
45         type uint32;
46         description
47             "An unsigned integer, indicating the maximum amount of
48             bandwidth that can be allocated to the streams reserved in
49             the RA class on the Port. The bandwidth value is represented
50             as a percentage of the portTransmitRate value on that Port
51             and expressed as a fixed-point number scaled by a factor of
52             1,000,000; i.e., 100,000,000 (the maximum value) represents
53             100%.";
54         reference
55             "Item d) in 6.8.4.8 of IEEE Std 802.1DD.";
56     }
57     leaf allocated-bandwidth {
58         type uint32;
59         config false;
60         description
61             "An unsigned integer, indicating the amount of bandwidth that has
62             been allocated to the streams reserved in the RA class on the
63             Port. The bandwidth value is represented as a percentage of the
64             portTransmitRate value on that Port and expressed as a fixed-
65             point number scaled by a factor of 1,000,000; i.e., 100,000,000
66             (the maximum value) represents 100%.";
67         reference
68             "Item e) in 6.8.4.8 of IEEE Std 802.1DD.";
69     }
70     leaf max-last-hop-latency {
71         type uint32;
72         description

```

```

1      "The value to be contained in the MaxLastHopLatency field of an
2      RA Class Descriptor sub-TLV associated with the RA class in the
3      RA attribute declared on the Port.";
4      reference
5      "Item f) in 6.8.4.8 of IEEE Std 802.1DD.";
6  }
7  }
8  list rap-priority-regeneration-override {
9      key "received-priority";
10     description
11     "Each table row contains a set of parameters for a received
12     priority value that is associated with an RA class ";
13     leaf received-priority {
14         type dot1qtypes:priority-type;
15         description
16         "Received priority value.";
17         reference
18         "6.9.4 of IEEE Std 802.1Q";
19     }
20     leaf regenerated-priority {
21         type dot1qtypes:priority-type;
22         description
23         "Priority regeneration value.";
24         reference
25         "6.9.4 of IEEE Std 802.1Q";
26     }
27 }
28 list rap-talker-announce-registration {
29     key "stream-id vid";
30     config false;
31     description
32     "Each table row in the table associated with a Port corresponds
33     to a registration of the Talker Announce attribute on that Port,
34     and contains a set of parameters";
35     uses rap-talker-announce;
36 }
37 list rap-talker-announce-declaration {
38     key "stream-id vid";
39     config false;
40     description
41     "Each table row in the Table associated with a Port corresponds
42     to a declaration of the Talker Announce attribute on that Port,
43     and contains the same set of parameters as in the Talker Announce
44     Registration";
45     uses rap-talker-announce;
46 }
47 list rap-listener-attach-registration {
48     key "stream-id vid";
49     config false;
50     description
51     "Each table row in the Table associated with a Port corresponds to
52     a registration of the Listener Attach attribute on that Port and
53     contains a set of parameters";
54     uses rap-listener-attach;
55     leaf reservation-age {
56         type uint32;
57         description
58         "A 32-bit unsigned integer, indicating the time, in seconds,
59         since a reservation associated with the Listener Attach
60         registration was successfully made, and set to zero when the
61         reservation is removed.";
62         reference
63         "Item e) in 6.8.4.3 of IEEE Std 802.1DD";
64     }
65     leaf ingress-status {
66         type enumeration {
67             enum not-propagated {
68                 value 0;
69             }
70             enum attach-ready {
71                 value 1;
72             }
73         }
74     }
75 }

```

```
1      enum attach-fail {
2          value 2;
3      }
4      enum attach-partial-fail {
5          value 3;
6      }
7  }
8  description
9      "The ingress status of the Listener Attach registration.";
10     reference
11         "item f) in 6.8.4.3 of IEEE Std 802.1DD";
12 }
13 }
14 list rap-listener-attach-declaration {
15     key "stream-id vid";
16     config false;
17     description
18         "Each table row in the Table associated with a Port corresponds to
19         a declaration of the Listener Attach attribute on that Port and
20         contains the same set of parameters except the reservationAge
21         parameter as in the Listener Attach Registration";
22     uses rap-listener-attach;
23 }
24 }
25 }
26
```

27

1 Annex A

2 (normative)

3 PICS proforma—<RAP implementations>¹¹

4 A.1 Introduction

5 Check if this matches the new structure.

6 The supplier of a protocol implementation that is claimed to conform to this standard shall complete the
7 following Protocol Implementation Conformance Statement (PICS) proforma.

8 I have not resolved whether this should be a PICS or a PCS, some tweaking may be required throughout.
9 More than one PICS/PCS may be required if distinct devices/functions are being called out. This file, once set
10 up, should be fairly static, so the actual PICS tables are in the next file.

11 A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of
12 which capabilities and options of the protocol have been implemented. The PICS can have a number of uses,
13 including use

- 14 a) By the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard
15 through oversight.
- 16 b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication
17 of the capabilities of the implementation, stated relative to the common basis for understanding
18 provided by the standard PICS proforma.
- 19 c) By the user—or potential user—of the implementation, as a basis for initially checking the
20 possibility of interworking with another implementation (note that, while interworking can never be
21 guaranteed, failure to interwork can often be predicted from incompatible PICSs).
- 22 d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for
23 conformance of the implementation.

24 A.2 Abbreviations and special symbols

25 A.2.1 Status symbols

26	M	mandatory
27	O	optional
28	O.n	optional, but support of at least one of the group of options labeled by the same numeral n
29		is required
30	X	prohibited
31	pred:	conditional-item symbol, including predicate identification: see A.3.4
32	¬	logical negation, applied to a conditional item's predicate

¹¹ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

1 **A.2.2 General abbreviations**

2	N/A	not applicable
3	PICS	Protocol Implementation Conformance Statement

4 **A.3 Instructions for completing the PICS proforma**

5 **A.3.1 General structure of the PICS proforma**

6 The first part of the PICS proforma, implementation identification and protocol summary, is to be completed
7 as indicated with the information necessary to identify fully both the supplier and the implementation.

8 The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each
9 containing a number of individual items. Answers to the questionnaire items are to be provided in the
10 rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or
11 by entering a value or a set or range of values. (Note that there are some items where two or more choices
12 from a set of possible answers can apply; all relevant choices are to be marked.)

13 Each item is identified by an item reference in the first column. The second column contains the question to
14 be answered; the third column records the status of the item—whether support is mandatory, optional, or
15 conditional: see also A.3.4. The fourth column contains the reference or references to the material that
16 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

17 A supplier may also provide (or be required to provide) further information, categorized as either Additional
18 Information or Exception Information. When present, each kind of further information is to be provided in a
19 further subclause of items labeled Ai or Xi, respectively, for cross-referencing purposes, where i is any
20 unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format
21 and presentation.

22 A completed PICS proforma, including any Additional Information and Exception Information, is the
23 Protocol Implementation Conformation Statement for the implementation in question.

24 NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to
25 describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering
26 some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of
27 the information.

28 **A.3.2 Additional information**

29 Items of Additional Information allow a supplier to provide further information intended to assist the
30 interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS
31 can be considered complete without any such information. Examples might be an outline of the ways in
32 which a (single) implementation can be set up to operate in a variety of environments and configurations, or
33 information about aspects of the implementation that are outside the scope of this standard but that have a
34 bearing on the answers to some items.

35 References to items of Additional Information may be entered next to any answer in the questionnaire and
36 may be included in items of Exception Information.

37 **A.3.3 Exception information**

38 It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any
39 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer

1 will be found in the Support column for this item. Instead, the supplier shall write the missing answer into
2 the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the
3 appropriate rationale in the Exception item itself.

4 An implementation for which an Exception item is required in this way does not conform to this standard.

5 NOTE—A possible reason for the situation described previously is that a defect in this standard has been reported, a
6 correction for which is expected to change the requirement not met by the implementation.

7 **A.3.4 Conditional status**

8 **A.3.4.1 Conditional items**

9 The PICS proforma contains a number of conditional items. These are items for which both the applicability
10 of the item itself, and its status if it does apply—mandatory or optional—are dependent on whether certain
11 other items are supported.

12 Where a group of items is subject to the same condition for applicability, a separate preliminary question
13 about the condition appears at the head of the group, with an instruction to skip to a later point in the
14 questionnaire if the “Not Applicable” answer is selected. Otherwise, individual conditional items are
15 indicated by a conditional symbol in the Status column.

16 A conditional symbol is of the form “**pred:** S” where **pred** is a predicate as described in A.3.4.2 below, and
17 S is a status symbol, M or O.

18 If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated
19 by the status symbol following the predicate: The answer column is to be marked in the usual way. If the
20 value of the predicate is false, the “Not Applicable” (N/A) answer is to be marked.

21 **A.3.4.2 Predicates**

22 A predicate is one of the following:

- 23 a) An item-reference for an item in the PICS proforma: The value of the predicate is true if the item is
24 marked as supported and is false otherwise.
- 25 b) A predicate-name, for a predicate defined as a boolean expression constructed by combining item-
26 references using the boolean operator OR: The value of the predicate is true if one or more of the
27 items is marked as supported.
- 28 c) The logical negation symbol “¬” prefixed to an item-reference or predicate-name: The value of the
29 predicate is true if the value of the predicate formed by omitting the “¬” symbol is false, and vice
30 versa.

31 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for
32 grouped conditional items, is indicated by an asterisk in the Item column.

1 A.4 PICS proforma—<RAP implementations>

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, e.g., name(s) and version(s) of machines and/or operating system names	

NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.

NOTE 2—The terms “Name” and “Version” should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).

A.4.2 Protocol summary

Identification of protocol specification	IEEE Std 802.1DD, Resource Allocation Protocol			
Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS	Amd.	:	Corr.	:
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: the answer “Yes” means that the implementation is not conformant).	No []		Yes []	

Date of Statement	
-------------------	--

1 I don't want to waste time speculating as to the contents of these tables, how many there are etc. so what
2 follows is a fairly raw extract from 802.1Q, with just a couple of tables and layout examples. All will need
3 overwriting.

A.5 Major capabilities

Item	Feature	Status	References	Support
RRI	Is a RAP instance implemented as a RAP Relay Instance?	O	5.4	Yes [] No []
REI	Is a RAP instance implemented as a RAP Relay Instance?	O	5.5	Yes [] No []
RMGI	Is a RAP instance implemented as a RAP/MSRP Gateway instance?	O	5.6	Yes [] No []
RNB	RAP Native Bridge implemented?	O	5.7	Yes [] No []
RCB	RAP Controlled Bridge implemented?	O	5.8	Yes [] No []
RBP	RAP Bridge Proxy implemented?	O	5.9	Yes [] No []
FRB	FRER-capable RAP Bridge implemented	O	5.8	Yes [] No []
RNE	RAP Native end station implemented?	O	5.9	Yes [] No []
RCE	RAP Controlled end station implemented?	O	5.10	Yes [] No []
REP	RAP End Station Proxy implemented?	O	5.13	Yes [] No []
FRT	FRER-capable RAP Talker implemented?	O	5.10	Yes [] No []
FRL	FRER-capable RAP Listener implemented?	O	5.11	Yes [] No []
RMG	RAP/MSRP Gateway implemented?	O	5.12	Yes [] No []

4

A.6 RAP Relay instance capabilities

Item	Feature	Status	References	Support
RRI-1	Does the implementation support the RPSI and the RAP Participant state machine?	RRI:M	5.4 item a), 6.7	Yes []
RRI-2	Does the implementation support the RAP Propagator state machine?	RRI:M	5.4 item b), 6.7	Yes []
RRI-3	Does the implementation support the Application Information TLV?	RRI:M	5.4 item c), 6.6.6.4	Yes []
RRI-4	Does the implementation support the RAP attributes and their TLV encodings?	RRI:M	5.4 item d), 6.4	Yes []
RRI-5	Does the implementation support the RAP management?	RRI:M	5.4 item e), Clause 7	Yes []

A.7 RAP End instance capabilities

Item	Feature	Status	References	Support
REI-1	Does the implementation support the RPSI and the RAP Participant state machine?	REI:M	5.5 item a), 6.7	Yes []
REI-2	Does the implementation support the RESI and the RAP Endpoint procedures?	REI:M	5.5 item b), 6.5	Yes []
REI-3	Does the implementation support the Application Information TLV?	REI:M	5.5 item c), 6.6.6.4	Yes []
REI-4	Does the implementation support the RAP attributes and their TLV encodings?	REI:M	5.5 item d), 6.4	Yes []
REI-5	Does the implementation support the RAP management?	REI:M	5.5 item e), Clause 7	Yes []

A.8 RAP/MSRP Gateway instance capabilities

Item	Feature	Status	References	Support
RMGI-1	Does the implementation support the RPSI and the RAP Participant state machine?	RMGI:M	5.6 item a), 6.7	Yes []
RMGI-2	Does the implementation support the RAP Propagator state machine?	RMGI:M	5.6 item b), 6.7	Yes []
RMGI-3	Does the implementation support the RAP/MSRP Converter?	RMGI:M	5.6 item c), 6.7	Yes []
RMGI-4	Does the implementation support the Application Information TLV?	RMGI:M	5.6 item d), 6.6.6.4	Yes []
RMGI-5	Does the implementation support the RAP attributes and their TLV encodings?	RMGI:M	5.6 item e), 6.4	Yes []
RMGI-6	Does the implementation support the RAP management?	RMGI:M	5.6 item f), Clause 7	Yes []

A.9 RAP Native Bridge capabilities

Item	Feature	Status	References	Support
RNB-1	Does the implementation include a single conformant VLAN Bridge component?	RNB:M	5.7 item a), [Q]:5.4	Yes []
RNB-2	Does the implementation conform to the required and optional behaviors for a Native relay system?	RNB:M	5.7 item b), [CS]:5.6, [CS]:5.7	Yes []
RNB-3	Does the implementation support Edge Control Protocol on each Port and associated management entities?	RNB:M	5.7 item c), [Q]:43, [Q]:12.27	Yes []
RNB-4	Does the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RNB:M	5.7 item d), [CS]:6.9.1, 6.6.6.3	Yes []
RNB-5	Does the implementation include a single conformant RAP Relay instance?	RNB:M	5.7 item c), 5.4	Yes []

A.10 RAP Controlled Bridge capabilities

Item	Feature	Status	References	Support
RCB-1	Does the implementation include a single conformant VLAN Bridge component?	RCB:M	5.8 item a), [Q]:5.4	Yes []
RCB-2	Does the implementation conform to the optional behaviors for a Controlled system?	RCB:M	5.8 item b), [CS]:5.10	Yes []

A.11 RAP Bridge Proxy capabilities

Item	Feature	Status	References	Support
RBP-1	Does the implementation conform to the required and optional behaviors for a Proxy system?	RBP:M	5.9 item a), [CS]:5.8, [CS]:5.9	Yes []
RBP-2	Does the implementation include, for each RAP Controlled Bridge it proxies for, one conformant RAP Relay instance?	RBP:M	5.9 item b), 5.4	Yes []

A.12 FRER-capable RAP Bridge capabilities

Item	Feature	Status	References	Support
FRB-1	Does the implementation conform to the requirements for either a RAP Native Bridge or a RAP Controlled Bridge?	FRB:M	5.8, 5.7, 5.8	Yes []
FRB-2	Does the implementation conform to the required and optional behaviors for a FRER C-component?	FRB:M	5.8 item a), [CB]:5.15	Yes []
FRB-3	Does the implementation support Active Destination MAC and VLAN Stream identification function?	FRB:M	5.8 item b), [CB]:6.6	Yes []

A.13 RAP Native end station capabilities

Item	Feature	Status	References	Support
RNE-1	Does the implementation conform to the required behaviors for a Native end system?	RNE:M	5.9 item a), [CS]:5.4	Yes []
RNE-2	Does the implementation support the operation of the Applicant state machine and the Registrar state machine?	RNE:M	5.9 item b), [CS]:8.3, [CS]:8.4	Yes []
RNE-3	Does the implementation support Edge Control Protocol and associated management entities?	RNE:M	5.9 item c), [Q]:43, [Q]:12.27	Yes []
RNE-4	Does the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RNE:M	5.9 item d), [CS]:6.9.1, 6.6.6.3	Yes []
RNE-5	Does the implementation include a single conformant RAP End instance?	RNE:M	5.9 item c), 5.5	Yes []

A.14 RAP Controlled end station capabilities

Item	Feature	Status	References	Support
RCE-1	Does the implementation conform to the optional behaviors for a Controlled system?	RCE:M	5.10 item a), [CS]:5.10	Yes []

A.15 RAP End Station Proxy capabilities

Item	Feature	Status	References	Support
REP-1	Does the implementation conform to the required and optional behaviors for a Proxy system?	REP:M	5.13 item a), [CS]:5.8, [CS]:5.9	Yes []
REP-2	Does the implementation include, for each RAP Controlled end station it proxies for, one conformant RAP End instance?	REP:M	5.13 item b), 5.5	Yes []

A.16 FRER-capable RAP Talker capabilities

Item	Feature	Status	References	Support
FRT-1	Does the implementation conform to the requirements for either a RAP Native end station or a RAP Controlled end station?	FRT:M	5.10, 5.9, 5.10	Yes []
FRT-2	Does the implementation conform to the required, recommended and optional behaviors for a Talker end system?	FRT:M	5.10 item a), [CB]:5.6, [CB]:5.7, [CB]:5.8	Yes []

A.17 FRER-capable RAP Listener capabilities

Item	Feature	Status	References	Support
FRL-1	Does the implementation conform to the requirements for either a RAP Native end station or a RAP Controlled end station?	FRL:M	5.11, 5.9, 5.10	Yes []
FRL-2	Does the implementation conform to the required, recommended and optional behaviors for a Talker end system?	FRL:M	5.11 item a), [CB]:5.9, [CB]:5.10, [CB]:5.11	Yes []

A.18 RAP/MSRP Gateway capabilities

Item	Feature	Status	References	Support
RMG-1	Does the implementation include a single conformant VLAN Bridge component?	RMG:M	5.12 item a), [Q]:5.4	Yes []
RMG-2	Does the implementation include a single conformant RAP/MSRP Gateway instance?	RMG:M	5.12 item b), 5.6	Yes []
RMG-3	On each Port that supports the operation of RAP, dose the implementation conform to the required and optional behaviors for a Native relay system?	RMG:M	5.12 item c).1), [CS]:5.6, [CS]:5.7	Yes []
RMG-4	On each Port that supports the operation of RAP, dose the implementation support Edge Control Protocol and associated management entities?	RMG:M	5.12 item c).2), [Q]:43, [Q]:12.27	Yes []
RMG-5	On each Port that supports the operation of RAP, dose the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RMG:M	5.12 item c).3), [CS]:6.9.1, 6.6.6.3	Yes []
RMG-6	On each Port that supports the operation of MSRP, dose the implementation include a conformant MSRP implementation for an end station?	RMG:M	5.12 item d), [Q]:5.18.3	Yes []

1 Annex B

2 (informative)

3 Resource allocation examples

4 This Annex provides some examples of resource allocation using the Resource Allocation Protocol (RAP)
5 specified in this standard. These examples are chosen only for the purpose of illustrating the operation of
6 resource allocation under different scenarios, and are not intended to constraint the range of applicability of
7 RAP in terms of all of the possible usage scenarios.

8 Note that, unless otherwise stated, the examples in this Annex are based on the following assumptions:

- 9 a) In Talker Announce propagation (6.3.4.1), no further conditions, other than those related to the
10 VLAN Context rule as defined in 6.3.5.1.1, that would prevent a Talker Announce registration from
11 being propagated to other Port(s), are met.
- 12 b) Resource allocation is successful without encountering any problems that would cause Talker
13 Announcement or Listener Attachment to fail (except the example in B.5)

14 B.1 Single-path stream example

15 This subclause gives an example of resource allocation for transmission of a stream along a single path
16 between the Talker and each Listener. Figure B-1 shows the VLAN topology configured in the network for
17 use as a VLAN Context in the resource allocation for the stream.

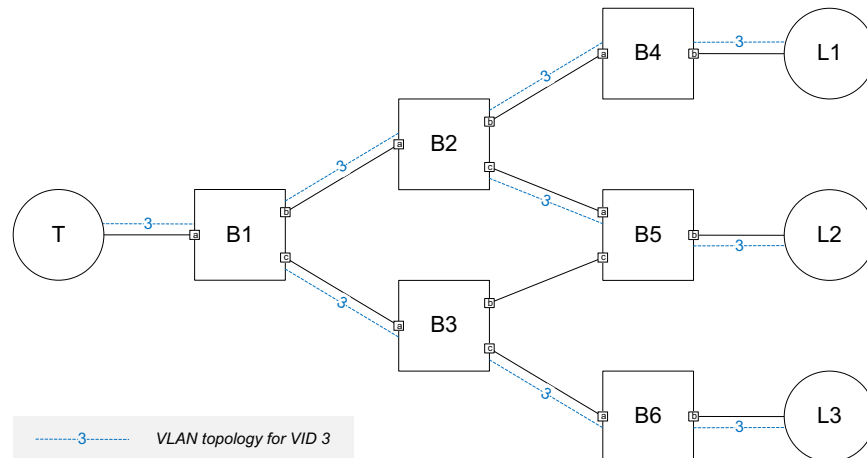


Figure B-1—Single-path stream example: topology and VLAN configuration

1 Figure B-2 illustrates the propagation of a Single-Context Talker Announcement [item a) in 6.3.5.1.1]
2 initiated by the Talker within the VLAN Context. Note that the Talker Announcement does not go through
3 the link between B3 and B5, as this link is not part of the VLAN Context in use.

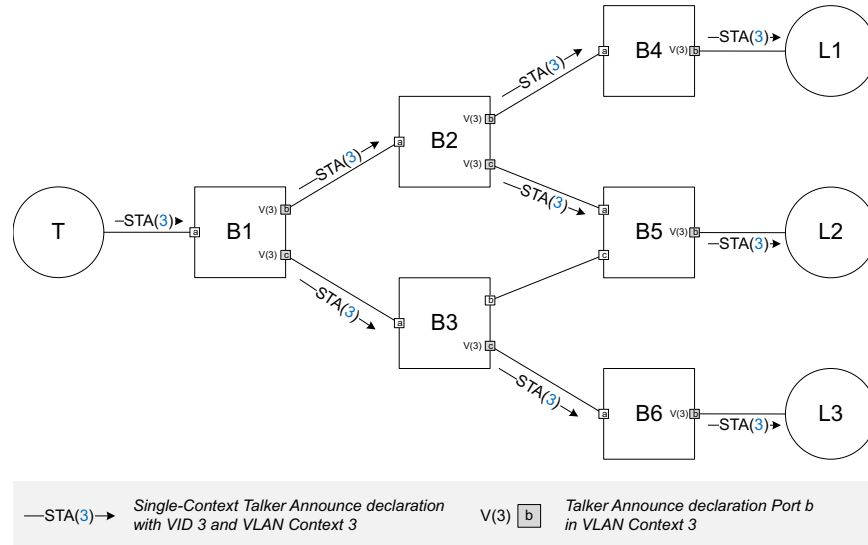


Figure B-2—Single-path stream example: Talker Announcement

4 As illustrated in Figure B-3, three Listeners participate in the Single-Context Listener Attachment [item a) in
5 6.3.4.1], which is propagated along the path previously traversed by the associated Talker Announcement in
6 reversed direction and is subject to Listener Attach merge (6.3.4.2) on both Port B2.a and Port B1.a.

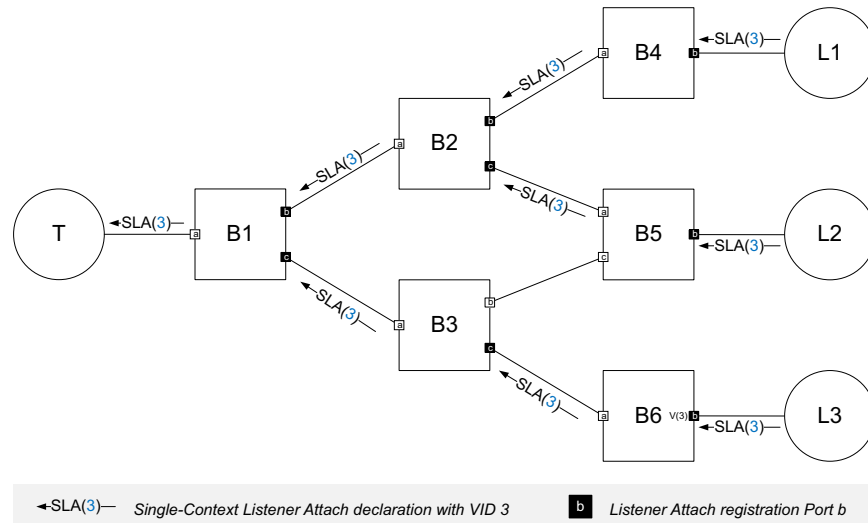


Figure B-3—Single-path stream example: Listener Attachment

1 Figure B-4 shows the single path established for transmission of the stream from the Talker to each Listener
2 after successful resource allocation.

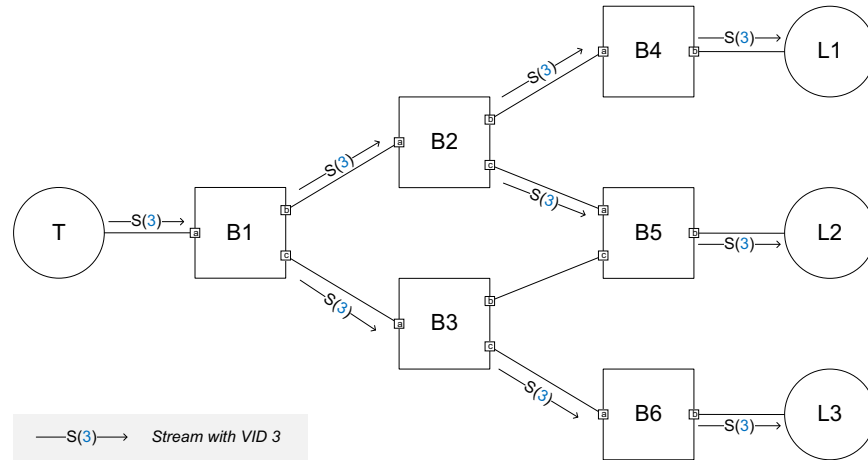


Figure B-4—Single-path stream example: established stream

3 B.2 Multi-path stream example 1

4 This subclause gives an example of resource allocation for transmission of a Compound Stream in a network
5 analogous to the one described in [CB]:C.1. As shown in Figure B-5, three VLAN topologies representing
6 an instance of redundant trees constitute a Redundancy Context used in the resource allocation. In this
7 example, FRER functionality is provided only in three FRER-capable end stations.

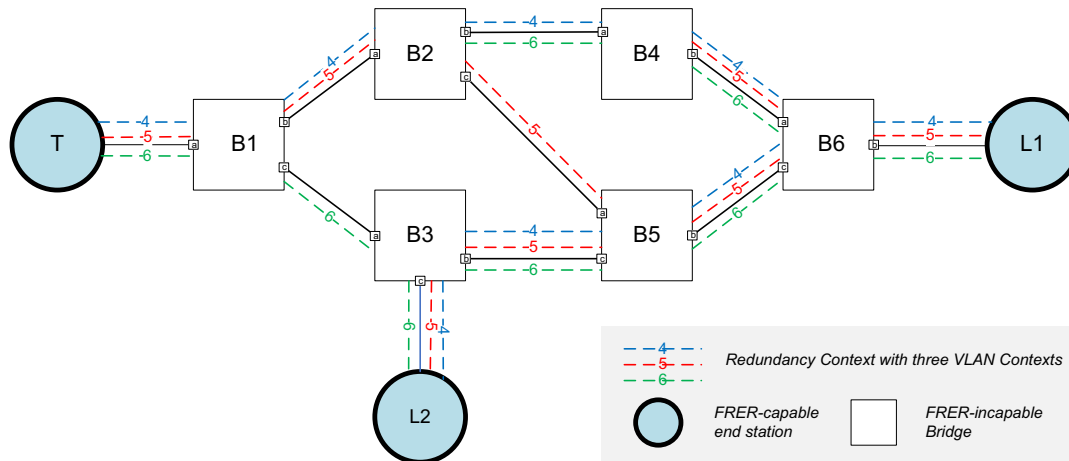
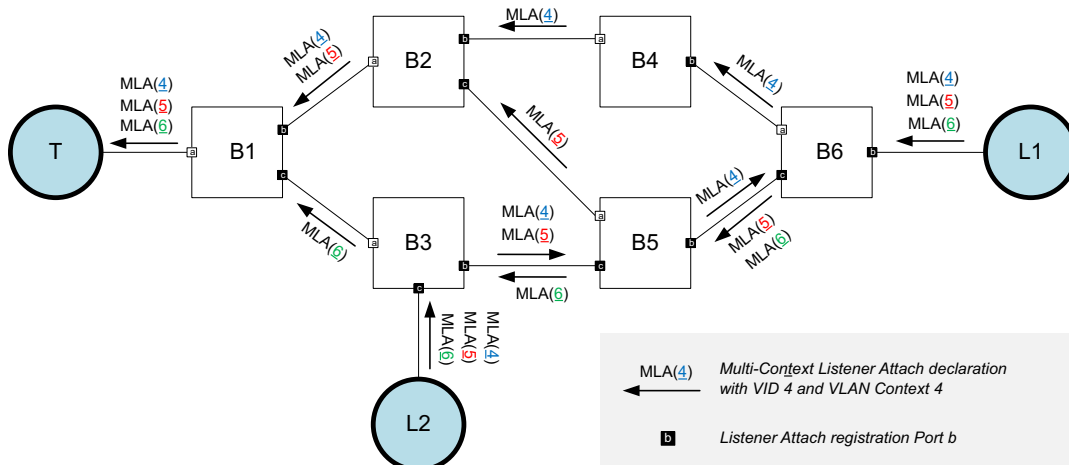


Figure B-5—Multi-path stream example 1: topology and VLAN configuration

8 As illustrated in Figure B-6, the FRER-capable Talker initiates a Multi-Context Talker Announcement [item
9 b) in 6.3.5.1.1] with three separate Talker Announce declarations, each using a different VID and VLAN
10 Context in the Redundancy Context to represent a Member Stream split from the Compound Stream. Since
11 there are no FRER-capable Bridges in the network, each of these Talker Announce declarations is
12 propagated and processed by the Bridges in the network independently.

Figure B-7 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker Announcement shown in Figure B-6. Each Listener makes three Listener Attach declarations, each with the same VID and VLAN Context as received in one of the three Talker Announce registrations. Each Listener Attach declaration made by a Listener is propagated along the path formed by the set of Bridge Ports that contain the associated Talker Announce registration. Listener Attach merge (6.3.4.2) occurs on the Bridge Ports B5.a and B6.a where two Listener Attach registrations propagated from the different Listeners in the same VLAN Context are merged into a joint Listener Attach declaration, indicating the need for the Bridge to multicast the Member Stream received on that Port.



123
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1 Figure B-8 shows the paths established and the FRER functions configured for redundant transmission of
2 the Compound Stream with three Member Streams after successful resource allocation.

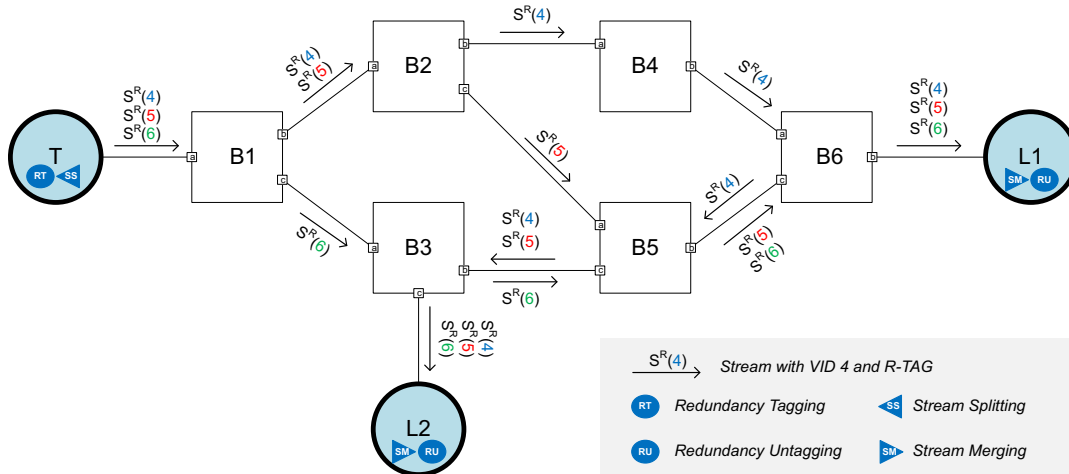


Figure B-8—Multi-path stream example 1: established stream

3 B.3 Multi-path stream example 2

4 This subclause gives an example of resource allocation for transmission of a Compound Stream in a network
5 analogous to the one described in [CB]:C.2. As shown in Figure B-9, this example has the same physical
6 topology as that in Figure B-5, but different placement of FRER-capable stations and VLAN configuration.
7 As both the Talker and Listener are not FRER-capable, three FRER-capable Bridges B1, B5 and B6 are
8 placed at the edge of the network as proxies to offer FRER functionality to the end stations. Additionally,
9 another FRER-capable Bridge B2 is placed within the network to split the VLAN topologies, in conjunction
10 with B1, into three disjoint paths. Note that the reason for using a different VLAN configuration in this
11 example is to ensure the VLAN membership configuration in each FRER-capable Bridge where stream
12 splitting is expected, e.g., B1 and B2, meet the condition described in b).) of 6.3.8.3.

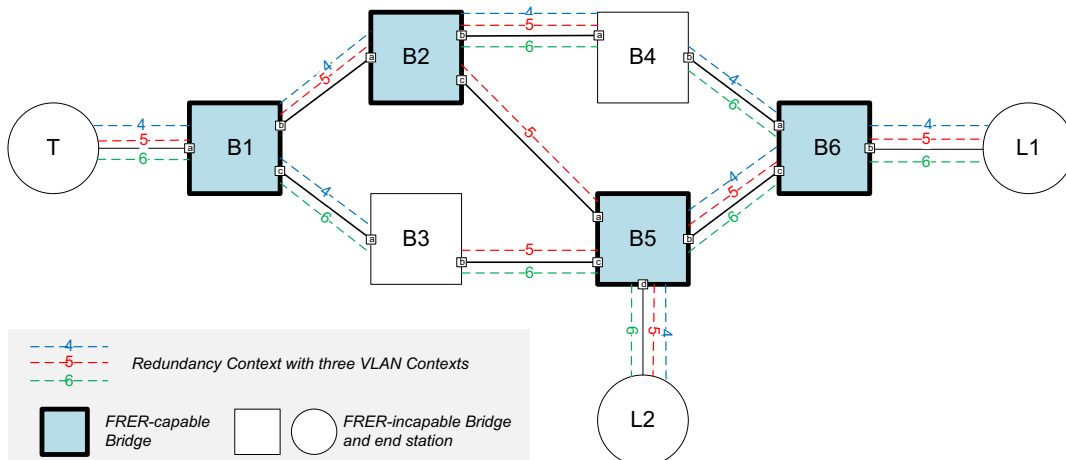


Figure B-9—Multi-path stream example 2: topology and VLAN configuration

1 As illustrated in Figure B-10, the Talker initiates a Multi-Context Talker Announcement by making a Talker
2 Announce declaration for all of the three VLAN Contexts and with RTagStatus (6.4.3.9.1) set FALSE. The
3 RTagStatus flag of the Talker Announcement is set TRUE by B1, which is responsible for redundancy
4 tagging in accordance with the rule described in a) of 6.3.8.2. Both B1 and B2 are responsible for applying
5 stream splitting to the stream, as all the conditions described in b) of 6.3.8.3 are met. The Talker
6 Announcement is subject to Talker Announce merge (6.3.5.3) on the Bridge Ports B5.b, B5.d and B6.b, each
7 corresponding to a potential stream merging port for the stream, in accordance with the rule described in a)
8 of 6.3.8.4. In addition, the RTagStatus flag of the Talker Announcement is set FALSE on B5.d and B6.b, as
9 the conditions for redundancy untagging are met on these two Ports, in accordance with the rule described in
10 b) of 6.3.8.2.

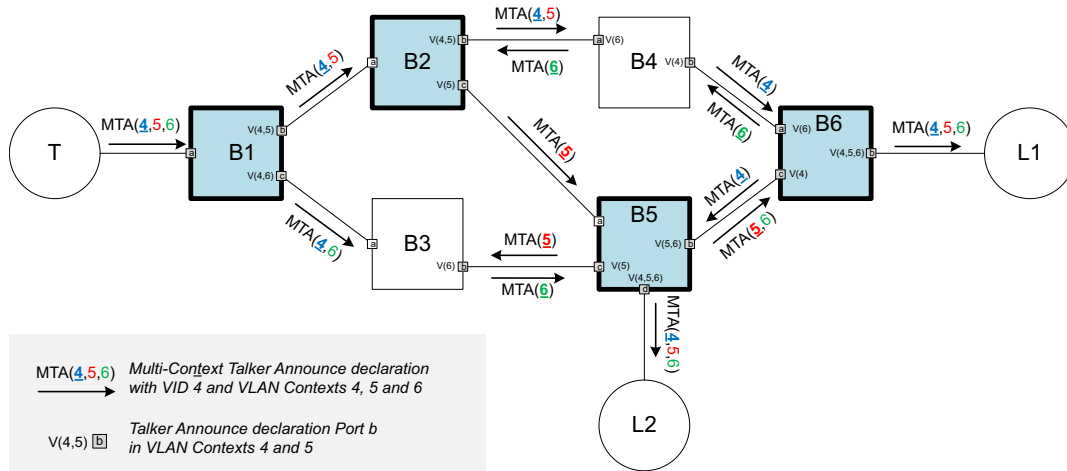


Figure B-10—Multi-path stream example 2: Talker Announcement

11 Figure B-11 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker
12 Announcement shown in Figure B-10. Each Listener makes a single Listener Attach declaration with the
13 same VLAN Contexts as received in the Talker Announce registration and a desired VID chosen from
14 among these VLAN Context IDs. Similar to that in Figure B-7, the Listener Attach declaration on both B5.a
15 and B6.a is merged from the two Listener Attach registrations propagated from different Listeners in the
16 same VLAN Context. Additionally in this example, Listener Attach merge (6.3.4.2) occurs on the Bridge
17 Ports B2.a and B1.a where two Listener Attach registrations containing the split VLAN Contexts are merged
18 into a single Listener Attach declaration with joint VLAN Contexts, as a reserve operation of splitting the
19 Talk Announcement previously performed on the same Port. Note that the Listener Attach declaration on
20 B2.a and B1.a takes a VID that is contained in the VLAN Context(s) of both of the Listener Attach
21 registrations merging into that Listener Attach declaration, because the multicast forwarding mechanism
22 used in stream splitting by FRER-capable Bridges requires stream transmission Ports to be in the member
23 set for a common VID.

1

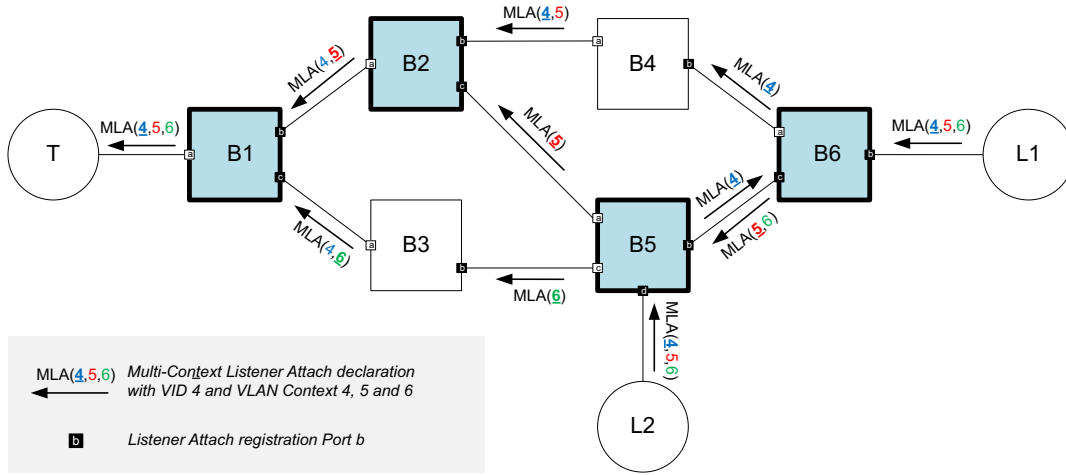


Figure B-11—Multi-path stream example 2: Listener Attachment

2 Figure B-12 shows the paths established and the FRER functions configured for redundant transmission of
3 the Compound Stream with three Member Streams after successful resource allocation.

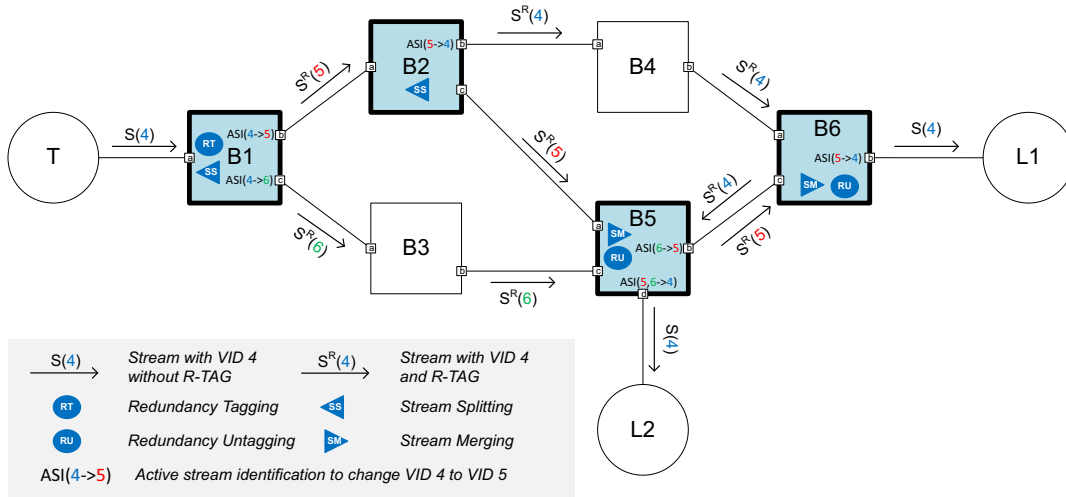


Figure B-12—Multi-path stream example 2: established stream

4 B.4 Multi-path stream example 3

5 This subclause gives an example of resource allocation for transmission of a Compound Stream in a network
6 analogous to the one described in [CB]:C.3. As shown in Figure B-13, FRER functionality is provided in
7 each end station and some of the Bridges. Four FRER-capable Bridges are connected as a ring, each acting
8 as both a stream splitting point and a stream merging point, to provide protection against multiple
9 simultaneous failures. A Redundancy Context with two VLAN topologies is configured in the network. In
10 order to meet the stream splitting condition described in b.) of 6.3.8.3, each FRER-capable Bridge has one
11 VLAN whose member set includes all the Bridge Ports. As a consequence, each of those Bridge Ports
12 marked as VLAN topology termination point needs be excluded from the member set for the indicated

- 1 VLAN and to have ingress filtering ([Q]:8.6.2) enabled for that VLAN, in order to ensure loop-free Talker
- 2 Announce propagation.

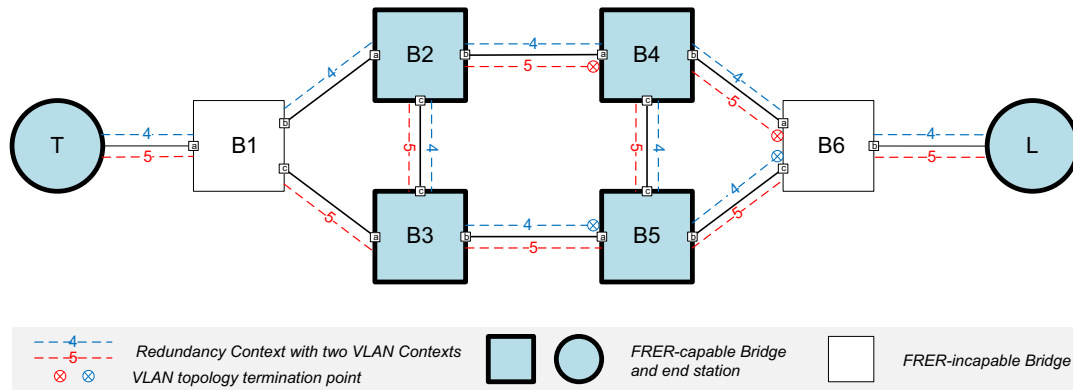


Figure B-13—Multi-path stream example 3: topology and VLAN configuration

3 As illustrated in Figure B-14, two Talker Announce declarations initiated by the FRER-capable Talker, each
4 with a distinct VLAN Context, are propagated by B1 separately. According to the VLAN configuration,
5 each FRER-capable Bridge propagates the Talker Announce registration on Port a with a single VLAN
6 Context to the other two Ports, and meanwhile merges on Port b two Talker Announce registrations
7 propagated from the other two Ports into a joint Talker Announce declaration with both VLAN contexts.
8 Note that on each Bridge Port that is excluded from the member set for one VLAN Context, ingress blocking
9 (6.3.5.1.3) ensures that the Talker Announce registration received with two VLAN Contexts is propagated
10 only with the other VLAN Context for which the Port in the member set.

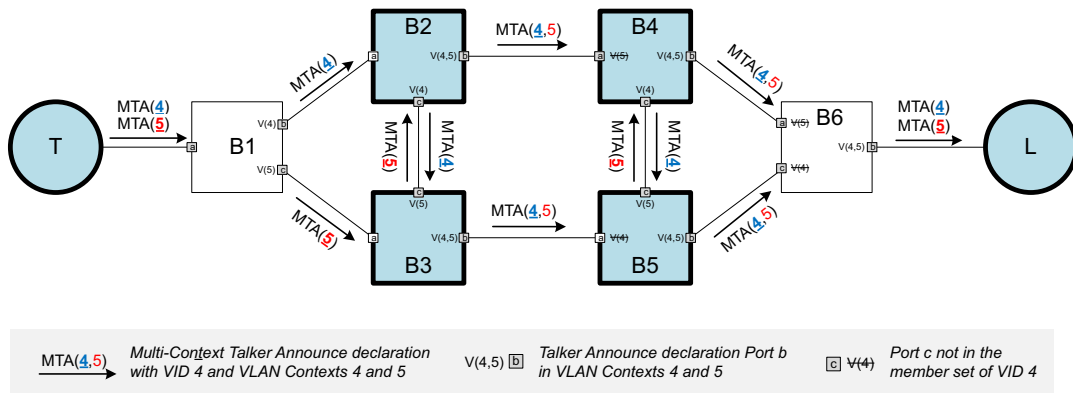


Figure B-14—Multi-path stream example 3: Talker Announcement

11 Figure B-15 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker
12 Announcement shown in Figure B-14. The FRER-capable Listener makes two Listener Attach declarations,
13 each with a different VLAN Context. Listener Attach merge occurs on Port a of each FRER-capable Bridge,
14 resulting in a joint Listener Attach declaration with the same VLAN Context as that contained in the

1 associated Talker Announce registration on the same Port. Finally, B1 passes two Listener Attach
2 declarations, each with a different VLAN Context, to the Talker.

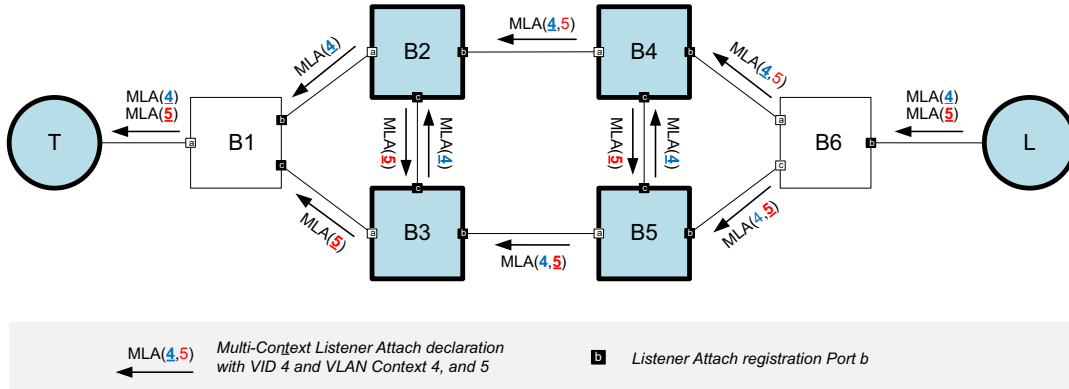


Figure B-15—Multi-path stream example 3: Listener Attachment

3 Figure B-16 shows the paths established and the required FRER functions configured for redundant
4 transmission of the Compound Stream with two Member Streams after successful resource allocation

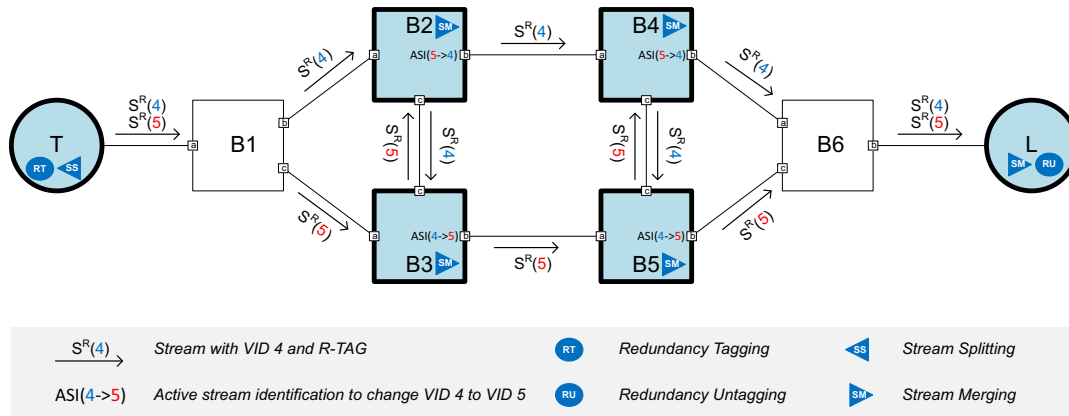


Figure B-16—Multi-path stream example 3: established stream

5 B.5 Example of status notification and failure diagnosis

6 This subclause uses the example network of B.4 to illustrate the status notification and failure diagnosis
7 mechanisms in resource allocation. As shown in Figure B-17, it is assumed that the Multi-Context Talker
8 Announcement encounters resource problems on three Bridge Ports.

9 The problem on Port B1.b causes the Talker Announce declaration on that Port to change to the Announce
10 Fail status [item b) in 6.3.5.2] and to attach the failure information about the problem on that Port to VLAN
11 Context 4. The failure information generated on B1.b remains attached to VLAN Context 4 in the
12 subsequent propagation of the Talker Announcement across the network toward the Listener, regardless of
13 the situations encountered on any downstream Port. Similarly, the problems on B3.b and B4.b lead to
14 Announce Fail in the Talker Announce declarations on those Ports. The failure information of VLAN
15 Context 5 is generated on B3.b, and is later merged along with that of VLAN Context 4 to the Talker
16 Announce declarations on B4.b and B5.b. The reason for the Announce Success status on both B2.b and
17 B5.b is that the Talker Announce declaration merges one Talker Announce registration propagated as

1 Announce Success, meaning that a reservation can be made for a stream on a merged path as long as at least
2 one of the paths merging into that merged path is available for reservation.

3 In the end, the Listener receives two Talker Announce declarations, which indicate that at least one path in
4 the network can be reserved for transmission of the Compound Stream as indicated by the Announce
5 Success status of one Talker Announce registration, even though there are problems detected at the locations
6 associated with both VLAN topologies as indicated in the supplied failure information.

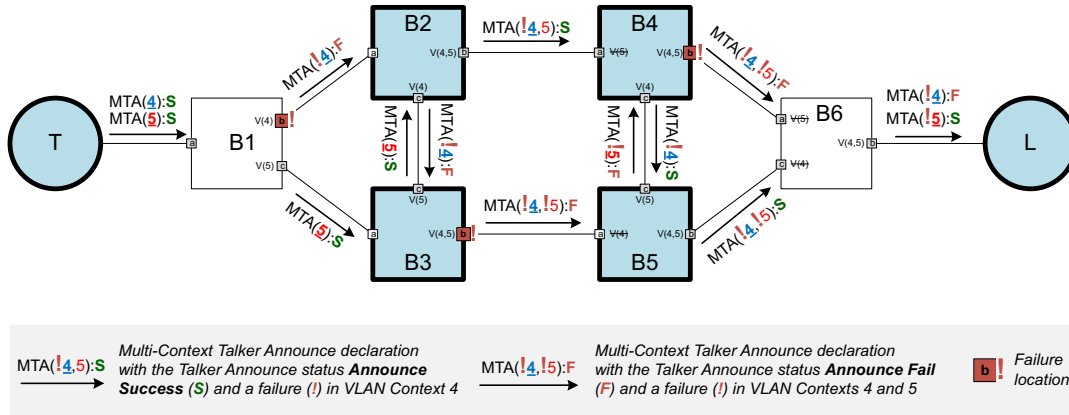


Figure B-17—Example Talker Announce status and failure information

7 Figure B-18 illustrates the Multi-Context Listener Attachment associated with the Talker Announcement
8 shown in Figure B-17. The Listener makes two Listener Attach declarations with Attach Ready contained in
9 the Listener Attach status (6.3.4.3), indicating it is ready to receive both Member Streams. As the Listener
10 Attachment is propagated along each stream path back to the Talker, a corresponding reservation is made on
11 each Port for a Listener Attach registration with the Attach Ready status and its associated Talker Announce
12 declaration with the Announce Success status. A Listener Attach registration for which a reservation cannot
13 be made is propagated as Attach Fail. On each Bridge Port where Listener Attach merge occurs in this
14 example, i.e., Port a or each FRER-capable Bridge, the Listener Attach declaration is made with Attach
15 Ready because one of the two Listener Attach registrations being merged on that Port is propagated as
16 Attach Ready. Note that, on those Ports with a terminated VLAN, such as B6.a and B6.c, the Listener Attach
17 declaration still contains the terminated VLAN (e.g., VLAN Context 5 on B6.a) in order to match the VLAN
18 Contexts contained in the associated Talker Announce registration on the same Port, but needs to set the
19 corresponding Path status to Unresponsive Path to indicate that Listener Attach declaration does not
20 originate from that VLAN Context.

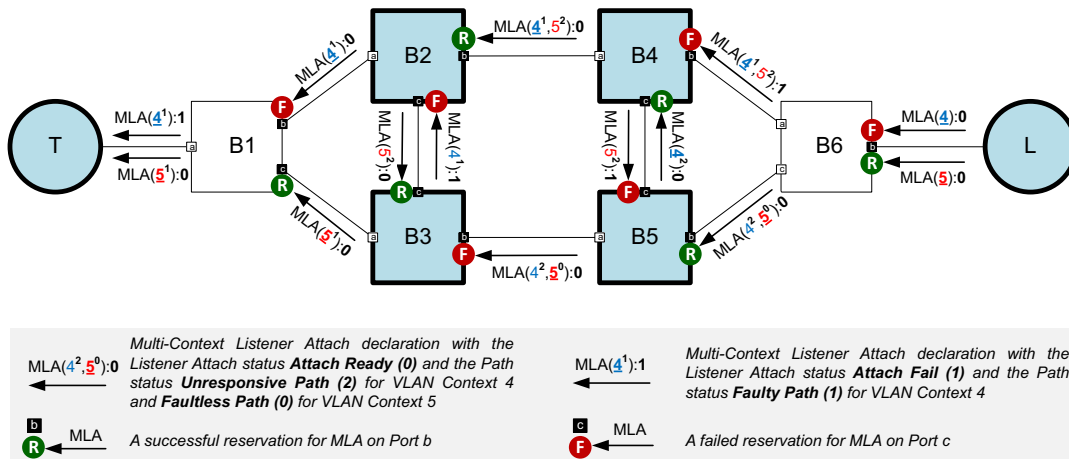


Figure B-18—Example Listener Attach status and Path status

1 Figure B-19 shows the single path established and the FRER functions configured for the Compound Stream
2 in the case of three problems in the network.

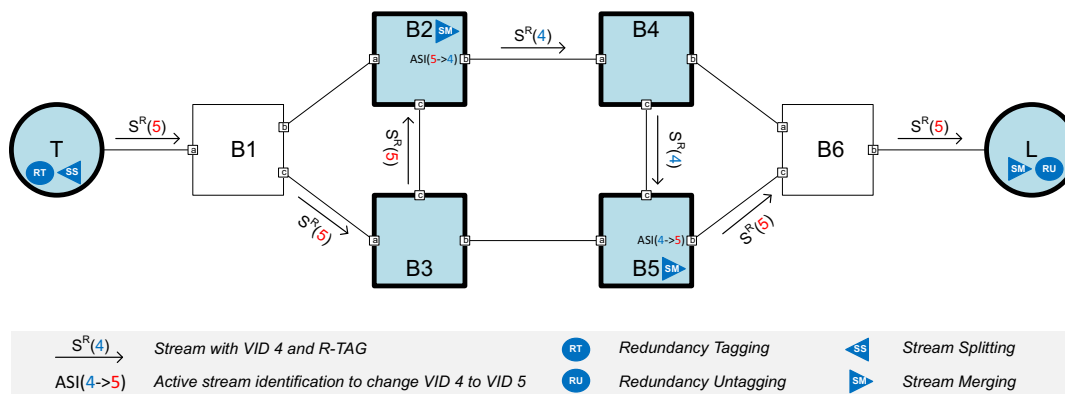


Figure B-19—Example partially established Compound Stream

¹ Annex C

² (informative)

³ Bibliography

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